EXAM PACK

EEE339

Power Electronics



Exam Pack

Session 2020–21 *Solutions and formula sheet of the Final Examination 2023 and Term Tests 1–2*

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Question Paper Solutions 2020–21 — pages 3--76 — 74 pages

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Prepared by Nur Mohammad

November 2026

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PAST PAPER SOLUTIONS

EEE339

Power Electronics

Question Paper Solutions

Session 2020–21

Final Examination 2023 (Q1–Q6) and Term Tests 1–2 (Q7–Q8)

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

Q7	Q8
20 marks	20 marks

EACH ANSWER



Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks); a part with its own time (a term test) uses that. Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
PART A				
Q1	(a) Distortion Factor and Total Harmonic Distortion	7	13 min	☐
	(b) Reverse Recovery Current and Time	6	11 min	
	(c) Peak Reverse Current and Storage Charge	6	11 min	
	(d) Ninth Harmonic Current from the THD	6	11 min	
Q2	(a) Two-Transistor Model of a Thyristor	7	13 min	☐
	(b) Half-Wave Controlled Rectifier with an RL Load	10	18 min	
	(c) Full-Wave Controlled Bridge with a Resistive Load	8	14 min	
Q3	(a) Three-Phase Uncontrolled Rectifier	8	14 min	☐
	(b) IGBT Structure, Punch-Through and Non-Punch-Through	7	13 min	
	(c) Advantages of DC Power Transmission	4	7 min	
	(d) Three-Phase Controlled Rectifier with an RL Load	6	11 min	
PART B				
Q4	(a) Linear DC Voltage Regulator	6	11 min	☐
	(b) Buck Converter Waveforms and Currents	8	14 min	
	(c) Boost Converter Design	6	11 min	
	(d) SEPIC and Cuk Converters	5	9 min	
Q5	(a) DC-DC Converters in a Stand-Alone Solar PV System	5	9 min	☐
	(b) Cascaded Multilevel Inverter and Pattern Swapping	6	11 min	
	(c) Bipolar PWM Inverter Design	10	18 min	
	(d) Voltage THD of a Square-Wave Inverter	4	7 min	
Q6	(a) Inverter with a Series LC Filter	7	13 min	☐
	(b) Cycloconverter	5	9 min	
	(c) TRIAC Structure and Power BJT Construction	8	14 min	
	(d) Comparison of Power Devices	5	9 min	
Term Test 1 · 20 marks · 30 min				
Q7	(a) Reverse Recovery Time	4	6 min	☐
	(b) Thyristor Characteristic, Holding and Latching Current	5	8 min	
	(c) Full-Wave Bridge Rectifier with an RL Load	5	8 min	
	(d) Nonlinear Load: Power, Power Factor and Distortion	6	9 min	
Term Test 2 · 20 marks · 30 min				
Q8	(a) Boost Converter Circuit and Waveforms	5	8 min	☐
	(b) Buck Converter Numerical	7.5	11 min	
	(c) Speed Control by Chopping the Field Supply	2.5	4 min	
	(d) Three-Phase Rectifier: RMS Load and Diode Currents	5	8 min	

Follows Hart, Power Electronics (distortion, rectifiers, converters, inverters); Rashid, Power Electronics, 4th ed. (devices, cycloconverters, HVDC); Rashid's Handbook (solar PV).

Shahjalal University of Science & Technology

Department of Electrical & Electronic Engineering

3rd Year 2nd Semester Final Examination – 2023

Course No: EEE 339 **Session:** 2020-21

Course Title: Power Electronics

Full Marks: 100 **Credits: 3.0** **Time: 3 Hours**

[Answer any four questions taking two from each part]

PART A

1. (a) Define distortion factor (DF) and total harmonic distortion (THD). Derive that $DF = \sqrt{\frac{1}{1+(THD)^2}}$ 7

   (b) Define reverse recovery current and reverse recovery time for diode switching. What is the importance of these parameters? 6

   (c) The manufacturer of a selected diode gives the rate of fall of the diode current di/dt=20 A/μs, and its reserve [*sic*: should be reverse] recovery time t_{rr}=5 μs. What value of peak reverse current and the storage charge do you expect? 6

   (d) A sinusoidal current of 10A rms at a 60 Hz fundamental frequency is contaminated with a ninth harmonic current. The current is expressed as $i(t) = 10\sqrt{2}\sin(2\pi 60t) + I_9\sqrt{2}\sin(18\pi 60t)$ A. Determine the value of the ninth harmonic rms current I9 if the THD is -i) 5 percent ii) 20 percent. 6

2. (a) Draw the two transistors model of a thyristor and explain the turn-on condition of a thyristor with equations from this model. 7

   (b) For the circuit of Fig. 2(b), the source is 120 V rms at 60 Hz, R=20 Ohm, L=0.04 H, and the delay angle is 45°. Determine (i) an expression for i(ωt) , (ii) the average current, (iii) the power absorbed by the load, and (iv) the power factor. 10

   (c) A full-wave controlled bridge rectifier has an ac input of 120 V rms at 60 Hz and a 20 Ω load resistor. The delay angle is 40°. Determine the average current of the load, the power absorbed by the load, the apparent power of the sourced [*sic*: should be source] and the input pf. 8

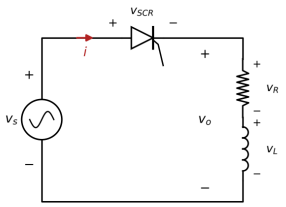


Fig 2(b) of the paper (redrawn)

3. (a) Draw the configuration of three phase uncontrolled rectifier. Show the source voltages, output voltage, diode currents and line currents with proper alignment. 8

   (b) Draw the cross section of an IGBT showing the layers. Differentiate between punch-through and non-punch through IGBTs. 7

   (c) Write down the advantages of DC power transmission. 4

   (d) A three phase controlled rectifier has an input voltage of 480 V rms at 60 Hz. The load of the rectifier is resistor of 10 Ω in series with 50 mH inductance.

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(i) Determine the delay angle required to produce an average current of 50A in the load
(ii) Determine the amplitude of the harmonics n=6 and n=12. For harmonics calculation, Fig. 3(d) can be utilized. 6

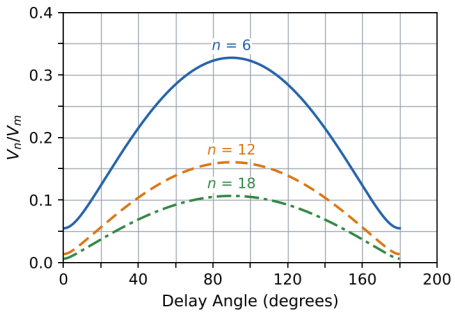


Fig. 3(d): Normalized output voltage harmonics as a function of delay angle for a three-phase rectifier (redrawn)

PART B

4. (a) Draw a typical circuit of a linear dc voltage regulator. What is the drawback of it? How can it be overcome? 6

(b) Draw the circuit diagram and the waveform of inductor current, diode current and capacitor current for a typical buck converter. Also derive the relation between inductor current and diode current from the waveforms. 8

(c) A boost converter has an input of [unclear: 5 V] and an output of 25 W at 15 V. The minimum inductor current must be no less than 50% of the average. The ripple output must be less than 1 percent. The switching frequency is [unclear: 300 kHz]. (i) Determine the duty ratio. (ii) Minimum inductor value (iii) Minimum capacitor value 6

(d) Differentiate between SEPIC converter and Ćuk converter. 5

5. (a) Briefly explain the types of switch mode DC-DC converters used in a stand-alone solar PV system. 5

(b) Draw the circuit of a three-level cascade multi source inverter. Why pattern swapping is important in multi-source inverter circuit? 6

(c) Design a bipolar PWM inverter that will produce a 75-V rms 60-Hz output from a 150 V dc source. The load is a series RL combination with R=12Ω and L=60 mH. Select the switching frequency such that the current THD is less than 10%. Assume the impedance at the carrier frequency is entirely inductive. 10

(d) Determine the voltage THD of a square wave inverter. 4

Normalized Fourier Coefficients V_n/V_{dc} for Bipolar PWM

	$m_a = 1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n = 1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n = m_f$	0.60	0.71	0.82	0.92	1.01	1.08	1.15	1.20	1.24	1.27
$n = m_f \pm 2$	0.32	0.27	0.22	0.17	0.13	0.09	0.06	0.03	0.02	0.00

The table printed under Q5(c)

6. (a) A 10 Ohm resistive load requires a 1000-Hz, 50-V rms sinusoidal voltage. The THD of the load voltage must be no more than 5 percent. An adjustable dc source is available. (i) Design an inverter for this application. (ii) What is maximum voltage across the capacitor? 7

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(b) What is cyclo-converter? Mention its applications. 5

(c) Explain the structure of TRIAC with diagram. Mention the constructional feature of a Power BJT. 8

(d) Compare the following device in terms of power capacity and switching speed: BJT, MOSFET, GTO, IGBT, MCT 5

Term Test 1 (its questions 1 to 4 are 7(a) to 7(d))

[Term Test-1, Marks: 20, Time: 30 minutes]

7. (a) Explain the reverse recovery time of diode with diagram. 4

(b) Draw the I-V curve of thyristor. Define Holding current and Latching current for thyristor 5

(c) With proper alignment, show the waveshape of output voltage, output current, diode currents and source current for a full wave bridge rectifier with sinusoidal source and R-L load. 5

(d) A sinusoidal voltage source of $v(t) = 240\sqrt{2}\sin(2\pi 60t)$ V is applied to a nonlinear load, resulting in a nonsinusoidal current which is expressed in Fourier series form as $i(t) = 8\sin(2\pi 60t) + 4\sin(4\pi 60t)$ A. Determine (a) the power absorbed by the load, (b) the power factor of the load, (c) the distortion factor of the load current, (d) the total harmonic distortion of the load current, (e) crest factor of the load current. 6

Term Test 2 (its questions 1 to 4 are 8(a) to 8(d))

[Term Test-2, Marks: 20, Time: 30 minutes]

8. (a) Draw the circuit diagram, equivalent circuit for switch on and off condition and the waveforms of the inductor current, diode current and load voltage for a typical boost converter. 5

(b) A buck converter has parameters, Vs=50 V, D=0.4, R=20 Ω, L=400 μH and C=100 μF. The switching frequency is 20 kHz. a) Determine the output voltage, b) Determine the average, minimum and maximum inductor current c) The output ripple voltage. 7.5

(c) Draw the circuit diagram to control the speed of a separately excited dc motor by chopping field current supply. Explain the operation using required equations. 2.5

(d) A three phase rectifier is supplied by 480 V rms line to line 60 Hz source. The RL load is 100 Ω resistor in series with 15 mH inductor. Determine the rms load current and rms diode currents. 5

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1. (a) Define distortion factor (DF) and total harmonic distortion (THD). Derive that $DF = \sqrt{\frac{1}{1+(THD)^2}}$ Marks: 7

   (b) Define reverse recovery current and reverse recovery time for diode switching. What is the importance of these parameters? Marks: 6

   (c) The manufacturer of a selected diode gives the rate of fall of the diode current di/dt=20 A/μs, and its reserve [*sic*: should be reverse] recovery time t_{rr}=5 μs. What value of peak reverse current and the storage charge do you expect? Marks: 6

   (d) A sinusoidal current of 10A rms at a 60 Hz fundamental frequency is contaminated with a ninth harmonic current. The current is expressed as $i(t) = 10\sqrt{2}\sin(2\pi 60t) + I_9\sqrt{2}\sin(18\pi 60t)$ A. Determine the value of the ninth harmonic rms current I9 if the THD is -i) 5 percent ii) 20 percent. Marks: 6

1.(a) Define distortion factor (DF) and total harmonic distortion (THD). Derive that $DF = \sqrt{\frac{1}{1+(THD)^2}}$ Marks: 7

1. Theory

1. What is a harmonic? (Core Idea)

The mains gives a smooth sine wave. Many power electronic loads do not draw a smooth sine wave of current. They draw current in pulses, so the current wave has a distorted shape.

A French mathematician, Fourier, showed that any repeating wave can be built by adding sine waves. The first has the same frequency as the wave. It is called the **fundamental**. The others have frequencies 2, 3, 4, … times the fundamental. They are called **harmonics**. For a 60 Hz wave, the 3rd harmonic has frequency 180 Hz and the 9th has frequency 540 Hz.

$$i(t) = I_0 + \sum_{n=1}^{\infty} \sqrt{2}\, I_n \sin(n\omega t + \phi_n)$$

where I_0 is the average (dc) part, I_n is the rms value of harmonic n, $\omega = 2\pi f$ and ϕ_n is the phase of harmonic n.

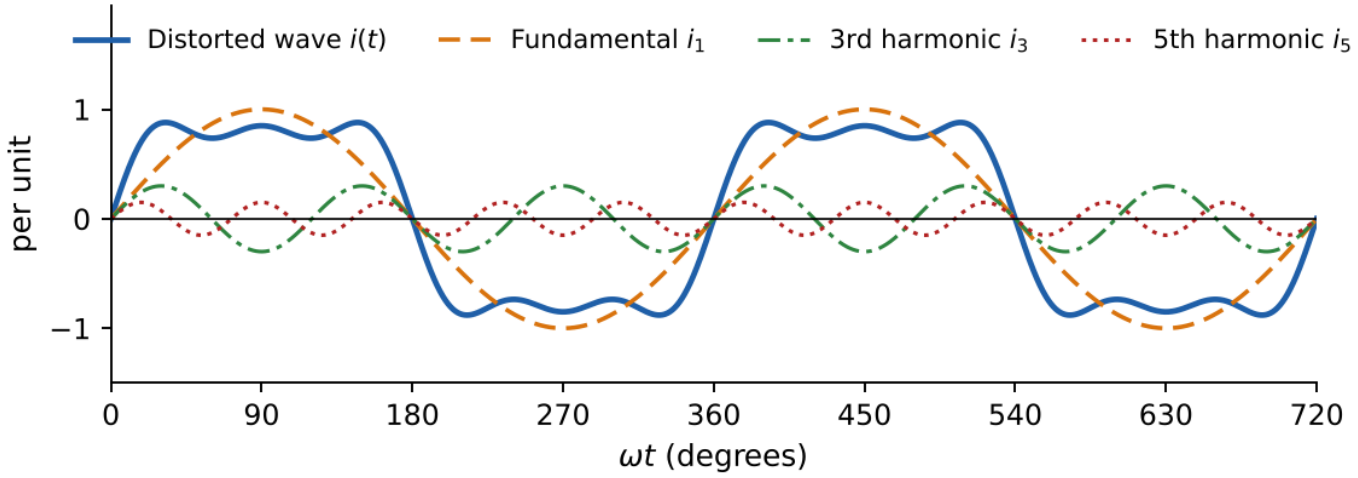


Figure 1: A distorted wave (solid blue curve) is the sum of its fundamental (dashed orange curve), a 3rd harmonic (dash-dot green curve) and a 5th harmonic (dotted red curve)

Reading the figure. Add the three thin curves at any angle and you get the height of the thick solid blue curve. The solid blue wave repeats at the same rate as the fundamental, but it is flatter on top. The harmonics are what change the shape.

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2. Why does the rms value add as a sum of squares?

The rms value tells us the heating effect of a current. To find it, we square the wave, take the average and then take the square root. When we square a sum of sines, we get two kinds of terms: each sine multiplied by itself, and each sine multiplied by a different one. Over one period, a product of two sines of different frequency averages to zero. Only the squares are left. So with no dc part

$$I_{rms}^2 = I_1^2 + I_2^2 + I_3^2 + \cdots = \sum_{n=1}^{\infty} I_n^2$$

3. Why do we need two measures?

A pure sine wave has only I_1. We want a number that says how far a wave is from that ideal. There are two natural ways to compare:

- Compare the harmonics with the fundamental. This gives the **THD**. A pure sine has THD = 0.
- Compare the fundamental with the whole wave. This gives the **distortion factor**. A pure sine has DF = 1.

They describe the same thing, so one can be found from the other. That link is what the question asks for. (A third number, the **crest factor**, compares the peak of the wave with its rms value; it is not needed here.)

4. Why does the distortion factor matter?

Only the fundamental current, with a sinusoidal source, can carry average power. Harmonic currents heat the wires but deliver no average power from a sinusoidal source. The power factor of such a load is

$$pf = \frac{P}{S} = \frac{V_1 I_1 \cos\theta_1}{V_1 I_{rms}} = \frac{I_1}{I_{rms}} \cos\theta_1 = DF \cos\theta_1$$

So distortion lowers the power factor, even if the current is exactly in phase with the voltage.

2. Answer

Distortion factor: the ratio of the rms value of the fundamental to the rms value of the whole wave (Hart, Eq. 2-65).

$$DF = \frac{I_1}{I_{rms}}$$

Total harmonic distortion: the ratio of the rms value of all the harmonics together to the rms value of the fundamental (Hart, Eq. 2-67).

$$THD = \frac{\sqrt{\sum_{n\neq 1} I_n^2}}{I_1}$$

Derivation of the relation.

Step 1: Write the current as a Fourier series

Take a periodic current with no dc part, as on an ac line. By Fourier's result it is a sum of its fundamental and harmonics, with I_n the rms value of harmonic n:

$$i(t) = \sum_{n=1}^{\infty} \sqrt{2}\, I_n \sin(n\omega t + \phi_n)$$

Step 2 (Basic): The rms value of a sum of harmonics

The rms value is the square root of the mean of i^2 over one period T. Squaring the sum gives two kinds of terms: each harmonic times itself, and each harmonic times a different one.

- A square term: $2I_n^2 \sin^2(n\omega t + \phi_n) = I_n^2[1 - \cos(2n\omega t + 2\phi_n)]$. The cosine completes whole cycles in T, so its mean is 0, and the mean of the term is I_n^2.

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- A cross term ($m \neq n$): $2I_m I_n \sin(m\omega t + \phi_m)\sin(n\omega t + \phi_n) = I_m I_n[\cos((m-n)\omega t + \phi_m - \phi_n) - \cos((m+n)\omega t + \phi_m + \phi_n)]$. Both cosines complete whole cycles in T (since $m-n$ and $m+n$ are nonzero whole numbers), so the mean is 0.

Only the square terms are left:

$$I_{rms}^2 = \frac{1}{T}\int_0^T i^2\,dt = \sum_{n=1}^{\infty} I_n^2 = I_1^2 + \sum_{n\neq 1} I_n^2$$

This is Hart's Eq. 2-64 with no dc term.

Step 3: Write the THD with the rms current

From Step 2, the harmonics together give $\sum_{n\neq 1} I_n^2 = I_{rms}^2 - I_1^2$. Put this under the square root of the THD:

$$THD = \frac{\sqrt{I_{rms}^2 - I_1^2}}{I_1}$$

(Hart, Eq. 2-68.)

Step 4: Square both sides

$$THD^2 = \frac{I_{rms}^2 - I_1^2}{I_1^2} = \frac{I_{rms}^2}{I_1^2} - 1$$

Step 5: Bring in the distortion factor

Add 1 to both sides:

$$1 + THD^2 = \frac{I_{rms}^2}{I_1^2}$$

The right side is $1/DF^2$, because $DF = I_1/I_{rms}$. So $DF^2 = 1/(1+THD^2)$, and taking the positive square root (DF is a ratio of rms values, so it is positive):

$$\boxed{DF = \sqrt{\frac{1}{1+(THD)^2}}}$$

This is Hart's Eq. 2-70.

Interpretation: a pure sine has THD = 0 and DF = 1. As the THD grows, DF falls below 1.

Steps to remember

Fourier series of $i(t)$ → rms² = sum of the harmonics' rms² (cross terms average to 0) → harmonics = $I_{rms}^2 - I_1^2$ → square the THD → add 1 → $1/DF^2$.

3. Keyword Explanation

1. Rms value

↩ back

The steady dc value that would give the same heating in a resistor.

2. Crest factor

↩ back

The peak of a wave divided by its rms value. A sine wave has $\sqrt{2}$.

1.(b) Define reverse recovery current and reverse recovery time for diode switching. What is the importance of these parameters? Marks: 6

1. Theory

1. What does a diode do? (Core Idea)

A diode lets current flow from anode to cathode and blocks it the other way. A simple view says that it switches at once. A real diode does not.

2. Why can a diode not turn off at once?

While a p-n diode conducts, the p side injects holes into the n side and the n side injects electrons into the p side. These extra carriers are stored near the junction, like water held in a sponge. The stored amount is called the **stored charge**.

To block, the junction must first be empty of those carriers. They must be removed, and some recombine by themselves. Until that is done, the diode still behaves like a closed path, so current can flow **backwards** for a short time.

3. What does the current look like?

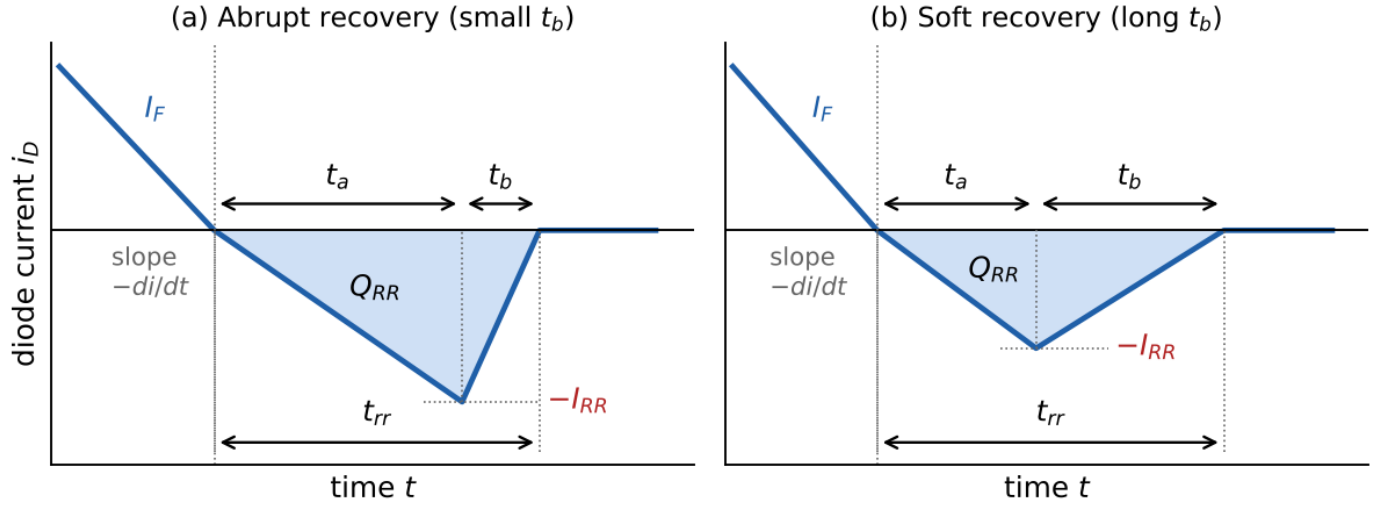


Figure 2: Reverse recovery of a power diode: (a) abrupt recovery, (b) soft recovery. The solid blue line is the diode current; the shaded area is the stored charge removed

Reading the figure. The current falls from I_F at a slope di/dt set by the outside circuit, and it passes through zero. During t_a the charge is taken out of the junction region, and the current keeps going negative. At the end of t_a it reaches its largest reverse value I_{RR}. During t_b the rest of the charge is removed and the current falls back to about zero. The shaded triangle is the charge Q_{RR}.

4. Why are two shapes shown?

The ratio $S = t_b/t_a$ is the **softness factor**. A small t_b (abrupt recovery) makes the current stop very quickly. Any stray inductance in the circuit then gives a large voltage spike, $v = L\,di/dt$. A soft-recovery diode avoids that.

2. Answer

Reverse recovery time (t_{rr}): the time from the instant the forward current crosses zero until the reverse current has decayed to about zero (the book takes a small fraction of the peak). It has two parts:

$$t_{rr} = t_a + t_b$$

Reverse recovery current (I_{RR}): the peak value of the reverse current during recovery. It flows because stored charge must be removed.

$$I_{RR} = t_a \frac{di}{dt}$$

Importance:

- **Switching frequency:** t_{rr} must be much shorter than the switching period, so it limits how fast the circuit can run. Fast-recovery diodes are used in converters; slow diodes are used at mains frequency.
- **Current stress:** I_{RR} adds to the current in the transistor that is turning on, so that device must be rated for it.
- **Switching loss:** voltage and current exist together during recovery, so power is lost at every switching. This loss grows with frequency.
- **Voltage spikes:** a sudden end of I_{RR} in a stray inductance gives a spike, and may call for a snubber.

1.(c) The manufacturer of a selected diode gives the rate of fall of the diode current di/dt=20 A/μs, and its reserve [*sic:* should be reverse] recovery time t_{rr}=5 μs. What value of peak reverse current and the storage charge do you expect? Marks: 6

1. Theory

1. Which numbers do we need? (Core Idea)

A diode conducts one way only, but it cannot stop at once. While it conducts, extra charge is stored around its junction. When the current reverses, that charge must first be removed, so a short reverse current flows. The time this takes is the reverse recovery time t_{rr}, the peak of that current is I_{RR}, and the charge removed is the storage charge Q_{RR}. 1(b) shows the full picture.

On a graph of current against time, the reverse current is a triangle, and the area under such a graph is a charge. The area of a triangle is half its base times its height. So

$$Q_{RR} \approx \frac{1}{2} I_{RR}\, t_{rr}$$

2. What simple assumption does the book make?

The current falls at a steady slope di/dt during t_a, so $I_{RR} = t_a\, di/dt$. For a diode with a small t_b, Rashid takes t_b as small compared with t_a, which gives $t_{rr} \approx t_a$. Then

$$I_{RR} \approx t_{rr} \frac{di}{dt}, \qquad Q_{RR} \approx \frac{1}{2} \frac{di}{dt} t_{rr}^2$$

Removing t_{rr} gives a form that uses only the charge and the slope:

$$I_{RR} = \sqrt{2 Q_{RR} \frac{di}{dt}}$$

2. Answer

Note on the paper

The paper prints “reserve recovery time”. It means reverse recovery time.

Given:

- $di/dt = 20$ A/μs $= 20 \times 10^6$ A/s
- $t_{rr} = 5$ μs $= 5 \times 10^{-6}$ s

Assumption: t_b is negligible, so $t_{rr} \approx t_a$ (Rashid).

Step 1: Storage charge

$$Q_{RR} = \frac{1}{2} \frac{di}{dt} t_{rr}^2 = 0.5(20 \times 10^6)(5 \times 10^{-6})^2 = 250 \times 10^{-6} \text{ C}$$

$$\boxed{Q_{RR} = 250\ \mu\text{C}}$$

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Step 2: Peak reverse current

$$I_{RR} = \sqrt{2Q_{RR}\frac{di}{dt}} = \sqrt{2(250\times10^{-6})(20\times10^{6})} = 100 \text{ A}$$

$$\boxed{I_{RR} = 100 \text{ A}}$$

Final Answer: peak reverse current 100 A, storage charge 250 μC.

1.(d) A sinusoidal current of 10A rms at a 60 Hz fundamental frequency is contaminated with a ninth harmonic current. The current is expressed as $i(t) = 10\sqrt{2}\sin(2\pi 60t) + I_9\sqrt{2}\sin(18\pi 60t)$ A. Determine the value of the ninth harmonic rms current I9 if the THD is -i) 5 percent ii) 20 percent. Marks: 6

1. Theory

1. What does the wave look like? (Core Idea)

A harmonic is a sine wave at a whole-number multiple of the main (fundamental) frequency. When harmonics are added to a sine, its shape is distorted. THD measures how much: it is the rms value of the harmonics divided by the rms value of the fundamental.

The given current has two parts: a 10 A rms fundamental at 60 Hz, and a 9th harmonic of rms value I_9. The second sine has angle $18\pi\,60\,t = 2\pi(9\times60)t$, so its frequency is 540 Hz.

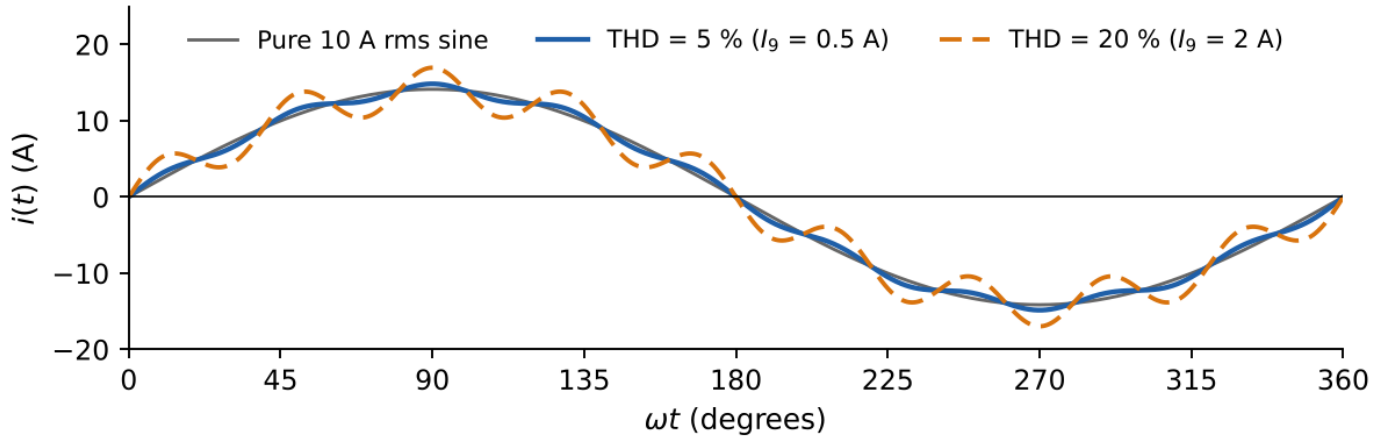


Figure 3: The pure 10 A rms sine (thin grey curve), and the current with THD = 5 % (solid blue curve) and THD = 20 % (dashed orange curve)

Reading the figure. A THD of 5 % adds only small ripples to the sine. A THD of 20 % is clearly visible: nine small waves ride on each period of the fundamental.

2. What does THD become with one harmonic?

With only one harmonic, the sum in the THD formula of 1(a) has one term:

$$THD = \frac{\sqrt{I_9^2}}{I_1} = \frac{I_9}{I_1}$$

So the required rms value is just $I_9 = THD \times I_1$.

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2. Answer

Given: $I_1 = 10$ A rms. Only the 9th harmonic is present, so $THD = I_9/I_1$ and $I_9 = THD\, I_1$.

Case i: THD = 5 %

$$I_9 = 0.05 \times 10 = 0.5 \text{ A}$$

$$\boxed{I_9 = 0.5 \text{ A rms}}$$

Case ii: THD = 20 %

$$I_9 = 0.20 \times 10 = 2 \text{ A}$$

$$\boxed{I_9 = 2 \text{ A rms}}$$

Final Answer: 0.5 A rms for 5 % THD and 2 A rms for 20 % THD.

3. Notes and Remarks

The total rms current hardly changes: $\sqrt{10^2 + 0.5^2} = 10.01$ A and $\sqrt{10^2 + 2^2} = 10.20$ A. A THD of 20 % raises the rms value by only 2 %.

4. Final Concise Answer

(a) $DF = I_1/I_{rms}$, $THD = \sqrt{I_{rms}^2 - I_1^2}/I_1$. Squaring gives $1 + THD^2 = (I_{rms}/I_1)^2 = 1/DF^2$, so $DF = \sqrt{1/(1+THD^2)}$.

(b) $t_{rr} = t_a + t_b$ is the time for the reverse current to die after the current crosses zero; I_{RR} is the peak reverse current. They limit the switching frequency and set the extra current stress, the switching loss and the voltage spikes.

(c) $Q_{RR} = 250$ μC, $I_{RR} = 100$ A.

(d) $I_9 = 0.5$ A rms for 5 %, $I_9 = 2$ A rms for 20 %.

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2. (a) Draw the two transistors model of a thyristor and explain the turn-on condition of a thyristor with equations from this model. Marks: 7

   (b) For the circuit of Fig. 2(b), the source is 120 V rms at 60 Hz, R=20 Ohm, L=0.04 H, and the delay angle is 45°. Determine (i) an expression for i(ωt) , (ii) the average current, (iii) the power absorbed by the load, and (iv) the power factor. Marks: 10

   (c) A full-wave controlled bridge rectifier has an ac input of 120 V rms at 60 Hz and a 20 Ω load resistor. The delay angle is 40°. Determine the average current of the load, the power absorbed by the load, the apparent power of the sourced [*sic*: should be source] and the input pf. Marks: 8

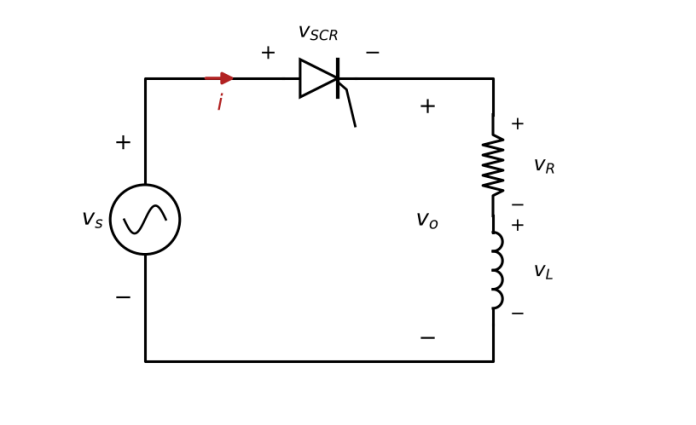


Figure 4: Fig 2(b) of the paper (redrawn)

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2.(a) Draw the two transistors model of a thyristor and explain the turn-on condition of a thyristor with equations from this model. Marks: 7

1. Theory

1. What is a thyristor? (Core Idea)

A thyristor, or silicon-controlled rectifier (SCR), is a controlled diode. It conducts from anode (A) to cathode (K), but only after it gets a small pulse of current at its third terminal, the gate (G). Once on, it stays on by itself, even after the gate pulse ends.

It is made of four silicon layers, p-n-p-n, so it has three junctions, J_1, J_2 and J_3.

2. Why does it block at first?

With the anode positive, J_1 and J_3 are forward biased, but J_2 in the middle is reverse biased. A reverse-biased junction passes only a tiny leakage current, so the device blocks.

3. Where do two transistors come from?

Imagine cutting the middle two layers along a slanted line, so each half keeps them. The upper half is then p-n-p and the lower half is n-p-n. These are two bipolar transistors, Q_1 and Q_2, joined so each one's collector feeds the other's base.

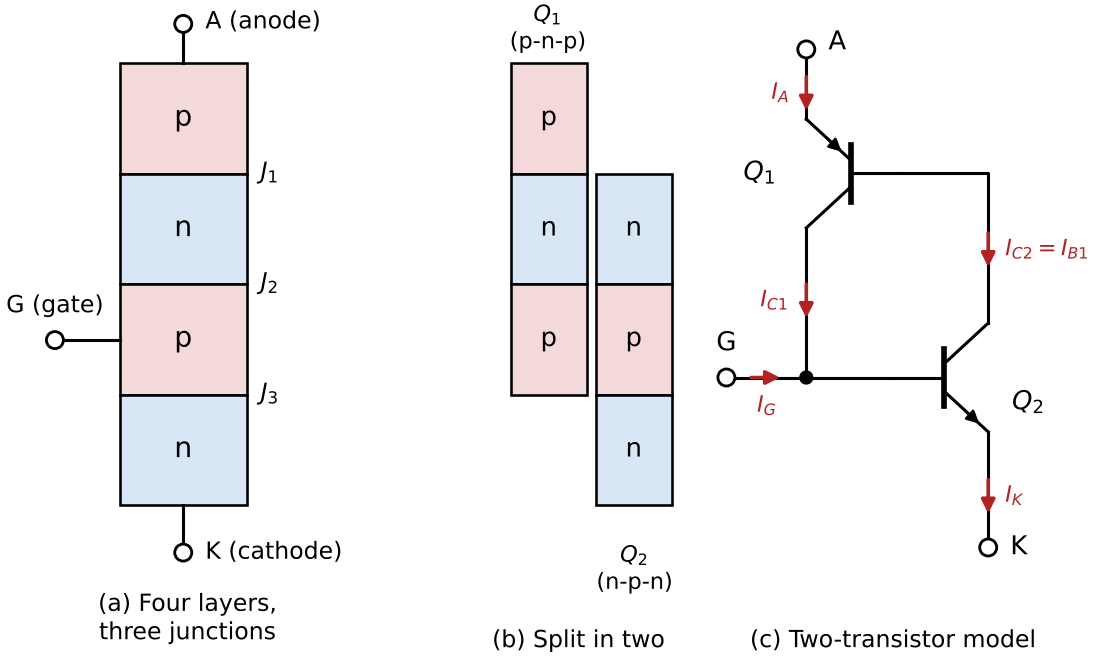


Figure 5: Thyristor: (a) the four layers, (b) the layers split in two, (c) the two-transistor model. Red arrows show the currents

Reading the figure. In (c), the current into the gate enters the base of Q_2. The collector of Q_2 pulls the base current of Q_1, and the collector current of Q_1 goes back into the base of Q_2. It is a loop in which each transistor drives the other.

4. What does a transistor's α mean?

In a bipolar transistor, nearly all the current entering at the emitter comes out at the collector. The fraction that does is α. There is also a small leakage current I_{CBO} that flows even with no signal:

$$I_C = \alpha I_E + I_{CBO}$$

For a transistor at very low current, α is small. As the current rises, α rises towards 1. This is the key to the turn-on.

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2. Answer

Model: the p-n-p-n layers are treated as a p-n-p transistor Q_1 and an n-p-n transistor Q_2, with the collector of each connected to the base of the other (figure above; Rashid, Fig. 9.4).

Equations from the model.

Step 1: Split the thyristor into two transistors

Cut the middle two layers in half along the device. The top three layers, p-n-p, form Q_1: its emitter is the anode. The bottom three layers, n-p-n, form Q_2: its emitter is the cathode. The middle n layer is both the base of Q_1 and the collector of Q_2, and the middle p layer is both the collector of Q_1 and the base of Q_2. The gate connects to the base of Q_2.

Step 2 (Basic): Collector current of one transistor

In a transistor, a fraction α (the common-base current gain) of the emitter current reaches the collector. The reverse-biased collector-base junction also passes a small leakage current I_{CBO}, even with no emitter current. So

$$I_C = \alpha I_E + I_{CBO}$$

(Rashid, Eq. 9.1).

Step 3: Apply it to each transistor

For Q_1 the emitter current is the anode current I_A; for Q_2 it is the cathode current I_K:

$$I_{C1} = \alpha_1 I_A + I_{CBO1}, \qquad I_{C2} = \alpha_2 I_K + I_{CBO2}$$

(Rashid, Eqs. 9.2 and 9.3).

Step 4: Kirchhoff's current law

At the middle junction, the current entering at the anode leaves through the two collectors, $I_A = I_{C1} + I_{C2}$. At the cathode, the current out is the anode current plus the gate current, $I_K = I_A + I_G$.

Step 5: Substitute and solve for the anode current

Put Step 3 into $I_A = I_{C1} + I_{C2}$, then replace I_K by $I_A + I_G$:

$$I_A = \alpha_1 I_A + I_{CBO1} + \alpha_2 (I_A + I_G) + I_{CBO2}$$

Collect the I_A terms on the left:

$$I_A[1 - (\alpha_1 + \alpha_2)] = \alpha_2 I_G + I_{CBO1} + I_{CBO2}$$

$$\boxed{I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}}$$

This is Rashid's Eq. 9.5.

Step 6: Read the turn-on condition

- With $I_G = 0$ and small currents, $\alpha_1 + \alpha_2 \ll 1$, so I_A is only a small leakage current. The thyristor is off.
- A gate current raises the current in Q_2 and so raises α_2 (a transistor's α grows with its current). More current then flows in Q_1, and α_1 rises.
- When $\alpha_1 + \alpha_2 \to 1$, the denominator tends to zero and I_A grows without limit. Only the outside circuit then limits it. This is regenerative action, and the thyristor is on.
- After this, the loop feeds itself, so the gate can be removed if I_A is above the latching current.

$$\boxed{\text{Turn-on condition: } \alpha_1 + \alpha_2 \approx 1}$$

Steps to remember

Split p-n-p-n into Q_1 (p-n-p) and Q_2 (n-p-n) $\to$ $I_C = \alpha I_E + I_{CBO}$ for each $\to$ $I_A = I_{C1} + I_{C2}$ and

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$I_K = I_A + I_G \rightarrow$ solve for $I_A \rightarrow$ it grows without limit as $\alpha_1 + \alpha_2 \rightarrow 1$.

3. Notes and Remarks

The same condition can be reached without a gate pulse: by a forward voltage above the breakover voltage, a very fast rise of voltage (dv/dt), a high temperature, or light. Only the gate method is used in normal control (light is used on purpose in light-activated SCRs).

2.(b) For the circuit of Fig. 2(b), the source is 120 V rms at 60 Hz, R=20 Ohm, L=0.04 H, and the delay angle is 45°. Determine (i) an expression for i(ωt) , (ii) the average current, (iii) the power absorbed by the load, and (iv) the power factor. Marks: 10

1. Theory

1. What does the circuit do? (Core Idea)

A rectifier turns ac into dc. With a diode, conduction starts as soon as the source goes positive. With a thyristor we can wait. The **delay angle** α is the angle after the zero crossing at which the gate pulse is given. A larger delay means less average voltage. This is how we control dc power from ac.

2. Why does an inductor change things?

An inductor resists a change of current. Its voltage is $v_L = L\,di/dt$. So when the source reaches zero, the current does not stop. The inductor gives back its stored energy and keeps the current flowing for a while into the negative half cycle. The thyristor stays on until the current reaches zero. The angle at which this happens is the **extinction angle** β.

3. What equation does the current obey?

While the thyristor conducts, Kirchhoff's voltage law gives

$$V_m \sin \omega t = Ri + L\frac{di}{dt}$$

The solution of such an equation has two parts:

- **Forced part:** the steady ac current the circuit would carry if the thyristor were not there. It has peak V_m/Z and lags by θ.
- **Natural part:** an exponential that decays with time constant $\tau = L/R$ and makes the current start from zero.

$$Z = \sqrt{R^2 + (\omega L)^2}, \qquad \theta = \tan^{-1}\frac{\omega L}{R}, \qquad \omega\tau = \frac{\omega L}{R}$$

$$i(\omega t) = \frac{V_m}{Z}\sin(\omega t - \theta) + A\,e^{-\omega t/\omega\tau}$$

The constant A comes from the starting condition. The current is zero when the thyristor fires, so $i(\alpha) = 0$:

$$A = -\frac{V_m}{Z}\sin(\alpha - \theta)\,e^{\alpha/\omega\tau}$$

4. How do we find β, the average and the power?

β is the angle that makes $i(\beta) = 0$. This equation has no neat solution, so we find it numerically. The average and rms values are then

$$I_o = \frac{1}{2\pi}\int_\alpha^\beta i\,d(\omega t), \qquad I_{rms} = \sqrt{\frac{1}{2\pi}\int_\alpha^\beta i^2\,d(\omega t)}$$

The inductor takes no average power, so all the power goes to the resistor, $P = I_{rms}^2 R$.

The power factor is P/S, and the source's apparent power is $S = V_{rms}I_{rms}$.

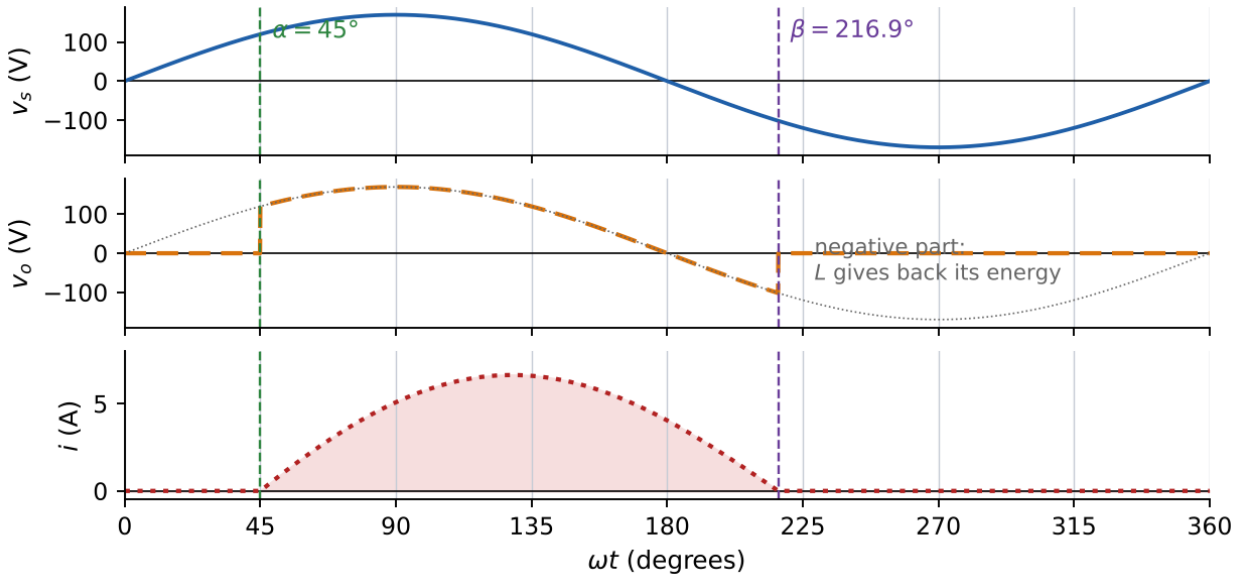


Figure 6: Waveforms for the numbers of this question: the source (solid blue curve), the output voltage (dashed orange curve) and the current (dotted red curve). The dashed green line marks the firing angle and the dashed purple line the extinction angle

Reading the figure. Nothing happens before 45°. Then the output follows the source. The current rises slowly, peaks late, near 128°, and is still flowing after 180°. It ends at 216.9°, and the output voltage is negative between 180° and 216.9°.

2. Answer

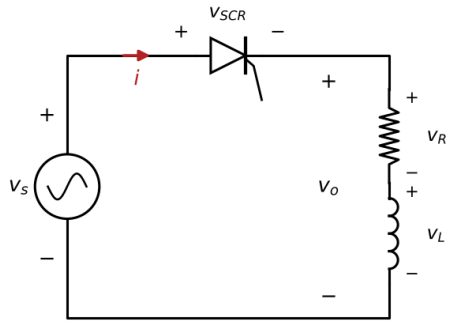


Figure 7: Repeated from the question: Fig 2(b) of the paper (redrawn)

Given:

- $V_s = 120$ V rms, so $V_m = 120\sqrt{2} = 169.7$ V
- $f = 60$ Hz, so $\omega = 377$ rad/s
- $R = 20\ \Omega$, $L = 0.04$ H
- $\alpha = 45° = 0.785$ rad

Step 1: Circuit constants

$$\omega L = 377(0.04) = 15.08\ \Omega$$

$$Z = \sqrt{20^2 + 15.08^2} = 25.05\ \Omega$$

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$$\theta = \tan^{-1}\frac{15.08}{20} = 37.0^\circ = 0.646 \text{ rad}$$

$$\omega\tau = \frac{\omega L}{R} = 0.754 \text{ rad}, \qquad \frac{V_m}{Z} = 6.78 \text{ A}$$

Step 2: Expression for the current

$$A = -6.78\sin(0.785 - 0.646)\, e^{0.785/0.754} = -2.67 \text{ A}$$

$$\boxed{i(\omega t) = 6.78\sin(\omega t - 0.646) - 2.67\, e^{-\omega t/0.754} \text{ A}, \quad \alpha \le \omega t \le \beta}$$

and $i = 0$ for the rest of the period. Setting $i(\beta) = 0$ and solving numerically gives

$$\beta = 3.785 \text{ rad} = 216.9^\circ$$

Step 3: Average current

Numerical integration of the expression from 0.785 to 3.785 rad gives

$$I_o = \frac{1}{2\pi}\int_{0.785}^{3.785} i\, d(\omega t) = 2.04 \text{ A}$$

$$\boxed{I_o = 2.04 \text{ A}}$$

Step 4: Power absorbed

$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_{0.785}^{3.785} i^2\, d(\omega t)} = 3.26 \text{ A}$$

$$P = I_{rms}^2 R = (3.26)^2(20) = 213 \text{ W}$$

$$\boxed{P = 213 \text{ W}}$$

Step 5: Power factor

$$pf = \frac{P}{S} = \frac{P}{V_s I_{rms}} = \frac{213}{(120)(3.26)} = 0.544$$

$$\boxed{pf = 0.544 \text{ lagging}}$$

Final Answer: $i(\omega t)$ as above with $\beta = 216.9^\circ$; $I_o = 2.04$ A; $P = 213$ W; $pf = 0.544$.

3. Notes and Remarks

The handwritten "$\beta = 217$" on the scanned paper agrees with the result here. It is someone's pencil note, not part of the printed question.

The average current also follows from the output voltage, since the inductor drops no average voltage: $V_o = \frac{V_m}{2\pi}(\cos\alpha - \cos\beta) = 40.7$ V and $V_o/R = 2.04$ A.

2.(c) A full-wave controlled bridge rectifier has an ac input of 120 V rms at 60 Hz and a 20 Ω load resistor. The delay angle is 40°. Determine the average current of the load, the power absorbed by the load, the apparent power of the sourced [*sic*: should be source] **and the input pf. Marks: 8**

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1. Theory

1. How does a bridge use both half cycles? (Core Idea)

A rectifier turns ac into dc. A thyristor is a controlled diode: it conducts forward, but only after a gate pulse, and it stops when its current falls to zero. Waiting for an angle α after each zero crossing before firing it controls the average output.

A single thyristor uses only the positive half cycle, as in 2(b), and wastes the negative one. A bridge of four thyristors turns the negative half up the other way, so the load gets current in the same direction in both half cycles.

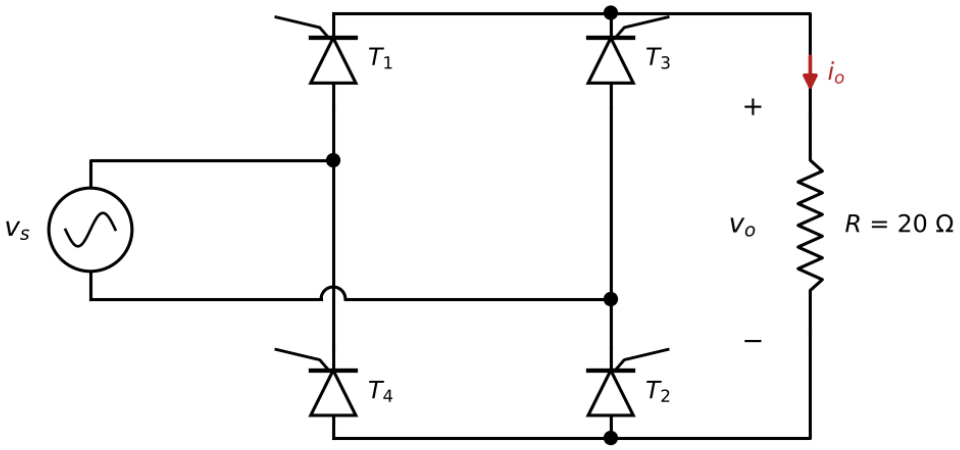


Figure 8: Full-wave controlled bridge. Only the marked joint is connected where the wires cross: the lower source wire hops over the left leg

Reading the figure. In the positive half cycle, T_1 and T_2 are fired: current goes from the source through T_1, down the load, and back through T_2. In the negative half, T_3 and T_4 are fired, and the current goes through the load in the same direction.

2. What is the output?

With a resistor, the current has the same shape as the voltage. So the current falls to zero at each zero crossing of the source, and each thyristor pair turns off there by itself. The next pair is fired α after that.

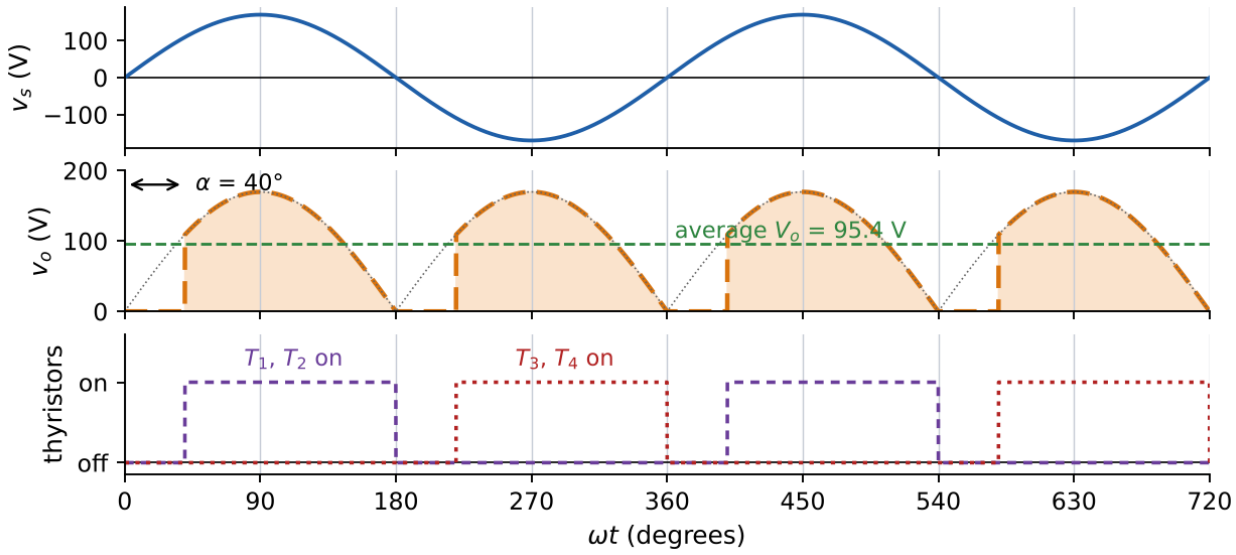


Figure 9: Source voltage (solid blue curve), output voltage (dashed orange curve) with its average (dashed green line), and which thyristor pair conducts (dashed purple and dotted red lines), for a delay of 40°

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Reading the figure. Each half cycle of the output is a piece of a sine from 40° to 180°. The thin dotted arch shows what is cut away by the delay.

3. Where do the formulas come from?

The output repeats every π rad, so average over π:

$$V_o = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin\omega t \, d(\omega t) = \frac{V_m}{\pi}(1+\cos\alpha)$$

For the rms value, use $\sin^2 x = (1-\cos 2x)/2$:

$$V_{rms} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi} V_m^2 \sin^2\omega t \, d(\omega t)} = \frac{V_m}{\sqrt{2}}\sqrt{1-\frac{\alpha}{\pi}+\frac{\sin 2\alpha}{2\pi}}$$

The load is a resistor, so $I_{rms} = V_{rms}/R$. The ac source's rms current is the same as the load's rms current, since squaring removes the sign.

2. Answer

Given: $V_s = 120$ V rms, $V_m = 169.7$ V, $R = 20\ \Omega$, $\alpha = 40° = 0.698$ rad.

Step 1: Average load current

$$V_o = \frac{169.7}{\pi}(1+\cos 40°) = 95.4 \text{ V}$$

$$I_o = \frac{95.4}{20} = 4.77 \text{ A}$$

$$\boxed{I_o = 4.77 \text{ A}}$$

Step 2: Power absorbed by the load

$$I_{rms} = \frac{169.7}{\sqrt{2}(20)}\sqrt{1-\frac{0.698}{\pi}+\frac{\sin 80°}{2\pi}} = 5.80 \text{ A}$$

$$P = I_{rms}^2 R = (5.80)^2(20) = 673 \text{ W}$$

$$\boxed{P = 673 \text{ W}}$$

Step 3: Apparent power of the source

$$S = V_s I_{rms} = (120)(5.80) = 696 \text{ VA}$$

$$\boxed{S = 696 \text{ VA}}$$

Step 4: Input power factor

$$pf = \frac{P}{S} = \frac{673}{696} = 0.967$$

$$\boxed{pf = 0.967}$$

Final Answer: $I_o = 4.77$ A, $P = 673$ W, $S = 696$ VA, $pf = 0.967$.

3. Notes and Remarks

The pf is below 1 even with a pure resistor. The delay makes the current non-sinusoidal and shifts its fundamental, so the source's current has a larger rms value than the power requires.

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4. Final Concise Answer

(a) Q_1 (p-n-p) and Q_2 (n-p-n) feed each other. $I_A = (\alpha_2 I_G + I_{CBO1} + I_{CBO2})/[1 - (\alpha_1 + \alpha_2)]$. The thyristor turns on when $\alpha_1 + \alpha_2 \approx 1$.

(b) $i(\omega t) = 6.78 \sin(\omega t - 0.646) - 2.67 e^{-\omega t/0.754}$ A for $45° \leq \omega t \leq 216.9°$; $I_o = 2.04$ A; $P = 213$ W; $pf = 0.544$.

(c) $I_o = 4.77$ A, $P = 673$ W, $S = 696$ VA, $pf = 0.967$.

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3. (a) Draw the configuration of three phase uncontrolled rectifier. Show the source voltages, output voltage, diode currents and line currents with proper alignment. Marks: 8

   (b) Draw the cross section of an IGBT showing the layers. Differentiate between punch-through and non-punch through IGBTs. Marks: 7

   (c) Write down the advantages of DC power transmission. Marks: 4

   (d) A three phase controlled rectifier has an input voltage of 480 V rms at 60 Hz. The load of the rectifier is resistor of 10 Ω in series with 50 mH inductance.

   (i) Determine the delay angle required to produce an average current of 50A in the load

   (ii) Determine the amplitude of the harmonics n=6 and n=12. For harmonics calculation, Fig. 3(d) can be utilized. Marks: 6

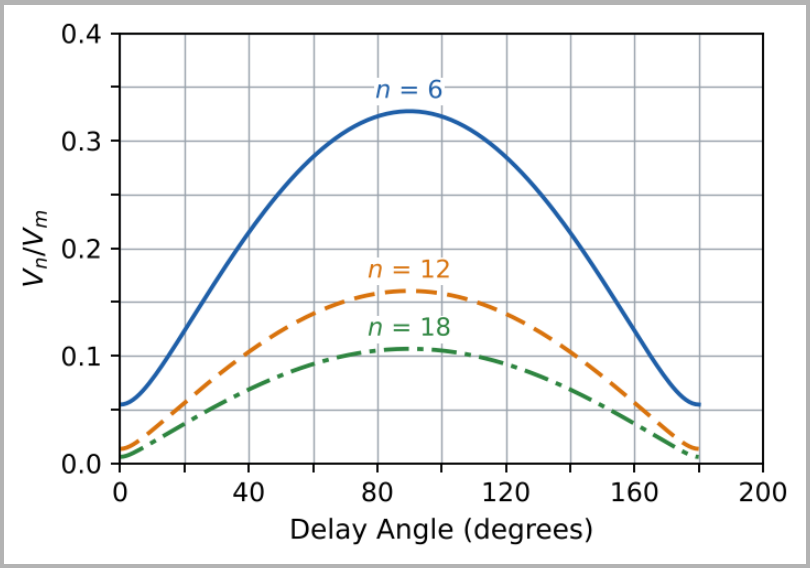


Figure 10: Fig. 3(d): Normalized output voltage harmonics as a function of delay angle for a three-phase rectifier (redrawn)

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3.(a) Draw the configuration of three phase uncontrolled rectifier. Show the source voltages, output voltage, diode currents and line currents with proper alignment. Marks: 8

1. Theory

1. What is three-phase ac? (Core Idea)

A three-phase source has three sine voltages of equal size, shifted 120° from each other. With V_m as the peak of one phase:

$$v_{an} = V_m \sin \omega t, \quad v_{bn} = V_m \sin(\omega t - 120°), \quad v_{cn} = V_m \sin(\omega t + 120°)$$

The voltage between two lines, for example $v_{ab} = v_{an} - v_{bn}$, is larger. It has peak $V_{m,L\text{-}L} = \sqrt{3} V_m$.

2. How is the rectifier built?

Each of the three lines joins the middle of a leg of two diodes. Diodes D_1, D_3, D_5 form the upper group. Their cathodes are joined at the positive output. Diodes D_4, D_6, D_2 form the lower group. Their anodes are joined at the negative output. The numbering follows the order of conduction.

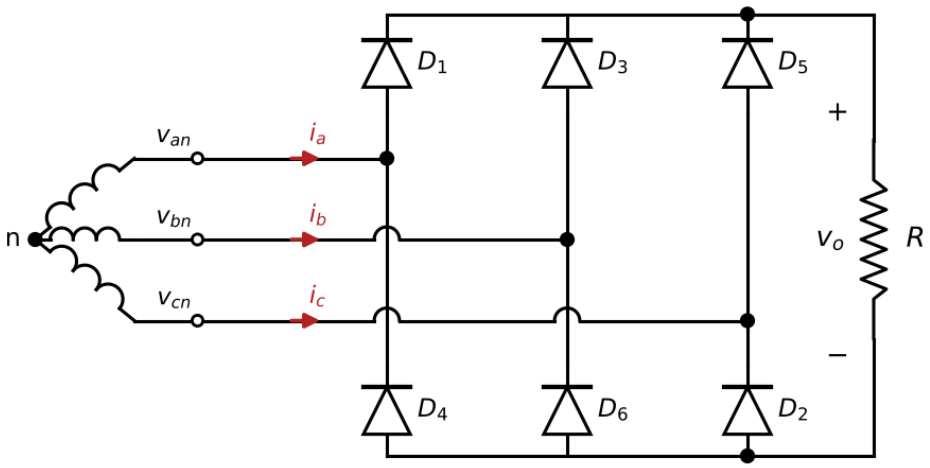


Figure 11: Three-phase diode bridge rectifier. Current arrows show the line currents; each line joins its own leg at the dot

3. Which diodes conduct?

The upper diodes all share one cathode. So the one whose anode voltage is highest conducts, and it holds the common cathode at that voltage. This reverse biases the other two. In the same way, in the lower group, the diode whose cathode is at the lowest voltage conducts.

So one diode from each group is always on, and the load sees the highest line-to-line voltage at every moment. Each pair lasts 60°, and each diode conducts for 120°.

ωt	Highest phase	Lowest phase	Diodes on	v_o
30° to 90°	a	b	D_1, D_6	v_{ab}
90° to 150°	a	c	D_1, D_2	v_{ac}
150° to 210°	b	c	D_3, D_2	v_{bc}
210° to 270°	b	a	D_3, D_4	v_{ba}
270° to 330°	c	a	D_5, D_4	v_{ca}
330° to 30°	c	b	D_5, D_6	v_{cb}

Table 1: Conduction sequence of the bridge

4. What is the average output?

One 60° piece of the output is the top of a line-to-line sine, from 60° to 120° of it:

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$$V_o = \frac{3}{\pi}\int_{\pi/3}^{2\pi/3} V_{m,L\text{-}L}\sin\omega t\, d(\omega t) = \frac{3V_{m,L\text{-}L}}{\pi} = 0.955\, V_{m,L\text{-}L}$$

The output has six pulses in each source period, so its ripple is at $6f$.

2. Answer

Configuration: the circuit is the six-diode bridge above. The upper diode with the highest anode voltage and the lower diode with the lowest cathode voltage conduct, so v_o is the largest line-to-line voltage. Each diode conducts for 120°, in the order $D_1, D_2, D_3, D_4, D_5, D_6$, one change every 60°.

Waveforms (resistive load, aligned on the same ωt axis):

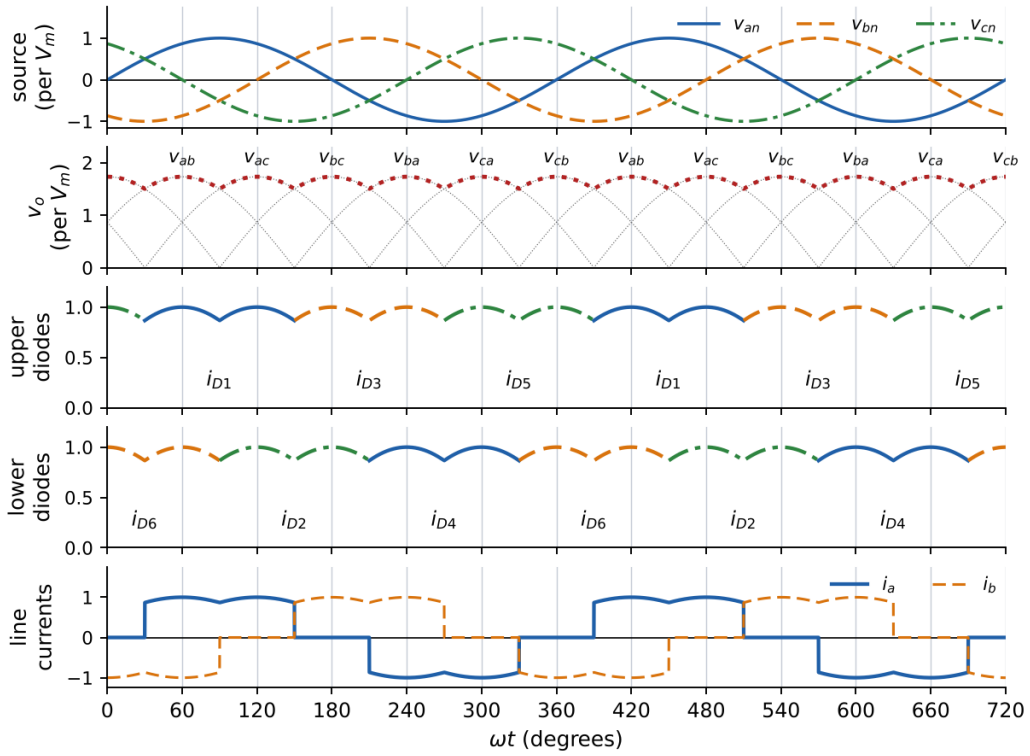


Figure 12: Aligned waveforms: source voltages, output voltage with the six line-to-line voltages as thin dotted curves, upper and lower diode currents, and the line currents. Currents are per unit of the peak load current

Reading the figure.

- **Top:** the three phase voltages; solid blue is v_{an}, dashed orange v_{bn}, dash-dot green v_{cn}.
- **Output:** the dotted red curve is v_o, the top of the line-to-line voltages. It never falls below 0.866 of its peak. The name of the conducting voltage is above each pulse.
- **Diode currents:** each diode carries two 60° pulses of load current, that is, 120° in all. In each panel, the line style shows the phase of that diode, and the current changes from one diode to the next every 120°.
- **Line current:** $i_a = i_{D1} - i_{D4}$. It is positive for 120°, zero for 60°, negative for 120° and zero for 60°. The dashed orange i_b is the same wave 120° later.

Key results:

$$V_o = \frac{3V_{m,L\text{-}L}}{\pi} = 0.955V_{m,L\text{-}L}, \qquad f_{ripple} = 6f$$

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3. Keyword Explanation

1. Line and phase voltage

↩ back

Phase voltage is from a line to the neutral. Line voltage is between two lines, and it is $\sqrt{3}$ times larger.

3.(b) Draw the cross section of an IGBT showing the layers. Differentiate between punch-through and non-punch through IGBTs. Marks: 7

1. Theory

1. Why was the IGBT invented? (Core Idea)

Two switches had opposite strengths. A power MOSFET is easy to drive: its gate takes almost no current. But at high voltage it has a large on-resistance. A BJT has a small on-voltage, but it needs a large base current. The insulated gate bipolar transistor (IGBT) joins them: the gate of a MOSFET with the conduction of a bipolar device.

2. How is it built, and how does it work?

An IGBT looks like a vertical MOSFET with one extra layer: a p layer at the collector side. A positive gate voltage forms a channel in the p body, so electrons flow from the n^+ emitter into the n^- drift region. These electrons reach the p collector, which then injects holes into the drift region. The drift region fills with both kinds of carrier, so its resistance falls a lot. This is **conductivity modulation**, and it is why the on-voltage is small.

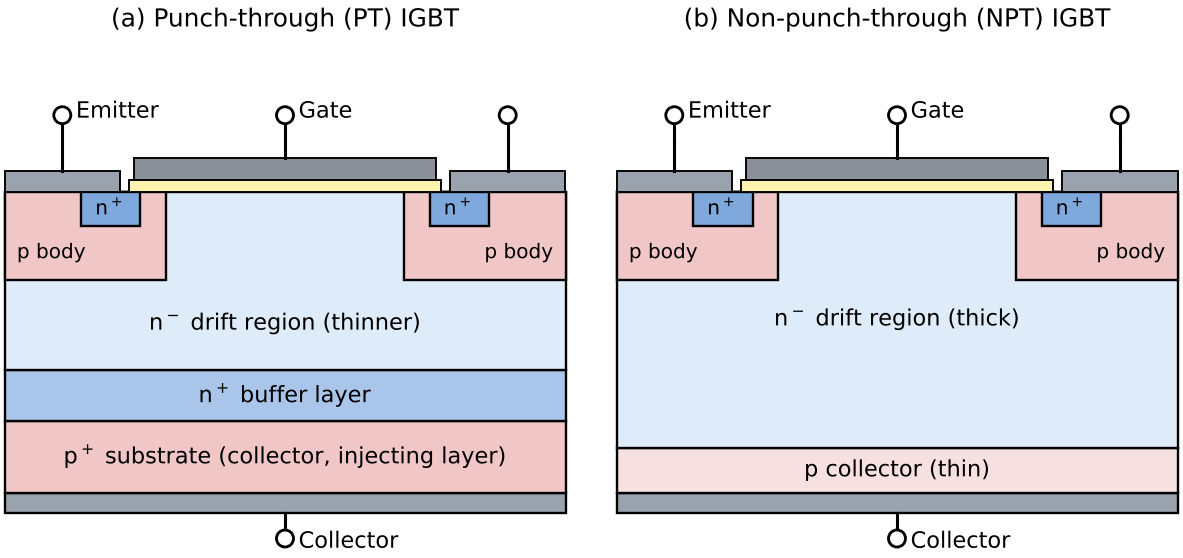


Figure 13: Cross-section of one IGBT cell: (a) punch-through, with an n+ buffer layer, (b) non-punch-through, with a thick drift region and no buffer

Reading the figure. The top of the two cells is the same. Below the gate, the p body wells and the n^+ emitters form the MOSFET part. The difference is at the bottom: (a) has an n^+ buffer layer between the drift region and the collector, while (b) has none, and has a thicker drift region and a thin p collector.

3. What does "punch-through" mean?

When the device blocks, a depletion region (an empty zone with a strong electric field) spreads through the n^- drift region from the junction with the p body. If the region is thin, the depletion region reaches its far end at full voltage. This reaching across is called punch-through. In a PT IGBT this is allowed, since the n^+ buffer stops the field. In an NPT IGBT the drift region is made thick enough so that the field never reaches the far end.

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4. What follows from this?

The buffer blocks the field, so a PT device can have a thin drift region, which gives a low on-voltage. But the buffer also spoils reverse blocking. In an NPT device, the thin, lightly doped collector injects fewer holes, so fewer carriers have to be removed at turn-off, and the on-voltage rises with temperature, so parallel devices share current by themselves.

2. Answer

Cross-section: see the figure above. From top to bottom: emitter metal and gate over oxide; n^+ emitter in a p body; n^- drift region; (n^+ buffer in a PT device); p collector; collector metal.

Feature	Punch-through (PT)	Non-punch-through (NPT)
Buffer layer	n^+ buffer present	no buffer
Drift region	thin, grown on the substrate	thick (the wafer itself)
Collector	thick heavily doped p^+ substrate	thin, lightly doped p layer
Blocking field	reaches the buffer (punches through)	never reaches the collector
Reverse blocking	small (asymmetric device)	large (nearly symmetric)
On-voltage	lower	somewhat higher
On-voltage with temperature	falls: paralleling is harder	rises: easy paralleling
Ruggedness	lower short-circuit strength	higher

Table 2: PT and NPT IGBTs compared

3.(c) Write down the advantages of DC power transmission. Marks: 4

1. Theory

1. Why does dc need power electronics? (Core Idea)

Power is generated and used as ac. To send it as dc, we need a rectifier at the sending end and an inverter at the receiving end. These are the same thyristor bridges studied in this paper, at very high power.

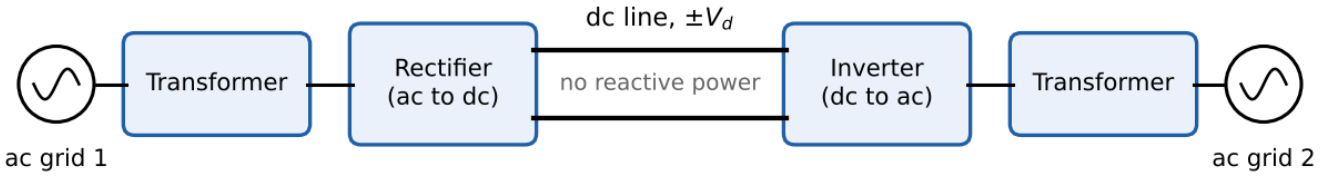


Figure 14: An HVDC link: a rectifier at one end and an inverter at the other, joined by a dc line

Reading the figure. Each ac grid connects through a transformer to a converter. Only the middle section is dc. The ac grids are joined only through the converters, so they need not run in step with each other.

2. Why can dc be cheaper when the line is long?

An ac line has inductance and capacitance. So its current carries a reactive part, which uses up the wire's capacity without delivering power. Dc has no such part. A dc line also needs only two conductors instead of three, and the insulation only has to stand the steady voltage, not the peak of a sine wave. Against this, the converter stations cost a lot.

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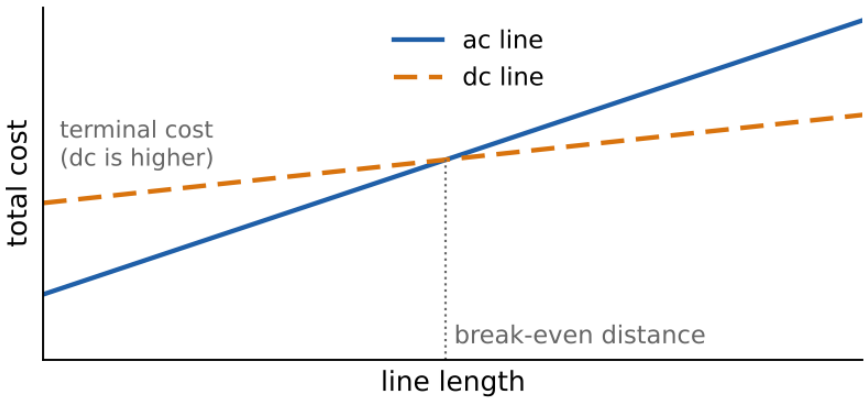


Figure 15: Total cost against line length (not to scale): the dc line (dashed orange) costs more at the terminals, but less per kilometre than the ac line (solid blue)

Reading the figure. The two lines cross at the break-even distance. Above it, dc is cheaper.

2. Answer

Advantages of dc transmission:

- **No reactive power:** line inductance and capacitance have no steady-state effect, so no charging current flows and no reactive compensation is needed.
- **Cheaper long lines:** two conductors, lighter towers and a narrower right of way. Beyond the break-even distance the whole system costs less.
- **Lower losses:** there is no skin effect and no dielectric loss in cables, and corona loss is lower.
- **No stability limit:** power is not limited by the angle between the two ends, so distance does not limit it.
- **Asynchronous links:** grids of different frequency, or not in step, can be joined.
- **Fast control:** the converters set the power flow quickly, and faults are not fed from one grid to the other.

3. Notes and Remarks

The drawbacks are the costly converters, the harmonics they make, and the need for dc circuit breakers.

3.(d) A three phase controlled rectifier has an input voltage of 480 V rms at 60 Hz. The load of the rectifier is resistor of 10 Ω in series with 50 mH inductance. (i) Determine the delay angle required to produce an average current of 50A in the load (ii) Determine the amplitude of the harmonics n=6 and n=12. For harmonics calculation, Fig. 3(d) can be utilized. Marks: 6

1. Theory

1. What changes from the diode bridge? (Core Idea)

A three-phase bridge has six devices, three joined to the positive output and three to the negative. With diodes, the load always sees the largest line-to-line voltage, and each pair conducts for 60° (3(a)). A thyristor is a diode that waits for a gate pulse.

Replace the six diodes by thyristors. Each thyristor can be held off for an angle α after the moment a diode would have turned on. Each 60° piece of the output then moves α later along the line-to-line sine.

2. What is the new average?

The piece now runs from $60° + \alpha$ to $120° + \alpha$ of a line-to-line sine:

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$$V_o = \frac{3}{\pi}\int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_m \sin\omega t\, d(\omega t) = \frac{3V_m}{\pi}\cos\alpha$$

Here, as in Hart, V_m is the peak **line-to-line** voltage. At $\alpha = 0$ this is the diode bridge's value.

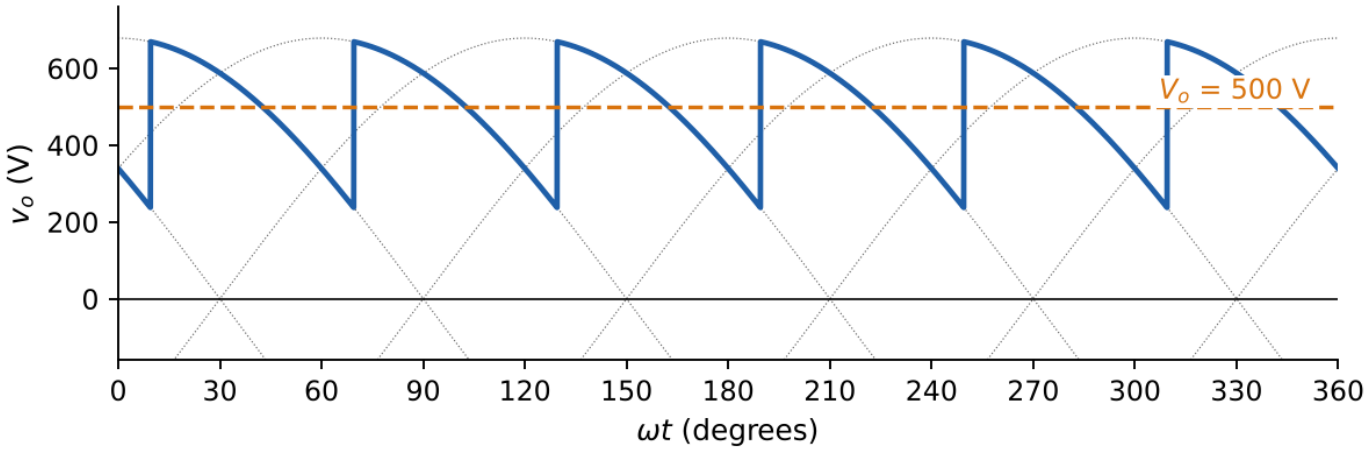


Figure 16: Output voltage (solid blue curve) of the controlled bridge at the delay angle found in this question. The thin dotted curves are the six line-to-line voltages, and the dashed orange line is the average

Reading the figure. The output no longer stays near the top of the sines. Each piece starts high and falls far down, so the ripple is much larger than for the diode bridge, though the frequency is still $6f$.

3. Where does the harmonic figure come from?

The output repeats every 60°, so it has only harmonics $n = 6, 12, 18, \ldots$ of the source frequency. Their sizes depend on α. Fourier analysis of one piece gives

$$V_n = \frac{3V_m}{\pi}\,\frac{2}{n^2-1}\sqrt{\cos^2\alpha + n^2\sin^2\alpha}$$

This is what Fig. 3(d) plots, as V_n/V_m. The ripple is smallest at $\alpha = 0$ and largest at $\alpha = 90°$.

4. How does the load respond?

For dc, an inductor is a plain wire, so the average current is $I_o = V_o/R$. For a harmonic of order n, the inductor has reactance $n\omega L$, so

$$Z_n = \sqrt{R^2 + (n\omega L)^2}, \qquad I_n = \frac{V_n}{Z_n}$$

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2. Answer

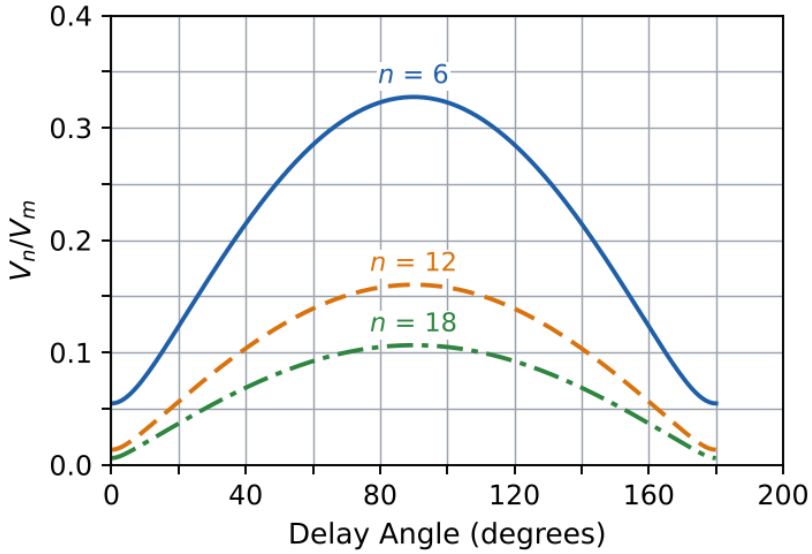


Figure 17: Repeated from the question: Fig. 3(d) of the paper (redrawn)

Given: $V_{L\text{-}L} = 480$ V rms, $R = 10\ \Omega$, $L = 50$ mH, $f = 60$ Hz, $I_o = 50$ A.
Assumption: 480 V is the line-to-line rms value, as is usual for a three-phase source. So $V_m = 480\sqrt{2} = 678.8$ V.

(i) Delay angle

The inductor drops no average voltage, so

$$V_o = I_o R = 50(10) = 500 \text{ V}$$

$$\cos\alpha = \frac{\pi V_o}{3V_m} = \frac{\pi(500)}{3(678.8)} = 0.771$$

$$\boxed{\alpha = 39.5^\circ}$$

(ii) Harmonics

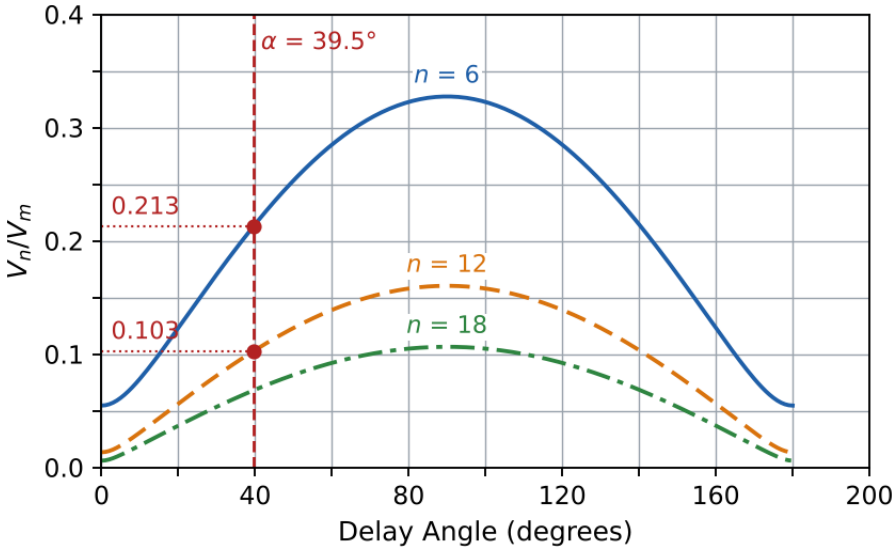


Figure 18: Reading Fig. 3(d) at the delay angle found: the dashed red line is at 39.5°, and the dotted red lines read the two curves

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At $\alpha = 39.5°$ the curves give $V_6/V_m \approx 0.213$ and $V_{12}/V_m \approx 0.103$:

$$V_6 = 0.213(678.8) = 144 \text{ V}, \qquad V_{12} = 0.103(678.8) = 70 \text{ V}$$

$$\boxed{V_6 \approx 144 \text{ V}, \quad V_{12} \approx 70 \text{ V}}$$

Current harmonics (the load data make these available):

$$Z_6 = \sqrt{10^2 + [6(377)(0.05)]^2} = 113.5\ \Omega, \qquad I_6 = \frac{144}{113.5} = 1.27 \text{ A}$$

$$Z_{12} = \sqrt{10^2 + [12(377)(0.05)]^2} = 226.4\ \Omega, \qquad I_{12} = \frac{69.6}{226.4} = 0.31 \text{ A}$$

Final Answer: $\alpha = 39.5°$; $V_6 \approx 144$ V and $V_{12} \approx 70$ V peak; $I_6 = 1.27$ A and $I_{12} = 0.31$ A peak.

3. Notes and Remarks

The values above are the exact formula's: $V_6 = 144.3$ V and $V_{12} = 69.6$ V. A hand reading of the printed graph, about 0.21 and 0.10, gives about 143 V and 68 V, which is also acceptable in the exam. The harmonic currents are small against 50 A dc, so the inductor makes the load current almost pure dc.

4. Final Concise Answer

(a) Six-diode bridge; each diode conducts 120°, in pairs for 60°; v_o is the largest line-to-line voltage, $V_o = 0.955 V_{m,L\text{-}L}$; the line current is $i_a = i_{D1} - i_{D4}$.

(b) A PT IGBT has an n^+ buffer and a thin drift region (lower on-voltage, poor reverse blocking, on-voltage falls with temperature). An NPT IGBT has no buffer, a thick drift region and a thin p collector (symmetric blocking, easy paralleling, more rugged).

(c) No reactive power, cheaper long lines, lower losses, no stability limit, asynchronous links, fast control.

(d) $\alpha = 39.5°$; $V_6 \approx 144$ V, $V_{12} \approx 70$ V.

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4. (a) Draw a typical circuit of a linear dc voltage regulator. What is the drawback of it? How can it be overcome? Marks: 6
   (b) Draw the circuit diagram and the waveform of inductor current, diode current and capacitor current for a typical buck converter. Also derive the relation between inductor current and diode current from the waveforms. Marks: 8
   (c) A boost converter has an input of [unclear: 5 V] and an output of 25 W at 15 V. The minimum inductor current must be no less than 50% of the average. The ripple output must be less than 1 percent. The switching frequency is [unclear: 300 kHz]. (i) Determine the duty ratio. (ii) Minimum inductor value (iii) Minimum capacitor value Marks: 6
   (d) Differentiate between SEPIC converter and Ćuk converter. Marks: 5

4.(a) Draw a typical circuit of a linear dc voltage regulator. What is the drawback of it? How can it be overcome? Marks: 6

1. Theory

1. What is a voltage regulator? (Core Idea)

Electronic circuits need a steady dc voltage, but the raw supply may be higher and may change. A regulator gives a fixed output from such an input. The simplest idea is to put something in series with the load that drops the extra voltage, like a resistor that adjusts itself.

2. How does a transistor do this?

A transistor can act as a variable resistor when it runs in its linear (active) region. The load current passes through it, and the voltage across it is whatever is left over:

$$V_{CE} = V_s - V_o$$

A feedback loop keeps the output right. A divider gives a fraction of V_o to an error amplifier, which compares it with a steady reference V_{ref}. If V_o falls, the amplifier drives the transistor harder, so V_{CE} drops and V_o rises back.

3. Where does the power go?

The transistor carries the load current with a voltage across it, so it turns power into heat:

$$P_{loss} \approx V_{CE} I_L = (V_s - V_o) I_L$$

The load receives $V_o I_L$, while the source gives $V_s I_L$. So the best possible efficiency is

$$\eta \approx \frac{V_o}{V_s}$$

For $V_s = 24$ V and $V_o = 5$ V this is only 21 %. At 1 A load, 19 W becomes heat in the transistor.

2. Answer

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Circuit:

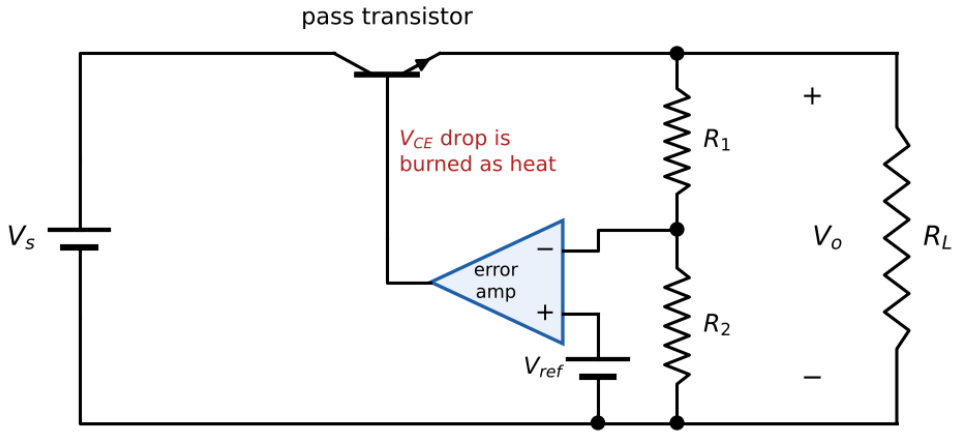


Figure 19: Linear regulator: a pass transistor in series with the load, with feedback from a resistor divider to an error amplifier

Reading the figure. The transistor operates in its linear region, and the error amplifier sets its base so the divider voltage equals V_{ref}. So $V_o = V_{ref}(1 + R_1/R_2)$.

Drawback: the transistor drops the whole difference $V_s - V_o$ while carrying the load current, so it wastes the power $P_{loss} = (V_s - V_o)I_L$ as heat. Its efficiency is about V_o/V_s, which is poor when V_s is much larger than V_o. It also needs a big heat sink.

How it is overcome: use the transistor as a **switch**, not a variable resistor. This is a switched-mode (dc-dc) converter. An ideal switch has either no voltage across it (on) or no current through it (off), so it dissipates almost no power. The output is the average of a pulsed voltage, set by the duty ratio, and an LC filter smooths it. Efficiencies above 90 % are normal. 4(b) shows the basic circuit.

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4.(b) Draw the circuit diagram and the waveform of inductor current, diode current and capacitor current for a typical buck converter. Also derive the relation between inductor current and diode current from the waveforms. Marks: 8

1. Theory

1. What is a buck converter? (Core Idea)

A buck converter steps a dc voltage down. A switch chops the input into pulses, and the average of the pulses is the output. If the switch is closed for a fraction D of each period T, the average is DV_s. D is the **duty ratio**.

2. Why do we need L, C and a diode?

A load wants smooth dc, not pulses. So an inductor and a capacitor form a low-pass filter. The inductor's current cannot jump. So when the switch opens, the inductor current needs another path. The diode gives that path, and so it is often called a freewheeling diode.

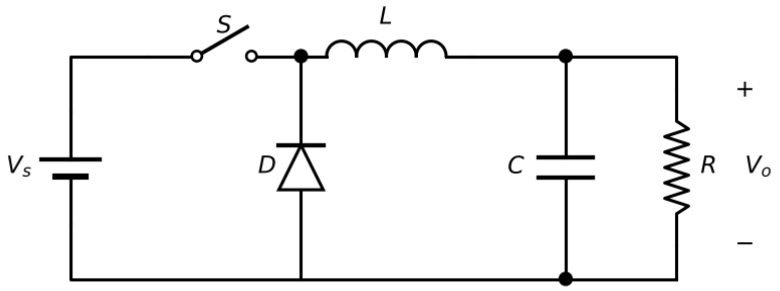


Figure 20: Buck converter (the on/off paths are shown in the next figure)

3. What happens in each interval?

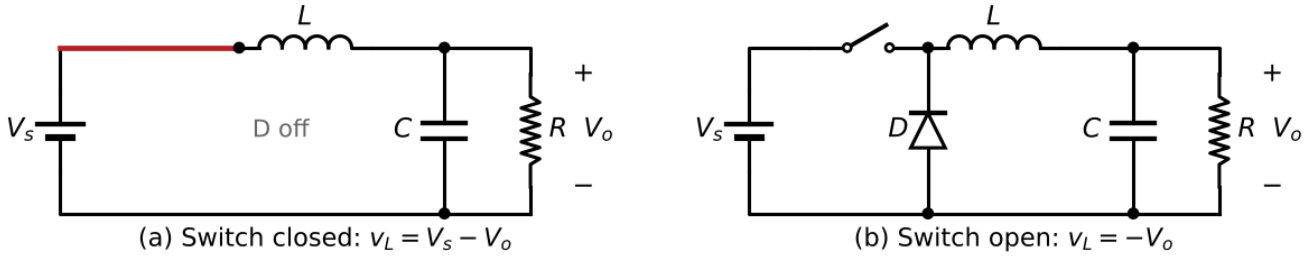


Figure 21: Equivalent circuits: (a) switch closed, the diode is reverse biased, (b) switch open, the diode carries the inductor current

Reading the figure. In (a) the source drives the inductor (the thick line is the closed switch), and the voltage across the inductor is $V_s - V_o$. It is positive, so the inductor current rises. In (b) the source is cut off, and the diode holds the left end of the inductor at ground. The voltage across the inductor is $-V_o$, so the current falls.

4. Why is the output DV_s?

In steady state, the inductor current ends each period where it began. Since $v_L = L\,di/dt$, the rise during the on time must equal the fall during the off time. So the average inductor voltage is zero:

$$(V_s - V_o)DT + (-V_o)(1 - D)T = 0 \quad \Rightarrow \quad V_o = DV_s$$

The current swing is

$$\Delta i_L = \frac{(V_s - V_o)DT}{L} = \frac{V_o(1 - D)}{Lf}$$

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2. Answer

Circuit: the buck converter above, with switch S, diode D, inductor L, capacitor C and load R.
Waveforms (continuous current):

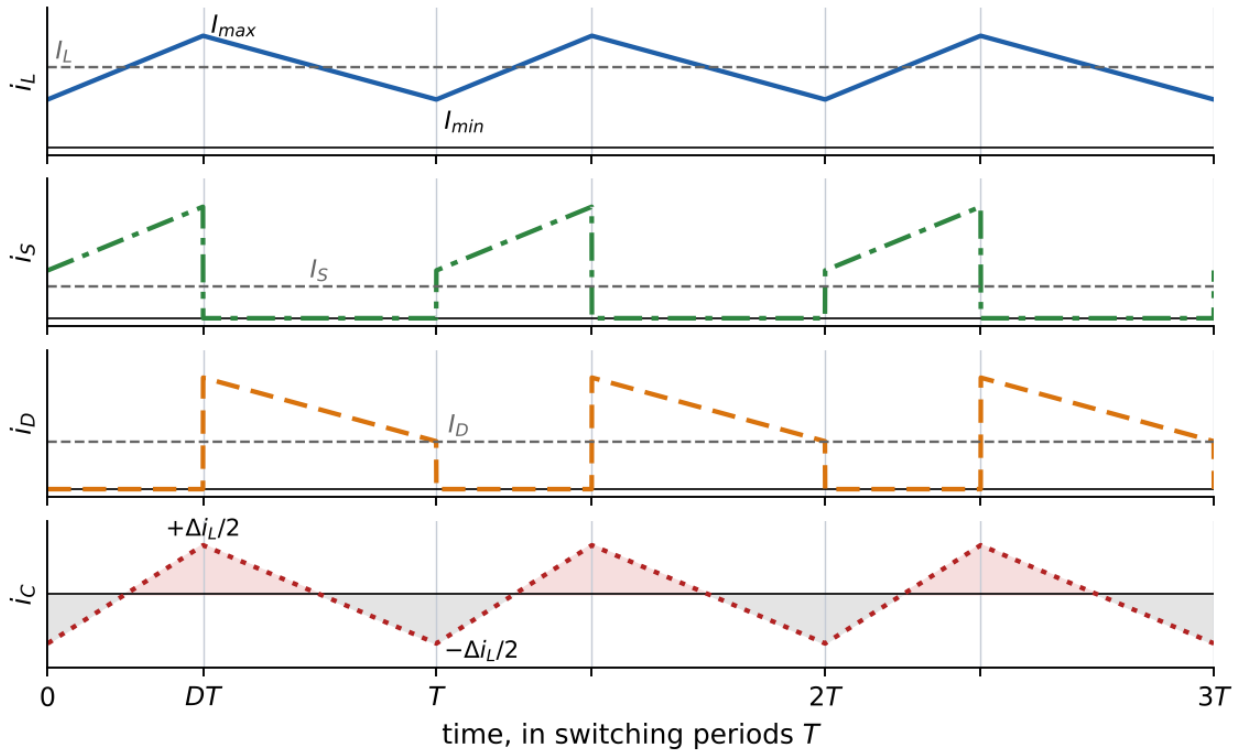


Figure 22: Buck converter currents: inductor current (solid blue), switch current (dash-dot green), diode current (dashed orange) and capacitor current (dotted red). Grey dashed lines are the averages

Reading the figure. The inductor current is a triangle around its average I_L. During $0 < t < DT$ it flows through the switch, and from DT to T it flows through the diode. The capacitor takes the difference between the inductor current and the steady load current, so it is the same triangle moved down to zero average.

Derivation of the relation. The waveforms follow from the two equivalent circuits in the Theory, so the derivation starts there.

Step 1: The inductor voltage in each interval

Assume steady state, continuous inductor current and a large capacitor, so that $v_o = V_o$ is constant.

- Switch closed ($0 < t < DT$): the diode is reverse biased, and the inductor sees $v_L = V_s - V_o$.
- Switch open ($DT < t < T$): the inductor current flows on through the diode, which holds the inductor's left end at 0 V, so $v_L = -V_o$.

Step 2: The inductor current waveform

Since $v_L = L\, di_L/dt$, a constant v_L gives a straight line. The current rises with slope $(V_s - V_o)/L$ while the switch is closed, and falls with slope $-V_o/L$ while it is open:

$$(\Delta i_L)_{closed} = \frac{V_s - V_o}{L} DT, \qquad (\Delta i_L)_{open} = -\frac{V_o}{L}(1 - D)T$$

In steady state the current ends each period where it began, so the two changes add to zero. This gives $V_o = DV_s$ (Hart, Eqs. 6-6 to 6-9). So i_L is a triangle between I_{min} and I_{max}.

Step 3: The average inductor current

At the output node, $i_L = i_C + I_o$. The average capacitor current is zero in steady state (otherwise its voltage would keep changing), so the averages give

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$$I_L = I_o = \frac{V_o}{R}$$

(Hart, Eq. 6-10). The triangle is centred on I_L, so $I_{max} = I_L + \Delta i_L/2$ and $I_{min} = I_L - \Delta i_L/2$ (Hart, Eqs. 6-11 and 6-12), and $I_L = (I_{max} + I_{min})/2$.

Step 4: The diode current waveform

While the switch is closed the diode is off; while it is open the diode carries the whole inductor current:

$$i_D = \begin{cases} 0, & 0 < t < DT \\ i_L, & DT < t < T \end{cases}$$

So i_D is the falling part of the triangle, from I_{max} down to I_{min}, and zero in between.

Step 5 (Basic): The area under a straight-line segment

A current that changes linearly from I_a to I_b over a time t_x encloses a trapezoid. Its area is the base times the mean of the two heights:

$$\int_0^{t_x} i\,dt = t_x \cdot \frac{I_a + I_b}{2}$$

Step 6: The average diode current

The average is the area under i_D over one period, divided by T. Only the open interval contributes; its base is $(1-D)T$ and its heights are I_{max} and I_{min} (Step 5):

$$I_D = \frac{1}{T}\left[(1-D)T \cdot \frac{I_{max} + I_{min}}{2}\right] = (1-D)\frac{I_{max} + I_{min}}{2}$$

With $I_L = (I_{max} + I_{min})/2$ from Step 3:

$$\boxed{I_D = (1-D)I_L}$$

Step 7: The switch current, as a check

The switch carries i_L only while it is closed, the rising part of the triangle from I_{min} to I_{max} over DT. By Step 5,

$$I_S = \frac{1}{T}\left[DT \cdot \frac{I_{min} + I_{max}}{2}\right] = DI_L$$

At every instant the inductor current flows through either the switch or the diode, so $I_S + I_D = DI_L + (1-D)I_L = I_L$, as it must.

Step 8: The capacitor current

From Step 3, $i_C = i_L - I_o$: the same triangle as i_L, moved down by $I_o = I_L$, so it swings between $-\Delta i_L/2$ and $+\Delta i_L/2$ with zero average.

The peak diode current equals the peak inductor current, $I_{max} = I_L + \Delta i_L/2$.

Steps to remember

v_L in each interval → straight-line i_L triangle → average capacitor current 0 gives $I_L = I_o = V_o/R$ → $i_D = i_L$ only in $(1-D)T$ → trapezoid area → $I_D = (1-D)I_L$ (and $I_S = DI_L$).

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4.(c) A boost converter has an input of [unclear: 5 V] and an output of 25 W at 15 V. The minimum inductor current must be no less than 50% of the average. The ripple output must be less than 1 percent. The switching frequency is [unclear: 300 kHz]. (i) Determine the duty ratio. (ii) Minimum inductor value (iii) Minimum capacitor value Marks: 6

1. Theory

1. What is a boost converter? (Core Idea)

A boost converter gives an output larger than its input. The inductor is on the input side. With the switch closed, the inductor is across the source and stores energy. When the switch opens, the inductor's voltage adds to the source, and together they push current through the diode into the output.

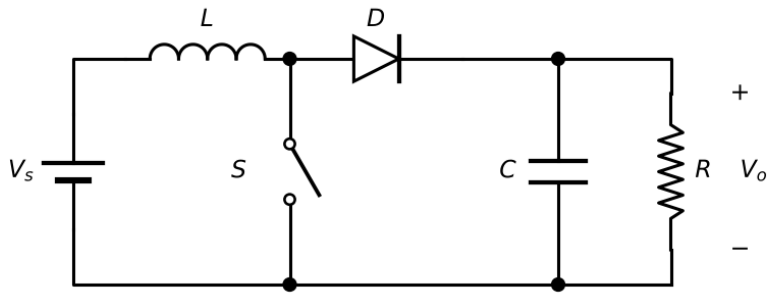


Figure 23: Boost converter

2. How large is the output?

The average inductor voltage is again zero. With the switch closed $v_L = V_s$, and with it open $v_L = V_s - V_o$:

$$V_s DT + (V_s - V_o)(1 - D)T = 0 \quad \Rightarrow \quad V_o = \frac{V_s}{1 - D}$$

3. How large is the inductor current?

The input power equals the output power (ideal parts). The source current is the inductor current:

$$V_s I_L = \frac{V_o^2}{R} \quad \Rightarrow \quad I_L = \frac{V_o^2}{V_s R}$$

4. What do L and C decide?

With the switch closed, the current rises by

$$\Delta i_L = \frac{V_s DT}{L} = \frac{V_s D}{Lf}$$

A bigger L makes this swing smaller. The inductor current's lowest value is $I_{min} = I_L - \Delta i_L/2$.
While the switch is closed, the diode is off and the capacitor alone feeds the load with the steady current V_o/R. It loses the charge $\Delta Q = (V_o/R)DT = C\Delta V_o$. So

$$\frac{\Delta V_o}{V_o} = \frac{D}{RCf}$$

A bigger C makes the ripple smaller.

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2. Answer

Note on the paper

One digit of the input voltage is blotted and could look like 65 V, and the letter before "Hz" is smudged. A boost converter's output cannot be below its input, so 65 V is impossible with 15 V out. The input is read as 5 V and the frequency as 300 kHz.

Given:

- $V_s = 5$ V, $V_o = 15$ V, $P_o = 25$ W
- $I_{min} \geq 0.5 I_L$
- $\Delta V_o / V_o \leq 1\,\% = 0.01$
- $f = 300$ kHz, so $T = 3.33$ μs

Step 1: Duty ratio

$$V_o = \frac{V_s}{1-D} \Rightarrow D = 1 - \frac{V_s}{V_o} = 1 - \frac{5}{15} = 0.667$$

$$\boxed{D = 0.667}$$

Step 2: Load and average inductor current

$$R = \frac{V_o^2}{P_o} = \frac{15^2}{25} = 9\ \Omega$$

$$I_L = \frac{V_o^2}{V_s R} = \frac{P_o}{V_s} = \frac{25}{5} = 5\ \text{A}$$

Step 3: Minimum inductor value

The minimum current is to be 50 % of 5 A, so

$$I_{min} = I_L - \frac{\Delta i_L}{2} = 2.5\ \text{A} \Rightarrow \Delta i_L = 5\ \text{A}$$

$$L = \frac{V_s D}{f \Delta i_L} = \frac{5(0.667)}{(300 \times 10^3)(5)} = 2.22 \times 10^{-6}\ \text{H}$$

$$\boxed{L_{min} = 2.22\ \mu\text{H}}$$

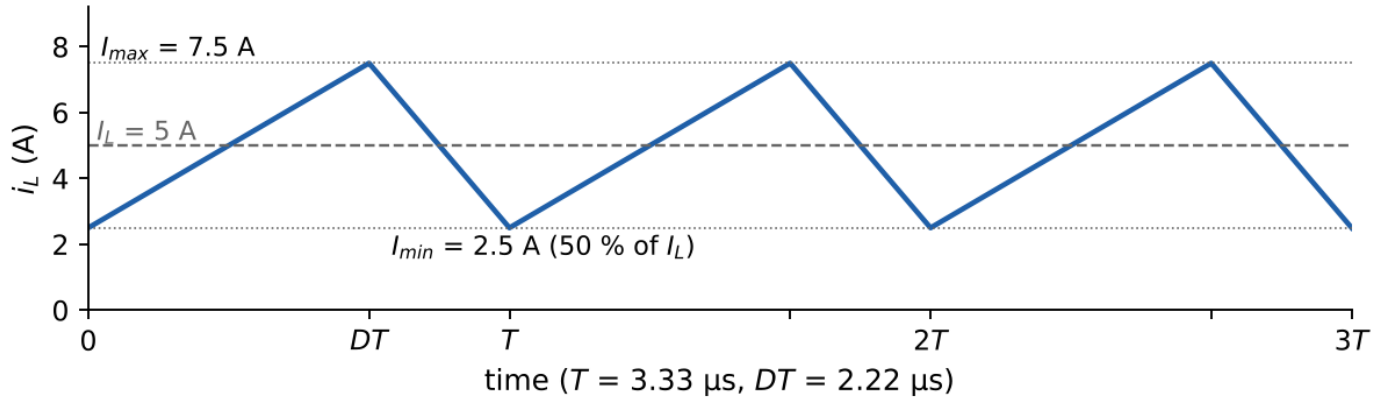


Figure 24: Inductor current with the minimum inductor: a swing from 2.5 A to 7.5 A around 5 A

Reading the figure. The lowest point just touches 2.5 A, that is, 50 % of the average. A larger L would make the triangle flatter, which also meets the requirement.

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Step 4: Minimum capacitor value

$$C = \frac{D}{Rf(\Delta V_o/V_o)} = \frac{0.667}{9(300\times10^3)(0.01)} = 24.7\times10^{-6}\ \text{F}$$

$$\boxed{C_{min} = 24.7\ \mu\text{F}}$$

Final Answer: $D = 0.667$, $L_{min} = 2.22\ \mu$H, $C_{min} = 24.7\ \mu$F.

4.(d) Differentiate between SEPIC converter and Ćuk converter. Marks: 5

1. Theory

1. Why are there more converters? (Core Idea)

A buck only steps down and a boost only steps up. Sometimes the input can be above or below the wanted output, as with a battery that runs down. Both the SEPIC (single-ended primary inductance converter) and the Ćuk converter can do both. Each has two inductors and a coupling capacitor C_1, which carries the energy from input to output.

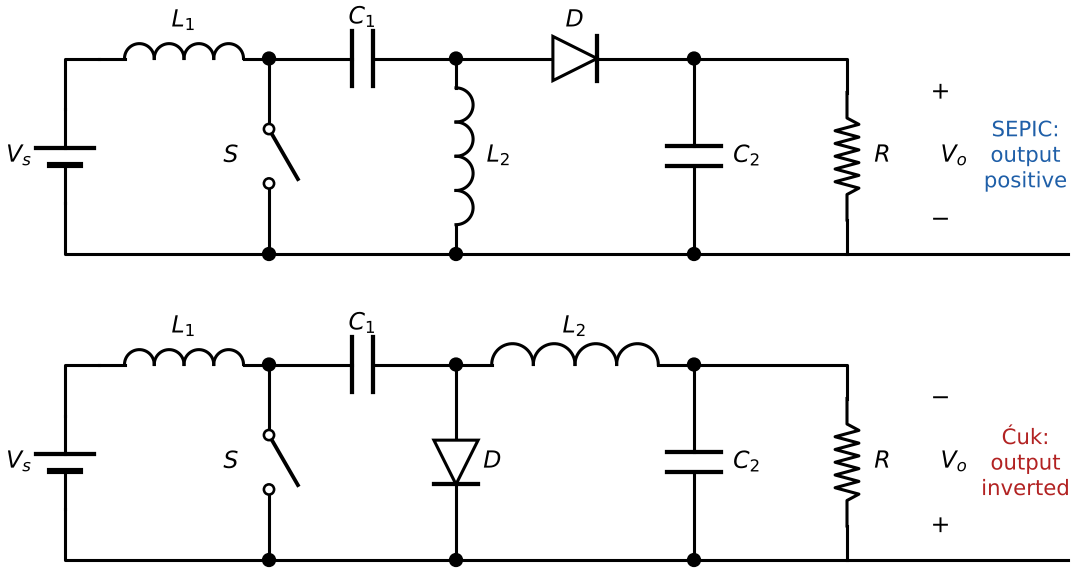


Figure 25: SEPIC (top) and Ćuk converter (bottom). Both have an inductor at the input. They differ in where the second inductor and the diode are

Reading the figure. The two circuits are the same up to C_1. In the SEPIC, L_2 goes to ground and the diode leads to the output. In the Ćuk, the diode goes to ground and L_2 leads to the output. This swap changes the polarity of the output and makes the output current smooth in the Ćuk.

2. Why does a Ćuk invert the output?

With the switch open, L_1's current charges C_1 through the diode to ground, so C_1 has its left plate positive. With the switch closed, the left plate is at ground, so the right plate is driven **negative**. That negative voltage is what reaches the load through L_2.

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2. Answer

Feature	SEPIC	Ćuk
Output voltage	$V_o = V_s \dfrac{D}{1-D}$	$V_o = -V_s \dfrac{D}{1-D}$
Output polarity	same as the input	inverted
Position of L_2	from C_1 to ground	in series with the output
Position of the diode	in series with the output	from C_1 to ground
Output current	pulsating (needs a larger C_2)	continuous (small output ripple)
Input current	continuous	continuous
Voltage on C_1	$V_{C1} = V_s$	$V_{C1} = V_s + \lvert V_o \rvert$

Table 3: The two converters compared

Both can step up ($D > 0.5$) or down ($D < 0.5$), and both move energy through C_1.

3. Final Concise Answer

(a) A pass transistor in series with the load, controlled by an error amplifier. Drawback: it wastes $(V_s - V_o)I_L$ as heat, so $\eta \approx V_o/V_s$. Cure: use a switched-mode converter.

(b) $i_D = i_L$ during $(1-D)T$ and zero otherwise, so $I_D = (1-D)I_L$; $i_C = i_L - I_o$; $I_L = I_o$.

(c) $D = 0.667$, $L_{min} = 2.22\ \mu$H, $C_{min} = 24.7\ \mu$F (with 5 V input and 300 kHz).

(d) SEPIC: positive output, pulsating output current, $V_{C1} = V_s$. Ćuk: inverted output, continuous output current, $V_{C1} = V_s + |V_o|$.

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5. (a) Briefly explain the types of switch mode DC-DC converters used in a stand-alone solar PV system. Marks: 5

(b) Draw the circuit of a three-level cascade multi source inverter. Why pattern swapping is important in multi-source inverter circuit? Marks: 6

(c) Design a bipolar PWM inverter that will produce a 75-V rms 60-Hz output from a 150 V dc source. The load is a series RL combination with R=12Ω and L=60 mH. Select the switching frequency such that the current THD is less than 10%. Assume the impedance at the carrier frequency is entirely inductive. Marks: 10

(d) Determine the voltage THD of a square wave inverter. Marks: 4

Normalized Fourier Coefficients V_n/V_{dc} for Bipolar PWM

	$m_a = 1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n = 1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n = m_f$	0.60	0.71	0.82	0.92	1.01	1.08	1.15	1.20	1.24	1.27
$n = m_f \pm 2$	0.32	0.27	0.22	0.17	0.13	0.09	0.06	0.03	0.02	0.00

Figure 26: The table printed under Q5(c)

5.(a) Briefly explain the types of switch mode DC-DC converters used in a stand-alone solar PV system. Marks: 5

1. Theory

1. What is a stand-alone PV system? (Core Idea)

A stand-alone system is not joined to the grid. Solar panels give dc power by day, a battery stores it, and the loads run from the battery, directly or through an inverter.

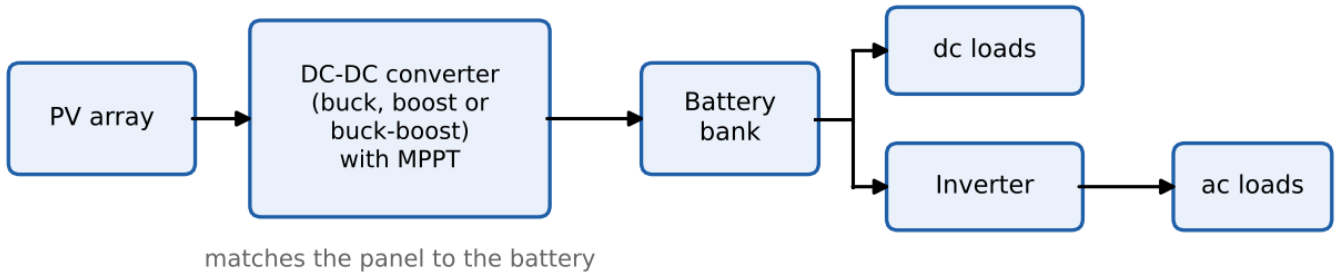


Figure 27: A stand-alone PV system. The dc-dc converter sits between the panels and the battery

Reading the figure. Power flows from left to right. The converter is the part that controls how the panel's power reaches the battery.

2. Why is a converter needed at all?

A panel's voltage and current depend on the sunlight and on what it is connected to. For each sun level, there is one operating point at which the panel gives the most power.

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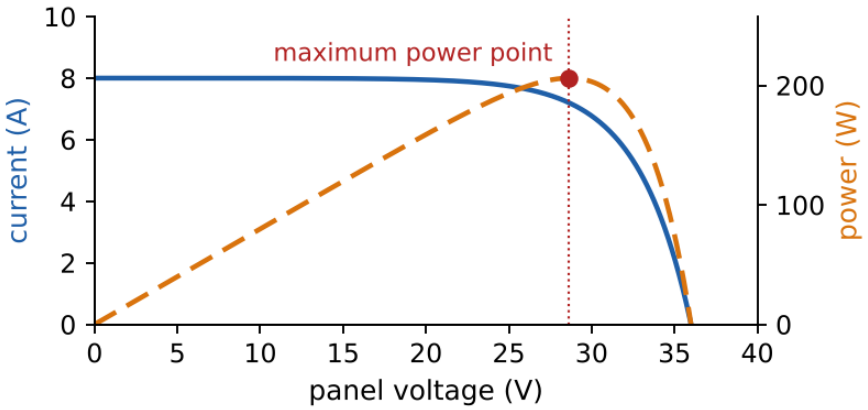


Figure 28: A panel's current (solid blue curve) and power (dashed orange curve) against voltage. The red dot is the maximum power point

Reading the figure. As the voltage rises, the current stays nearly flat and then falls fast. Power is voltage times current, so it has one peak, at the knee of the current curve.

The battery fixes its own voltage. If the panel were wired straight to it, the panel would have to work at the battery's voltage, which is usually not the peak. A dc-dc converter sits between them and lets the two voltages differ. Its duty ratio is adjusted by a controller until the panel works at its peak. This is **maximum power point tracking (MPPT)**.

2. Answer

Switch-mode dc-dc converters are used between the PV array and the battery to charge the battery and to track the maximum power point. The type depends on the voltages:

- **Buck converter:** used when the array voltage is above the battery voltage. $V_o = DV_s$. It is simple and efficient, and it is the usual charge controller.
- **Boost converter:** used when the array voltage is below the battery or dc-bus voltage. $V_o = V_s/(1-D)$.
- **Buck-boost converter (also SEPIC or Ćuk):** used when the array voltage can be above or below the battery voltage, as it changes with sun and temperature. $|V_o| = V_s D/(1-D)$. MPPT then works in all conditions.

In each case the controller changes D to keep the array at its maximum power point and to control the charging of the battery.

5.(b) Draw the circuit of a three-level cascade multi source inverter. Why pattern swapping is important in multi-source inverter circuit? Marks: 6

1. Theory

1. What is an inverter? (Core Idea)

An inverter turns dc into ac. The basic cell is a **full bridge**, or H-bridge, of four switches around a dc source. Closing S_1 and S_2 puts $+V_{dc}$ across the output. Closing S_3 and S_4 gives $-V_{dc}$. Closing both top switches (or both bottom ones) gives zero. The figure in 5(c) shows one bridge.

2. Why several levels?

A sine wave is smooth. A two-level wave jumps between $+V_{dc}$ and $-V_{dc}$, so it has many harmonics. If the output can take several levels, it can climb in steps and follow a sine much better. High-power, high-voltage systems also like this, since each switch sees only its own V_{dc}.

3. How does a cascade make levels?

Connect the outputs of several H-bridges in series. Each cell has its own independent dc source, such as a battery or a PV panel, and each can give $+V_{dc}$, 0 or $-V_{dc}$. The output is their sum:

$$v_o = v_1 + v_2 + v_3$$

With three cells, the output can be $\pm 3V_{dc}$, $\pm 2V_{dc}$, $\pm V_{dc}$ or 0. That is $2k+1 = 7$ voltage levels for $k = 3$ cells.

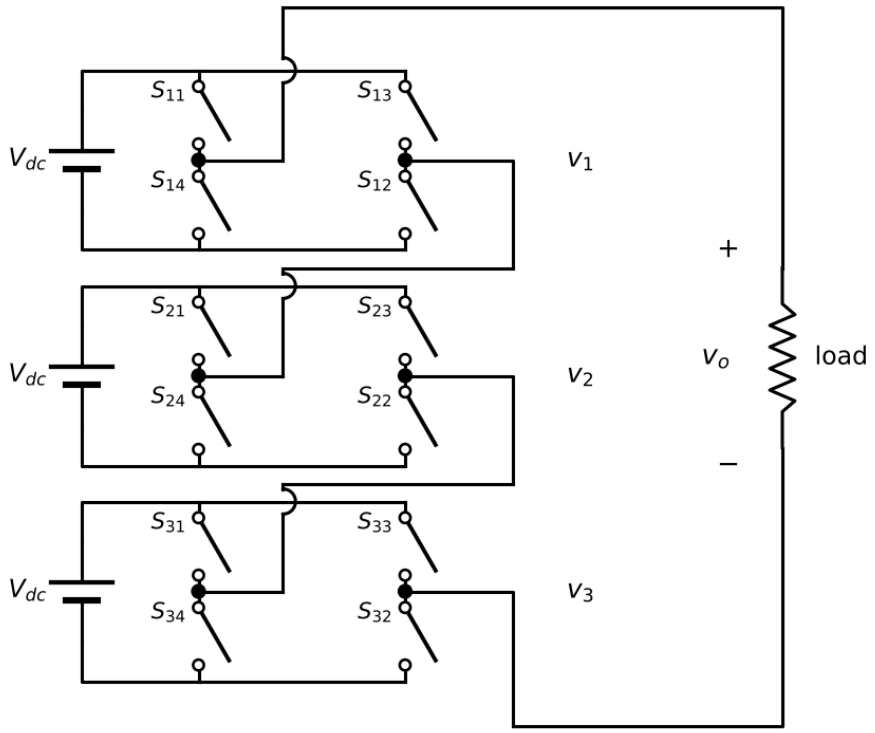


Figure 29: Three cells in cascade: each cell's right terminal joins the next cell's left terminal. v_1, v_2 and v_3 are each cell's left terminal minus its right terminal

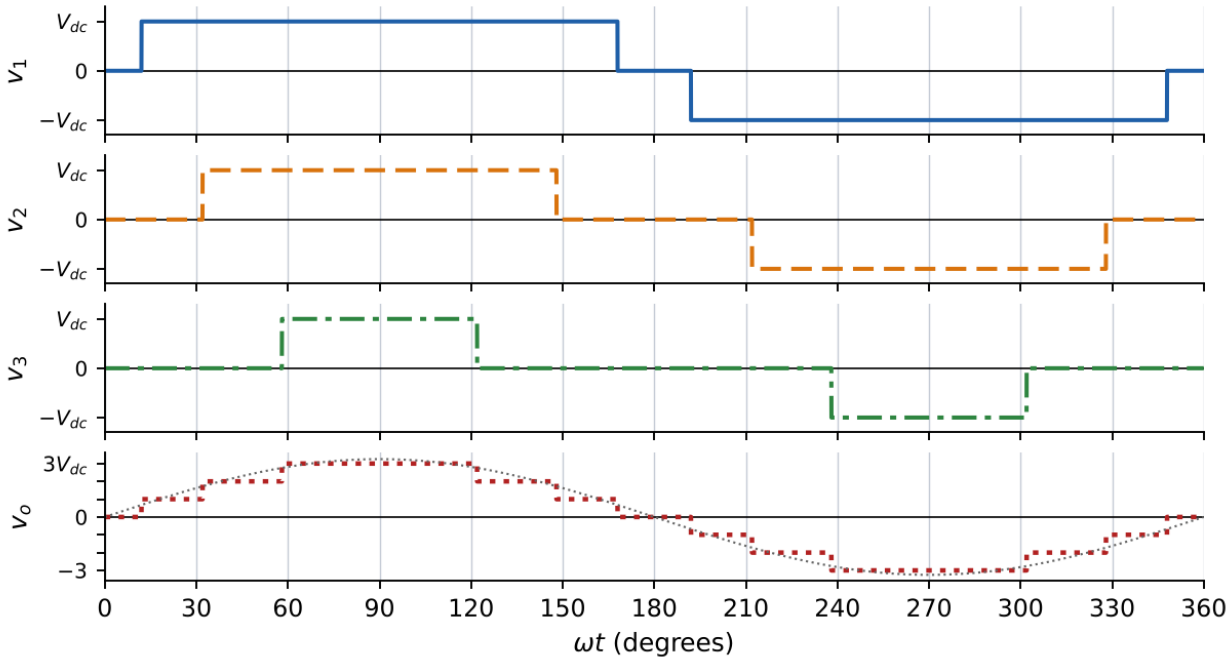


Figure 30: Each cell switched at its own angle (12°, 32°, 58° as an example) and the stepped sum, with a sine (thin dotted curve) for comparison

Reading the figures. Cell 1 (solid blue) turns on first and stays on longest. Cell 3 (dash-dot green) turns on last. The dotted red staircase v_o is the sum of the three, and it follows the thin dotted sine.

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The choice of the three angles sets the harmonics.

4. What is the problem with this pattern?

Look at the widths. Cell 1 conducts far longer than cell 3 in every half cycle. So it delivers more charge and more energy. If the sources are batteries, the first one runs down early, and if they are PV panels, they are unequally loaded.

2. Answer

Circuit: three H-bridges in series, each with its own source V_{dc} (figure above). The output is $v_o = v_1 + v_2 + v_3$, with steps $\pm 3V_{dc}$, $\pm 2V_{dc}$, $\pm V_{dc}$, 0.

Pattern swapping: in a fixed switching pattern, the cell with the widest pulse carries the load current for the longest time. So the sources are used unequally: one battery discharges much faster than the others.

To avoid this, the pulse patterns are **swapped between the cells** from one half cycle to the next, so that each source takes the wide, middle and narrow pulses in turn.

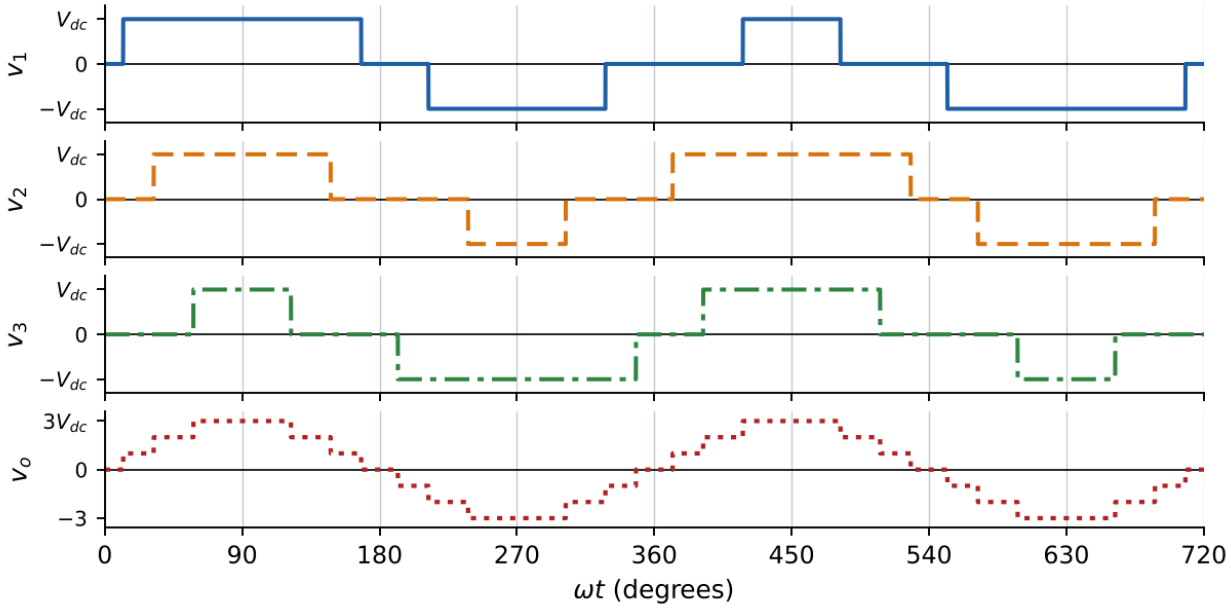


Figure 31: Pattern swapping: the three pulse widths rotate among the cells in each half cycle, and the sum v_o (dotted red) stays the same stepped wave

Why it is important:

- every source gives the same average power and energy over a few cycles;
- batteries discharge equally, and last longer;
- the switches share the conduction loss evenly;
- the output voltage does not change, because only the cell that makes each step changes.

3. Notes and Remarks

Assumption: the course slides call this circuit a "3-level inverter": three cascaded cells with v_o up to $3V_{dc}$. Counted by distinct voltages, it has seven levels. If "three-level" were read as three output voltages ($+V_{dc}$, 0, $-V_{dc}$), a single H-bridge would do it, and there would be nothing to swap, so the three-cell reading is the one the question means.

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5.(c) Design a bipolar PWM inverter that will produce a 75-V rms 60-Hz output from a 150 V dc source. The load is a series RL combination with R=12Ω and L=60 mH. Select the switching frequency such that the current THD is less than 10%. Assume the impedance at the carrier frequency is entirely inductive. Marks: 10

1. Theory

1. What is pulse-width modulation? (Core Idea)

The output of a bridge can only be $+V_{dc}$ or $-V_{dc}$. But if we switch much faster than the wanted 60 Hz, and change the pulse widths along the cycle, the **average** over a short time follows a sine. The load's inductance smooths the current, so it mostly sees that sine.

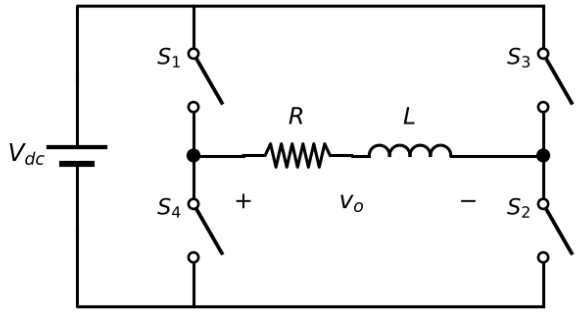


Figure 32: Full-bridge inverter with the RL load between the two leg midpoints

2. How are the pulses made?

Two signals are compared: a sine **reference** at the wanted frequency, and a fast triangle **carrier**. In **bipolar** switching:

- when the reference is above the carrier, S_1 and S_2 are closed: $v_o = +V_{dc}$;
- when the reference is below the carrier, S_3 and S_4 are closed: $v_o = -V_{dc}$.

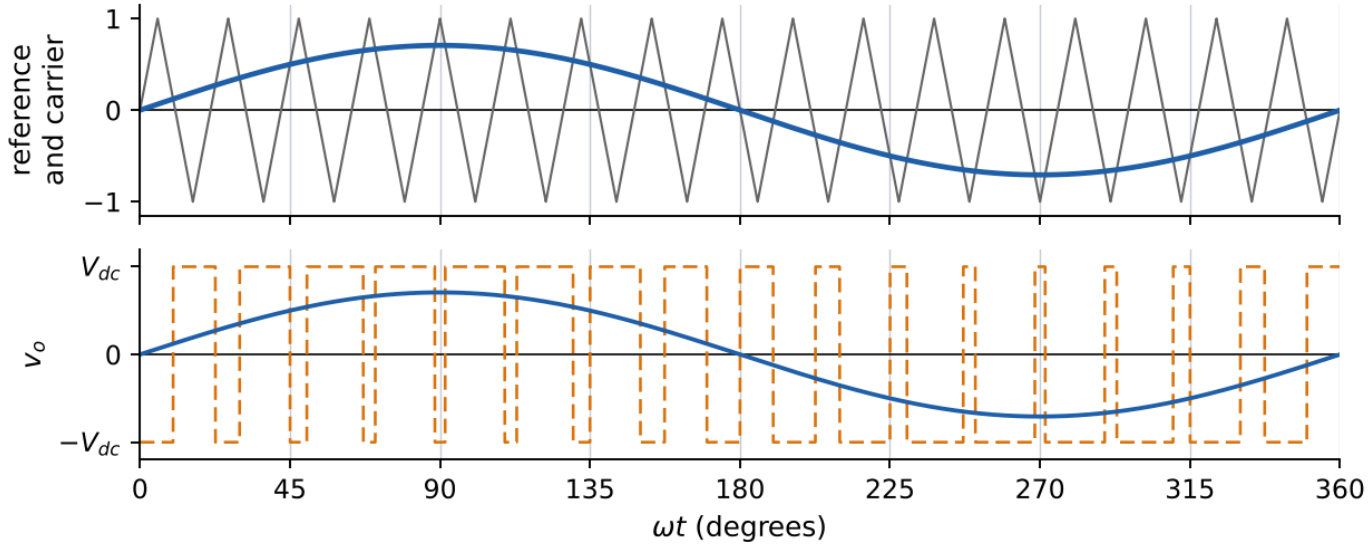


Figure 33: Bipolar PWM: reference (thick solid blue curve) and carrier (thin grey), and the output (thin dashed orange pulses) with its fundamental (solid blue curve), for the design values of this question

Reading the figure. Near 90° the reference is high, so the pulses are mostly at $+V_{dc}$. Near 270° they are mostly at $-V_{dc}$. The fundamental is the sine hidden inside the pulses.

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3. What are m_a and m_f?

$$m_a = \frac{V_{sine}}{V_{tri}}, \qquad m_f = \frac{f_{carrier}}{f_{reference}}$$

The **amplitude modulation ratio** m_a sets the size of the fundamental. For $m_a \leq 1$:

$$V_1 = m_a V_{dc} \text{ (peak)}$$

The **frequency modulation ratio** m_f sets where the harmonics sit. They cluster at m_f, at $m_f \pm 2$, and around multiples of m_f. Choosing m_f as an odd integer keeps the wave symmetrical, so there are no even harmonics. The table in the paper gives the sizes.

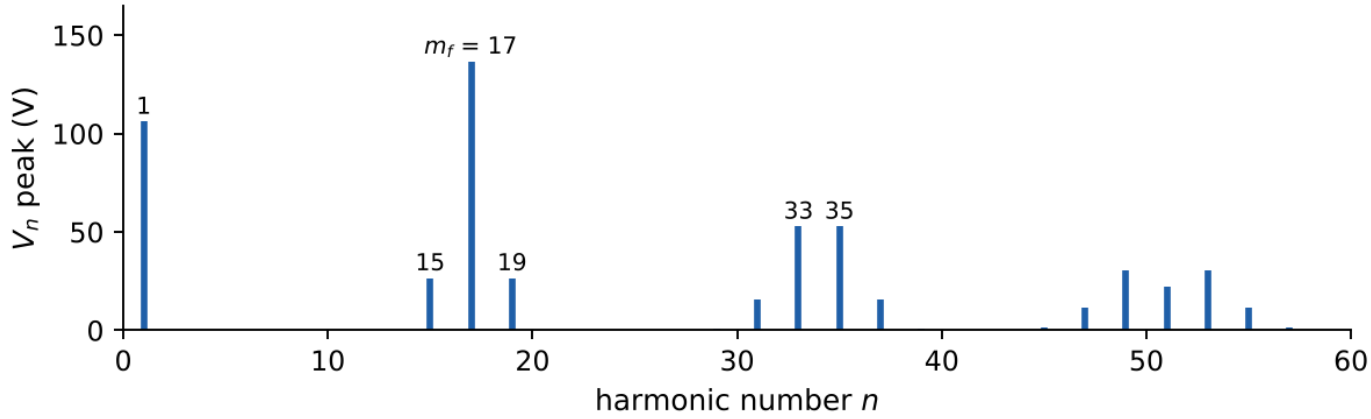


Figure 34: Harmonic spectrum of the output for the design values: a fundamental of 106 V, and clusters near 17, 34 and 51

Reading the figure. Nothing appears below about harmonic 15. The largest harmonic is at m_f, and it is even larger than the fundamental.

4. Why does a higher carrier frequency lower the current THD?

The voltage harmonics do not get smaller when m_f rises. But they move to a higher frequency, where the inductor's reactance $n\omega L$ is larger, so the harmonic **currents** $I_n = V_n/(n\omega L)$ are smaller. So the design is to pick m_f just high enough. Higher than that only adds switching loss.

2. Answer

Given: $V_{dc} = 150$ V, $V_{o,rms} = 75$ V, $f = 60$ Hz, $R = 12\ \Omega$, $L = 60$ mH, current THD below 10 %.

Normalized Fourier Coefficients V_n/V_{dc} for Bipolar PWM

	$m_a = 1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n = 1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n = m_f$	0.60	0.71	0.82	0.92	1.01	1.08	1.15	1.20	1.24	1.27
$n = m_f \pm 2$	0.32	0.27	0.22	0.17	0.13	0.09	0.06	0.03	0.02	0.00

Figure 35: Repeated from the question: the table printed under Q5(c)

Step 1: Amplitude modulation ratio

$$V_1 = 75\sqrt{2} = 106.1 \text{ V (peak)}, \qquad m_a = \frac{V_1}{V_{dc}} = \frac{106.1}{150} = 0.707$$

$$\boxed{m_a = 0.707}$$

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Step 2: Fundamental current

$$\omega L = 377(0.06) = 22.6\ \Omega$$

$$Z_1 = \sqrt{12^2 + 22.6^2} = 25.6\ \Omega, \qquad I_1 = \frac{106.1}{25.6} = 4.14\ \text{A (peak)}$$

Step 3: Allowed harmonic current

$$THD = \frac{\sqrt{\sum I_n^2}}{I_1} < 0.10 \Rightarrow \sqrt{\sum I_n^2} < 0.414\ \text{A}$$

Step 4: Harmonic voltages from the table

Take the column $m_a = 0.7$ (the nearest):

$$V_{m_f} = 0.92(150) = 138\ \text{V}, \qquad V_{m_f \pm 2} = 0.17(150) = 25.5\ \text{V}$$

Step 5: Choose the frequency ratio

The impedance at these frequencies is taken as inductive, so $I_n = V_n/(n\omega L)$. The first trial comes from $n = m_f$ alone:

$$m_f > \frac{138}{22.6(0.414)} = 14.7$$

Now check the odd integers, with the two side harmonics included:

m_f	I_{m_f-2} (A)	I_{m_f} (A)	I_{m_f+2} (A)	THD
15	0.087	0.407	0.066	10.2 %
17	0.075	0.359	0.059	9.0 %

Table 4: Current THD for two values of the frequency ratio

$m_f = 15$ just fails, so $m_f = 17$ is chosen.

$$f_{carrier} = m_f f = 17(60) = 1020\ \text{Hz}$$

$$\boxed{m_f = 17, \quad f_{carrier} = 1020\ \text{Hz}}$$

Final Answer (design): full bridge with $V_{dc} = 150$ V, bipolar PWM with $m_a = 0.707$ and a carrier of 1020 Hz ($m_f = 17$). The current THD is about 9 %.

3. Notes and Remarks

A numerical check of the true PWM wave, counting every harmonic and the load's resistance, gives 9.3 % for $m_f = 17$ and 10.6 % for $m_f = 15$. So the choice stands. The extra part comes from the cluster near $2m_f$, which the table does not list.

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5.(d) Determine the voltage THD of a square wave inverter. Marks: 4

1. Theory

1. What is a square-wave inverter? (Core Idea)

The simplest way to run an H-bridge is to close S_1, S_2 for half the period and S_3, S_4 for the other half. The output is a square wave of height $\pm V_{dc}$.

2. What is in a square wave?

The rms value of the wave is simple: its square is V_{dc}^2 everywhere, so $V_{rms} = V_{dc}$. Fourier analysis shows only odd harmonics:

$$v_o(t) = \sum_{n\,\text{odd}} \frac{4V_{dc}}{n\pi} \sin n\omega t$$

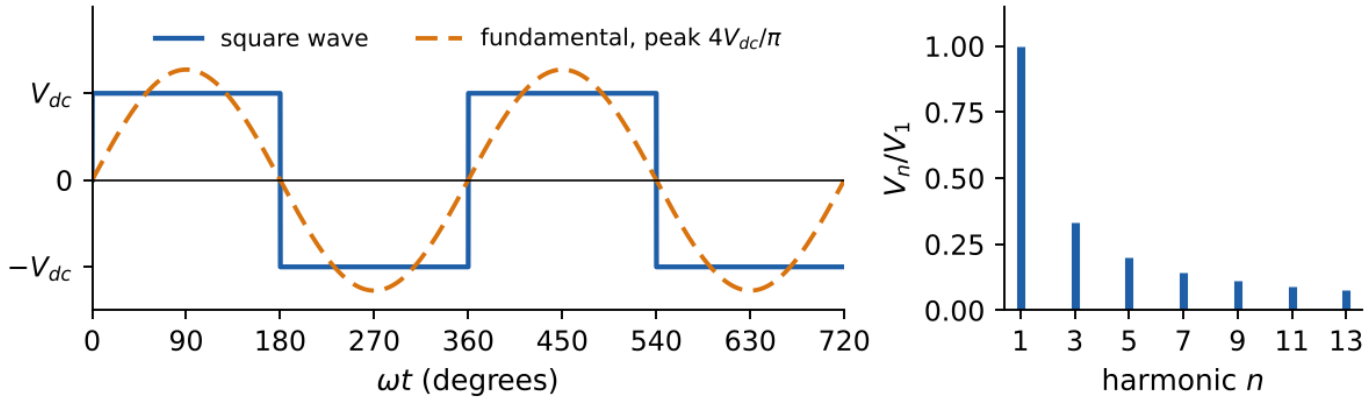


Figure 36: A square wave (solid blue) and its fundamental (dashed orange, taller than the wave), with the harmonic sizes at the right

Reading the figure. The fundamental has peak $4V_{dc}/\pi = 1.27V_{dc}$. That is higher than the square wave itself; the harmonics pull it back down. The nth harmonic is $1/n$ of the fundamental.

2. Answer

Step 1: The output wave

The square-wave inverter gives $v_o = +V_{dc}$ for $0 < \omega t < \pi$ and $v_o = -V_{dc}$ for $\pi < \omega t < 2\pi$, repeating every period.

Step 2 (Basic): Fourier series of the square wave

The wave has zero average and is odd ($v_o(-t) = -v_o(t)$), so its Fourier series has only sine terms, $v_o = \sum b_n \sin n\omega t$, with

$$b_n = \frac{1}{\pi}\int_0^{2\pi} v_o \sin(n\omega t)\, d(\omega t) = \frac{1}{\pi}\left[\int_0^{\pi} V_{dc} \sin n\theta\, d\theta - \int_{\pi}^{2\pi} V_{dc} \sin n\theta\, d\theta\right]$$

Each integral is $\int \sin n\theta\, d\theta = -\cos n\theta/n$:

$$b_n = \frac{V_{dc}}{n\pi}\left[(1 - \cos n\pi) + (\cos 2n\pi - \cos n\pi)\right] = \frac{V_{dc}}{n\pi} \cdot 2(1 - \cos n\pi)$$

For odd n, $\cos n\pi = -1$ and $b_n = 4V_{dc}/n\pi$; for even n, $\cos n\pi = 1$ and $b_n = 0$. So

$$v_o(t) = \sum_{n\,\text{odd}} \frac{4V_{dc}}{n\pi} \sin n\omega t$$

(Hart, Eq. 8-16).

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Step 3: The fundamental (rms)

The fundamental is the $n = 1$ term, with peak $4V_{dc}/\pi$. Its rms value is the peak divided by $\sqrt{2}$:

$$V_{1,rms} = \frac{4V_{dc}}{\pi\sqrt{2}} = 0.900V_{dc}$$

Step 4: The total rms

The wave is $\pm V_{dc}$ at every instant, so $v_o^2 = V_{dc}^2$ always, and its mean is V_{dc}^2:

$$V_{rms} = V_{dc}$$

Step 5: The THD

All the harmonics together have rms $\sqrt{V_{rms}^2 - V_{1,rms}^2}$ (the rms values of the harmonics add as squares, as in 1(a)). Divide by the fundamental:

$$THD = \frac{\sqrt{V_{rms}^2 - V_{1,rms}^2}}{V_{1,rms}} = \frac{\sqrt{V_{dc}^2 - (0.900V_{dc})^2}}{0.900V_{dc}} = 0.483$$

$$\boxed{THD = 48.3\ \%}$$

The answer does not depend on V_{dc}: every square wave has the same shape, so the same THD.

Steps to remember

$\pm V_{dc}$ wave $\rightarrow$ odd sine series, $b_n = 4V_{dc}/n\pi \rightarrow V_1 = 0.9V_{dc}$ rms $\rightarrow V_{rms} = V_{dc} \rightarrow THD = \sqrt{V_{rms}^2 - V_1^2}/V_1 = 48.3\,\%$.

3. Final Concise Answer

(a) Buck (array above battery voltage), boost (array below it) and buck-boost, SEPIC or Ćuk (either way), each adjusted by the duty ratio for maximum power point tracking and charging.

(b) Three H-bridges in series, each with its own source. Pattern swapping passes the wide and narrow pulses from cell to cell, so all sources are used equally.

(c) $m_a = 0.707$, $m_f = 17$, $f_{carrier} = 1020$ Hz; current THD about 9 %.

(d) THD = 48.3 %.

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6. (a) A 10 Ohm resistive load requires a 1000-Hz, 50-V rms sinusoidal voltage. The THD of the load voltage must be no more than 5 percent. An adjustable dc source is available. (i) Design an inverter for this application. (ii) What is maximum voltage across the capacitor? Marks: 7

(b) What is cyclo-converter? Mention its applications. Marks: 5

(c) Explain the structure of TRIAC with diagram. Mention the constructional feature of a Power BJT. Marks: 8

(d) Compare the following device in terms of power capacity and switching speed: BJT, MOSFET, GTO, IGBT, MCT Marks: 5

6.(a) A 10 Ohm resistive load requires a 1000-Hz, 50-V rms sinusoidal voltage. The THD of the load voltage must be no more than 5 percent. An adjustable dc source is available. (i) Design an inverter for this application. (ii) What is maximum voltage across the capacitor? Marks: 7

1. Theory

1. What is the idea? (Core Idea)

An inverter turns dc into ac. The simplest one is a full bridge of four switches, which connects the dc source to the load first one way, then the other. This makes a square wave of height $\pm V_{dc}$ (5(d)), whose fundamental has peak $4V_{dc}/\pi$.

The load wants a clean sine, and the square wave is easy to make. Its harmonics are at 3, 5, 7, … times the fundamental. So a filter that passes the fundamental and blocks everything above it will do. The paper says an adjustable dc source is available, so the bridge need not control the size of the voltage: the dc source does that.

2. What filter is used?

A series L and C between the bridge and the load. Their reactances have opposite signs. At one frequency, called the **resonant frequency**, they cancel exactly. There the filter is a plain wire.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

We tune this to the wanted 1000 Hz, so the fundamental passes with no loss. At harmonic n the net reactance is

$$X_n = n\omega_0 L - \frac{1}{n\omega_0 C} = \omega_0 L\left(n - \frac{1}{n}\right)$$

using $1/(\omega_0 C) = \omega_0 L$ at resonance.

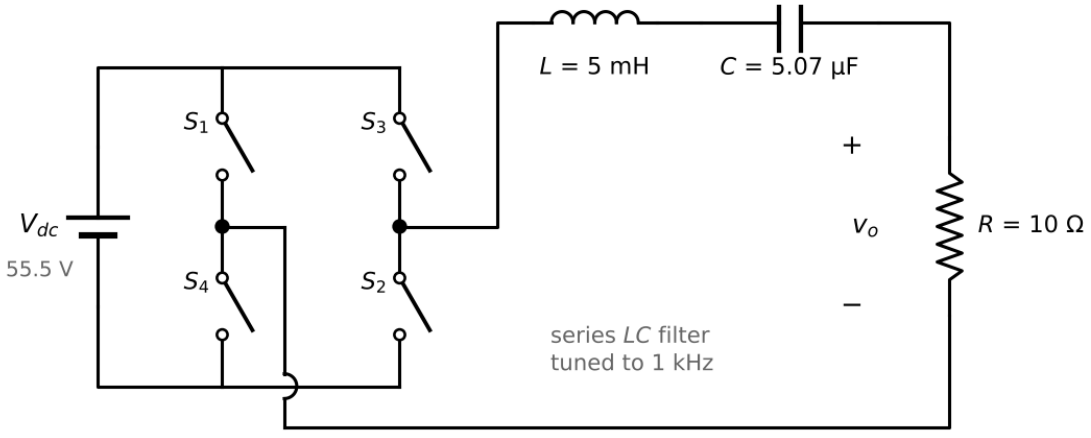


Figure 37: Inverter with the series LC filter. Values are the design values found below

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3. How much of each harmonic reaches the load?

The filter and the load form a voltage divider:

$$\frac{V_{o,n}}{V_n} = \frac{R}{\sqrt{R^2 + X_n^2}}$$

The bridge's own harmonic is $V_n = V_1/n$. So, compared with the fundamental, the load sees

$$\frac{V_{o,n}}{V_1} = \frac{1}{n}\frac{R}{\sqrt{R^2 + X_n^2}}$$

The third harmonic is the largest, so it sets the size of the filter. The THD counts all of them.

4. Why does the capacitor see a high voltage?

The whole load current flows through the capacitor. So its voltage is $I_1/(\omega_0 C)$. At resonance this can be much larger than the source voltage, and the inductor's voltage is equally large and opposite. They cancel in the circuit, but each part must be rated for it.

2. Answer

Given: $R = 10\ \Omega$, $f = 1000$ Hz, $V_{o,rms} = 50$ V, THD ≤ 5 %.

Design choice: a full-bridge inverter giving a square wave, with a series LC filter tuned to 1000 Hz, and the dc source set for the wanted output.

(i) Design

Step 1: DC source. At resonance the whole fundamental reaches the load:

$$V_1 = 50\sqrt{2} = 70.7 \text{ V (peak)}$$

$$\frac{4V_{dc}}{\pi} = 70.7 \Rightarrow \boxed{V_{dc} = 55.5 \text{ V}}$$

Step 2: First size of the filter, from the third harmonic alone. We need $V_{o,3}/V_1 \leq 0.05$:

$$\frac{1}{3}\frac{10}{|Z_3|} \leq 0.05 \Rightarrow |Z_3| \geq 66.7\ \Omega \Rightarrow X_3 \geq 65.9\ \Omega$$

$$X_3 = \omega_0 L\left(3 - \frac{1}{3}\right) \Rightarrow \omega_0 L \geq 24.7\ \Omega$$

Step 3: Check all harmonics. With $\omega_0 L = 24.7\ \Omega$ the higher harmonics push the THD to 5.4 %. Summing all of them, the limit of 5 % needs $\omega_0 L \geq 26.7\ \Omega$, that is, $L \geq 4.25$ mH. Choose the round value

$$\boxed{L = 5 \text{ mH}}$$

Step 4: Capacitor.

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(2\pi 1000)^2 (5 \times 10^{-3})} = 5.07\ \mu\text{F}$$

$$\boxed{C = 5.07\ \mu\text{F}}$$

Then $\omega_0 L = 1/(\omega_0 C) = 31.4\ \Omega$.

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Step 5: Check the THD.

n	X_n (Ω)	$V_{o,n}/V_1$
3	83.8	3.95 %
5	151	1.32 %
7	215	0.66 %

Table 5: Load-voltage harmonics of the design

$$THD = 4.26\ \% < 5\ \%$$

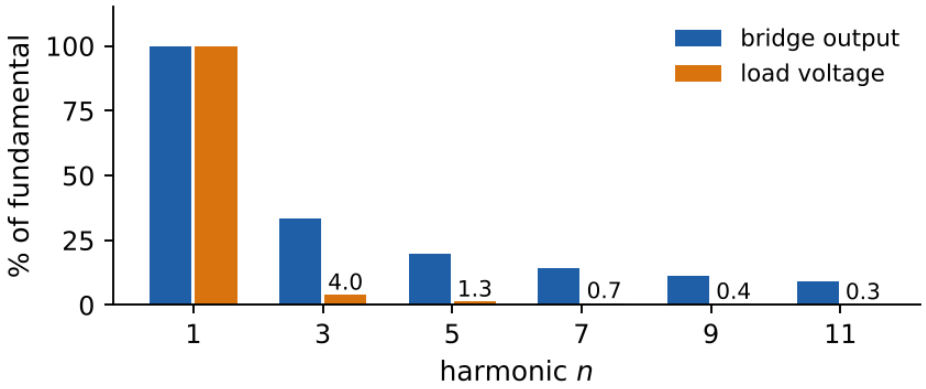


Figure 38: Harmonics of the bridge output (left bar of each pair, blue) and of the load voltage (right bar, orange), as a percentage of the fundamental

Reading the figure. The fundamental passes at full size. The third falls from 33 % to 4 %, and the rest almost vanish.

(ii) Maximum capacitor voltage

The load's peak current is the fundamental's:

$$I_1 = \frac{V_1}{R} = \frac{70.7}{10} = 7.07\ \text{A (peak)}$$

$$V_{C,max} \approx I_1 \frac{1}{\omega_0 C} = 7.07(31.4) = 222\ \text{V}$$

Summing all the harmonics in the time domain raises it only to 223 V.

$$\boxed{V_{C,max} \approx 223\ \text{V}}$$

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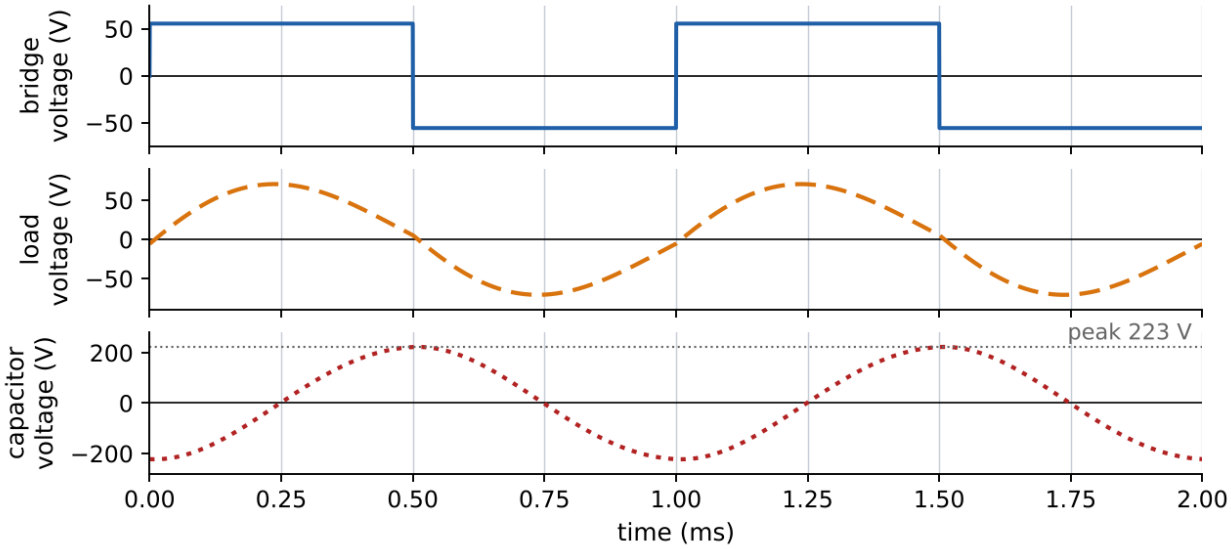


Figure 39: Bridge voltage (solid blue), load voltage (dashed orange) and capacitor voltage (dotted red) of the design

Reading the figure. The 55.5 V square wave becomes a nearly clean 70.7 V peak sine on the load, and the capacitor swings to about 223 V, four times the dc supply.

Final Answer: full bridge, $V_{dc} = 55.5$ V, $L = 5$ mH, $C = 5.07$ μF, THD 4.26 %; capacitor peak voltage about 223 V.

3. Notes and Remarks

Other L, C pairs are also correct if they are tuned to 1 kHz and $L \geq 4.25$ mH. A larger L gives a lower THD but a higher capacitor voltage, since $V_C = I_1 \omega_0 L$.

6.(b) What is cyclo-converter? Mention its applications. Marks: 5

1. Theory

1. What does it do? (Core Idea)

Sometimes we need ac at a lower frequency than the mains, for example to run a very large slow motor. One way is ac to dc to ac. A cycloconverter does it in one stage: it joins the load to the source directly through thyristors, and picks pieces of the source wave to build a lower-frequency wave.

2. How is it built?

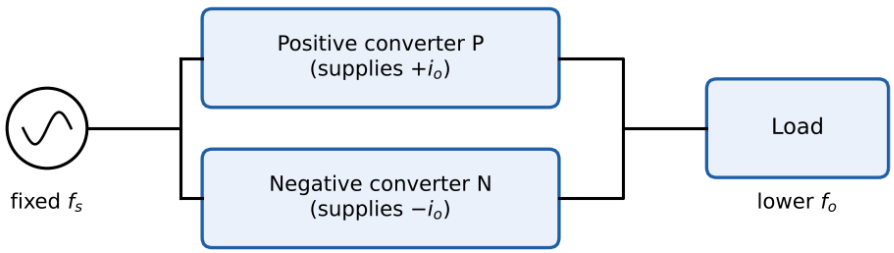


Figure 40: Single-phase cycloconverter: two converters, P and N, in anti-parallel, feeding one load

Reading the figure. One controlled rectifier gives positive output and the other gives negative output. They work in turn, so the load gets both polarities.

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3. How are the pieces chosen?

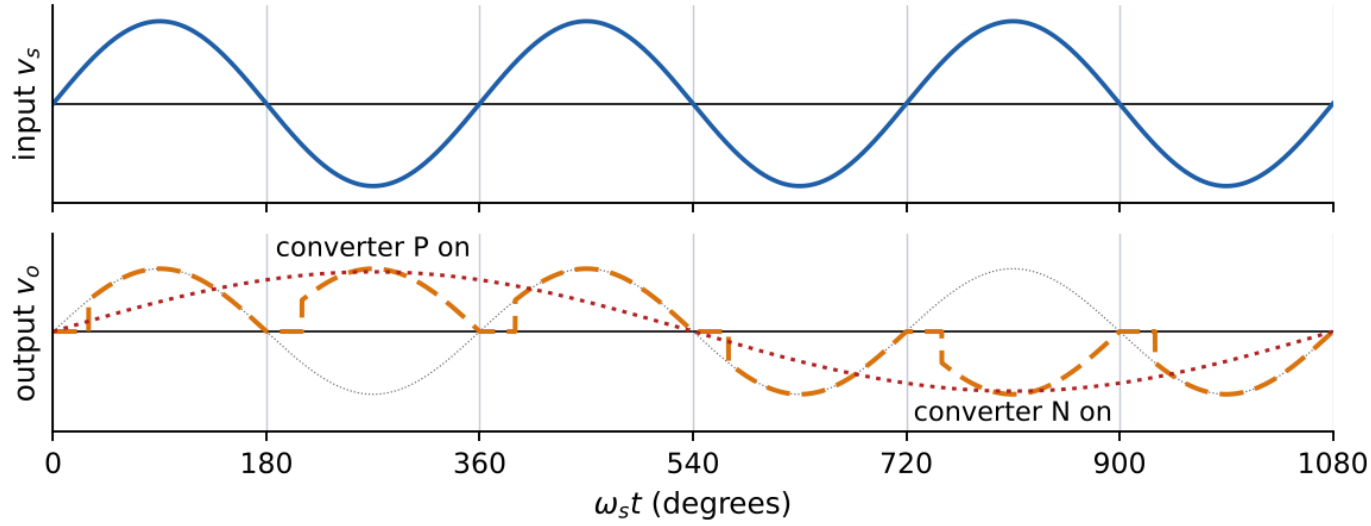


Figure 41: Input (solid blue curve) and output (dashed orange curve) for an output of one third of the source frequency. The dotted red curve is the output's fundamental

Reading the figure. Converter P runs for three half cycles of the source, then converter N runs for three. So the output has one third of the source frequency. A real unit changes the delay angle along the way, so the pieces follow the dotted red sine more closely.

The thyristors turn off by the source's own voltage reversal (natural, or line, commutation), so no extra turn-off circuit is needed. This is also why the output frequency must stay well below the source's: about one third or less.

2. Answer

Cycloconverter: a direct ac-to-ac frequency changer. It converts ac power at one frequency into ac power at another, usually lower, frequency without an intermediate dc link. Two controlled converters (positive and negative) in anti-parallel are line commutated and take turns in feeding the load, with their delay angles varied to shape the output.

Applications:

- low-speed, high-power ac motor drives: rolling mills, ball mills, cement kilns, mine hoists
- ship propulsion
- variable-speed constant-frequency (VSCF) power in aircraft
- large synchronous motor starting and traction

6.(c) Explain the structure of TRIAC with diagram. Mention the constructional feature of a Power BJT. Marks: 8

1. Theory

1. Why a TRIAC? (Core Idea)

A thyristor conducts only one way. To control ac we would need two thyristors in anti-parallel. A TRIAC ("triode for ac") puts both into one chip with one gate. It can conduct both ways.

2. How can one chip do that?

A thyristor needs p-n-p-n from anode to cathode. A TRIAC has two such paths side by side, in opposite directions, and shares the middle layers:

- with main terminal 2 (MT2) positive: p2, n1, p1, n2 on the right side of the figure, conducting down to up;

- with MT1 positive: p1, n1, p2, n3 on the left side, conducting up to down.

The extra small n4 region beside the gate allows the gate to fire with either polarity.

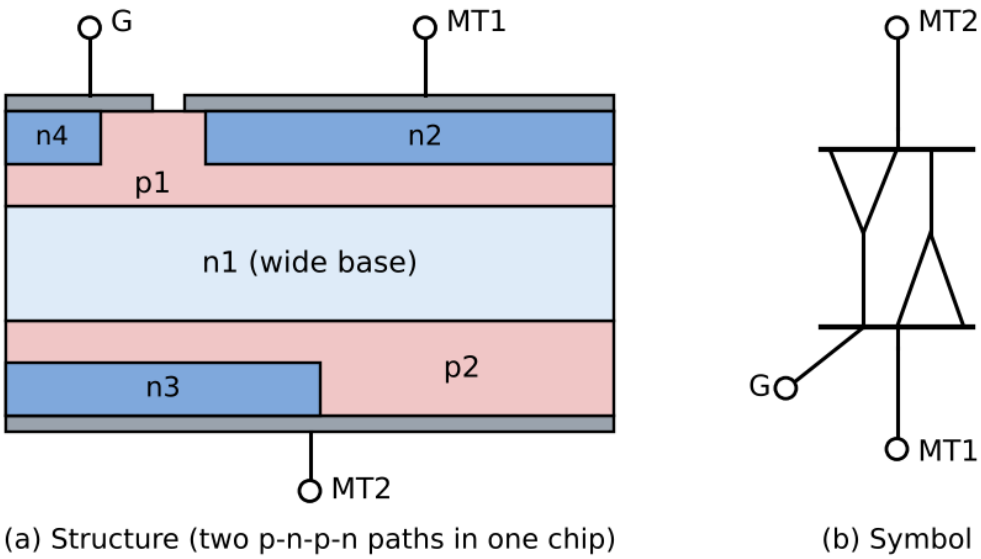


Figure 42: TRIAC: (a) layers, with the gate and MT1 contacts both touching p1, (b) symbol, two opposite thyristors

Reading the figure. The two n regions in the outer p layers sit at opposite ends, so the left side of the chip forms one p-n-p-n and the right side forms the opposite one. MT1 and MT2 are named "main terminals", since neither is always the anode.

3. How is a power BJT different from a small transistor?

A small transistor has a thin structure and a low voltage rating. To block high voltage, the collector needs a thick, lightly doped layer, in which the depletion region can spread. So a power BJT is built **vertically**, with current flowing through the thickness of the chip.

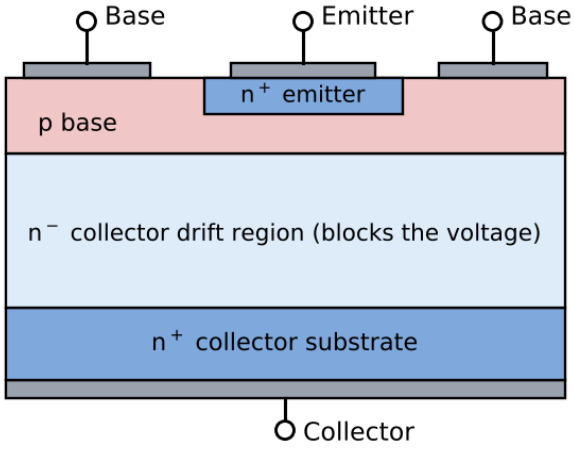


Figure 43: Vertical structure of a power BJT (n-p-n)

Reading the figure. Current flows from the collector at the bottom up to the emitter at the top. The n^- layer is thick and sets the voltage rating.

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2. Answer

TRIAC structure: a five-layer, three-terminal device, equal to two thyristors in anti-parallel on one chip with a common gate (figure above). The terminals are MT1, MT2 and the gate G. The layers are p1, n1, p2 with n regions diffused into the outer p layers: n2 under MT1, n3 under MT2 on the opposite side, and n4 under the gate. MT1 connects to both n2 and p1, the gate to both n4 and p1, and MT2 to both p2 and n3. One path p2-n1-p1-n2 conducts when MT2 is positive, and the other path p1-n1-p2-n3 conducts when MT1 is positive. It can be fired by a gate pulse of either polarity.

Constructional features of a power BJT:

- **Vertical structure:** n^+-p-n^--n^+, with the collector at the bottom of the chip and a large area for current.
- **Thick, lightly doped n^- drift region:** it blocks the high voltage. A thicker layer gives a higher voltage, but a lower gain and a slower device.
- **Interdigitated emitter:** emitter and base are narrow, comb-like fingers, so the current spreads evenly and does not crowd at the emitter edges.
- **Low current gain:** typically 5 to 20, so a Darlington pair is often used.
- **Heavily doped n^+ substrate:** it carries the current to the collector contact with little resistance.

6.(d) Compare the following device in terms of power capacity and switching speed: BJT, MOSFET, GTO, IGBT, MCT Marks: 5

1. Theory

1. What limits a power switch? (Core Idea)

We want a switch that blocks high voltage, carries high current and switches fast. These pull against each other. A device that uses both holes and electrons (bipolar) conducts well, because its body fills with carriers. But those carriers take time to remove, so it is slow. A device that uses one carrier (the MOSFET) is fast, but it has a high on-resistance at high voltage.

- **GTO:** a thyristor that the gate can also turn off. Highest power, slowest.
- **BJT:** current-driven bipolar transistor.
- **MOSFET:** voltage-driven, one carrier type. Fastest, lowest power.
- **IGBT:** MOSFET gate with bipolar conduction. The usual middle choice.
- **MCT:** MOS-controlled thyristor: a thyristor with MOS gates for on and off.

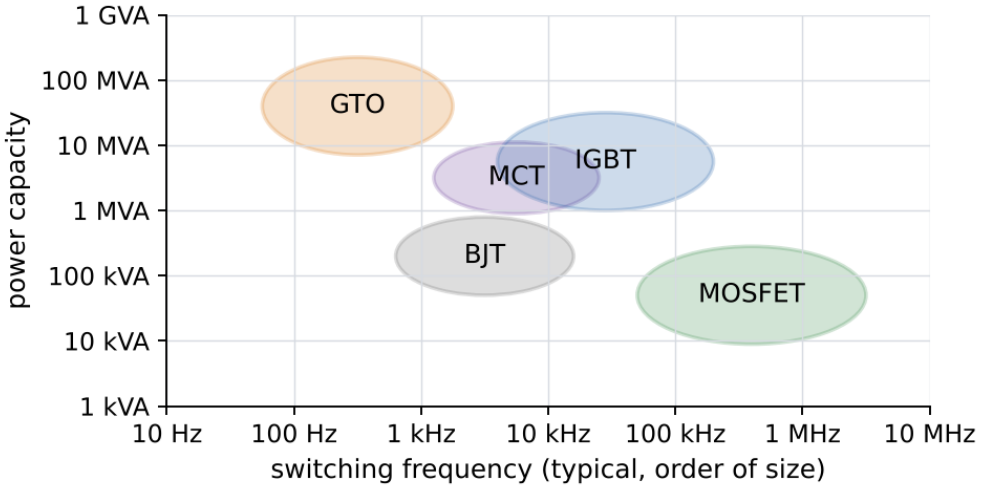


Figure 44: Rough map of power capacity against switching frequency (typical orders of size, not limits)

Reading the figure. Moving right, the devices get faster, and moving up they carry more power. No device is at the top right.

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2. Answer

Device	Type	Power capacity	Switching speed
GTO	bipolar, current gate	highest (MVA range)	slow (about 1 kHz)
MCT	MOS-gated thyristor	high	medium (10 to 20 kHz)
IGBT	MOS gate, bipolar conduction	medium to high	medium to fast (up to about 50 kHz)
BJT	bipolar, base current	medium	medium (up to about 10 kHz)
MOSFET	unipolar, voltage gate	lowest	fastest (above 100 kHz)

Table 6: Power devices compared (typical figures)

Order: power capacity, GTO > MCT > IGBT > BJT > MOSFET; switching speed, MOSFET > IGBT > MCT > BJT > GTO.

3. Final Concise Answer

(a) Full-bridge square-wave inverter with a series LC filter tuned to 1 kHz: $V_{dc} = 55.5$ V, $L = 5$ mH, $C = 5.07$ μF, THD $= 4.26$ %. $V_{C,max} \approx 223$ V.

(b) A direct ac-to-ac frequency changer with no dc link. Used in large low-speed motor drives, ship propulsion, VSCF aircraft power and traction.

(c) TRIAC: two anti-parallel p-n-p-n paths in one chip with one gate, terminals MT1, MT2, G. Power BJT: vertical n^+-p-n^--n^+ structure, thick drift region, interdigitated emitter, low gain.

(d) Power: GTO > MCT > IGBT > BJT > MOSFET. Speed: MOSFET > IGBT > MCT > BJT > GTO.

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7. (a) Explain the reverse recovery time of diode with diagram. Marks: 4
   (b) Draw the I-V curve of thyristor. Define Holding current and Latching current for thyristor Marks: 5
   (c) With proper alignment, show the waveshape of output voltage, output current, diode currents and source current for a full wave bridge rectifier with sinusoidal source and R-L load. Marks: 5
   (d) A sinusoidal voltage source of $v(t) = 240\sqrt{2}\sin(2\pi 60t)$ V is applied to a nonlinear load, resulting in a nonsinusoidal current which is expressed in Fourier series form as $i(t) = 8\sin(2\pi 60t) + 4\sin(4\pi 60t)$ A. Determine (a) the power absorbed by the load, (b) the power factor of the load, (c) the distortion factor of the load current, (d) the total harmonic distortion of the load current, (e) crest factor of the load current. Marks: 6

7.(a) Explain the reverse recovery time of diode with diagram. Marks: 4

1. Theory

1. What is a diode? (Core Idea)

A diode is a one-way valve for current. It is made by joining p-type silicon (with mobile holes) to n-type silicon (with mobile electrons). Current flows from the p side (anode) to the n side (cathode), and is blocked the other way. In a simple picture it changes from conducting to blocking at once. A real diode needs a short time.

2. Why does it need time?

While the diode conducts, holes cross into the n side and electrons cross into the p side. They do not vanish at once, so a cloud of extra carriers builds up around the junction.

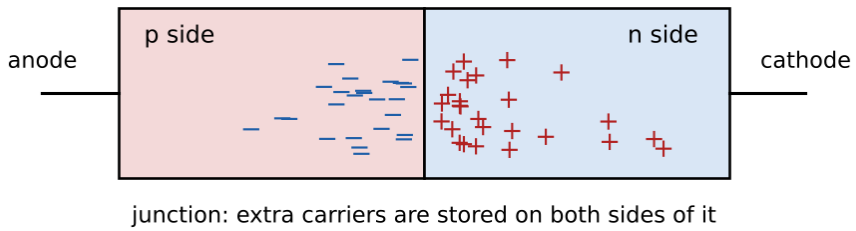


Figure 45: Stored carriers in a conducting diode: electrons (blue minus signs) on the p side and holes (red plus signs) on the n side

Reading the figure. The extra carriers are crowded near the junction and thin out with distance. This is the stored charge.

A diode can block only when the region at the junction is empty of carriers. So when the outside circuit tries to reverse the current, those carriers first flow back out. Until they have gone, the diode conducts **backwards**.

3. What does this look like in time?

The forward current falls at the rate di/dt set by the circuit, and it passes through zero. Current then flows in reverse and reaches a peak I_{RR}. By then the junction region is clear, and the diode begins to block, so the reverse current falls back to zero. The two stages are t_a and t_b.

2. Answer

When a conducting diode is switched off, the stored charge must be removed before it can block, so a reverse current flows for a short time.

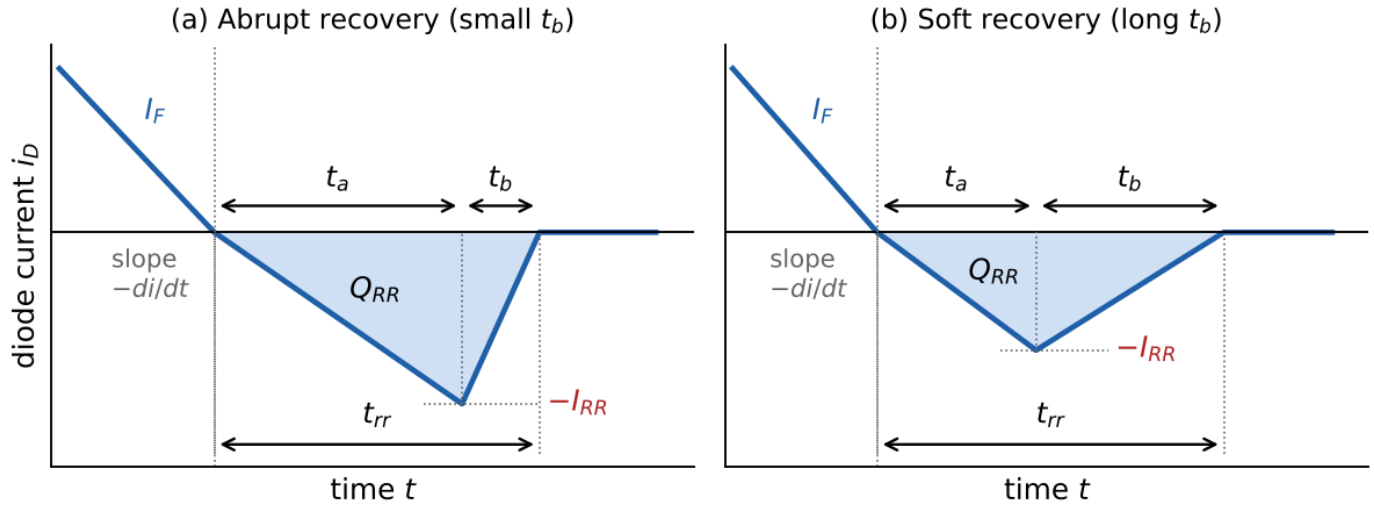


Figure 46: Reverse recovery current of a diode: (a) abrupt recovery, (b) soft recovery

Reverse recovery time t_{rr} is the time from the current's zero crossing until the reverse current has decayed to about zero:

$$t_{rr} = t_a + t_b$$

where t_a is the time to remove the charge from the junction region (the current reaches its peak I_{RR} at its end), and t_b is the time to remove the charge from the bulk. t_{rr} limits the switching frequency.

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7.(b) Draw the I-V curve of thyristor. Define Holding current and Latching current for thyristor Marks: 5

1. Theory

1. What are the three states? (Core Idea)

A thyristor, or SCR, is a four-layer p-n-p-n switch with three terminals: anode, cathode and gate. It conducts from anode to cathode, but only after it is fired by a gate pulse or a very high voltage. Once on, it stays on by itself: two parts of its structure feed each other (the two-transistor model of 2(a)).

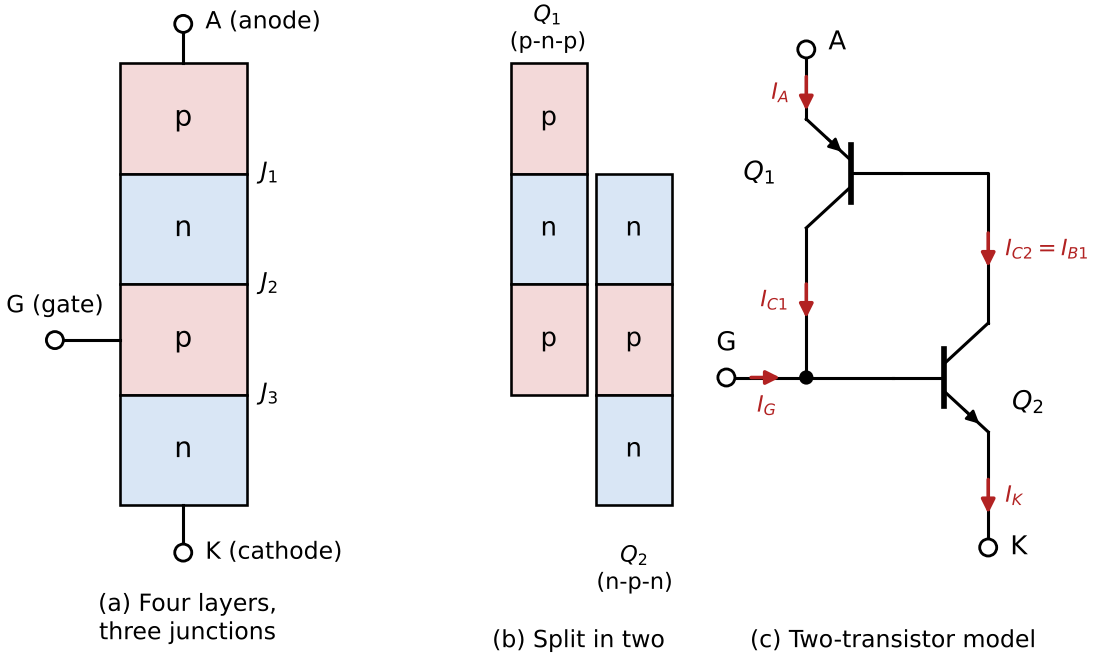


Figure 47: The two-transistor view of the thyristor

Reading the figure. In (c), each transistor feeds the base of the other. This loop is what keeps the device on, and it needs a minimum current to keep going.

A thyristor has three states:

- **Reverse blocking:** the anode is negative. Only a small leakage flows.
- **Forward blocking:** the anode is positive, but the device has not been fired. Only a small leakage flows.
- **Conducting:** after firing, a large current flows with a small drop of about 1 to 2 V.

2. Why are there two small currents?

Regeneration in the model needs enough current to keep $\alpha_1 + \alpha_2$ near 1. Starting the loop needs a bit more current than keeping it going. So there are two thresholds. The one for starting is the latching current, and the one for keeping is the holding current. The first is larger, about two to three times.

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2. Answer

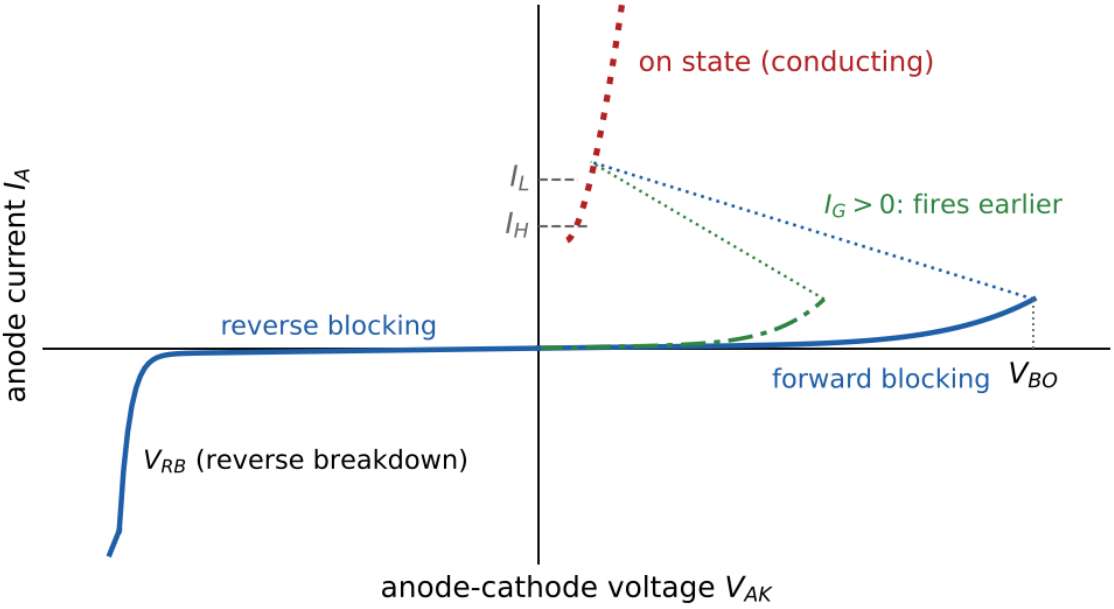


Figure 48: I-V curve of a thyristor

Reading the figure. With zero gate current, the forward blocking curve (solid blue) rises until the breakover voltage V_{BO}, and the device then jumps (thin dotted line) to the on state (dotted red). A gate current fires it at a lower voltage. Below the holding level, the device falls back to blocking. On the left is the reverse blocking curve, ending in reverse breakdown.

Latching current (I_L): the minimum anode current that must flow, just after firing, for the thyristor to stay on when the gate signal is removed.

Holding current (I_H): the minimum anode current below which a conducting thyristor turns off.

$I_L > I_H$. The first applies at turn-on and the second during conduction.

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7.(c) With proper alignment, show the waveshape of output voltage, output current, diode currents and source current for a full wave bridge rectifier with sinusoidal source and R-L load. Marks: 5

1. Theory

1. How does the bridge work? (Core Idea)

A diode conducts one way only. An ac source reverses every half cycle, and a load that wants dc needs current in one direction. Four diodes in a bridge steer the source to the load so that this happens. In the positive half cycle, D_1 and D_2 conduct. In the negative half cycle, D_3 and D_4 conduct. So the load always sees $|v_s|$.

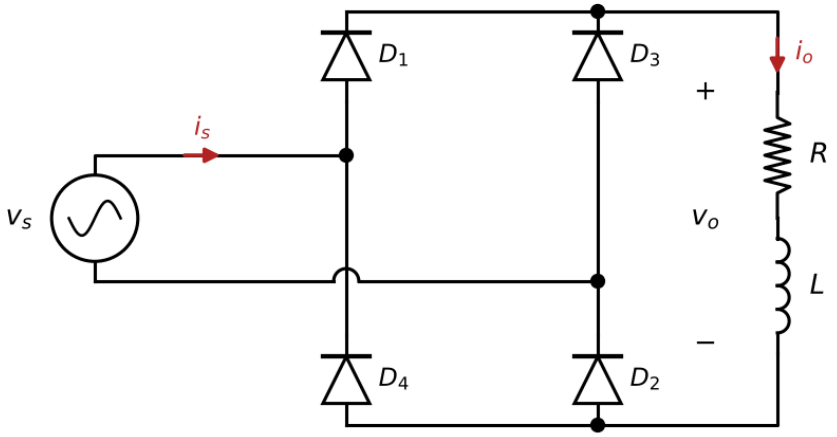


Figure 49: Diode bridge with an RL load. The lower source wire hops over the left leg

2. What does the inductor do?

The output voltage does not depend on the load here, since a pair of diodes conducts at all times. Its average is $V_o = 2V_m/\pi$. The inductor smooths the current. Its reactance rises with frequency, so the ac ripple (at 2ω and above) is much reduced, and the dc part is unchanged: $I_o = V_o/R$. The diodes then carry almost square pulses, and the source current is nearly a square wave.

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2. Answer

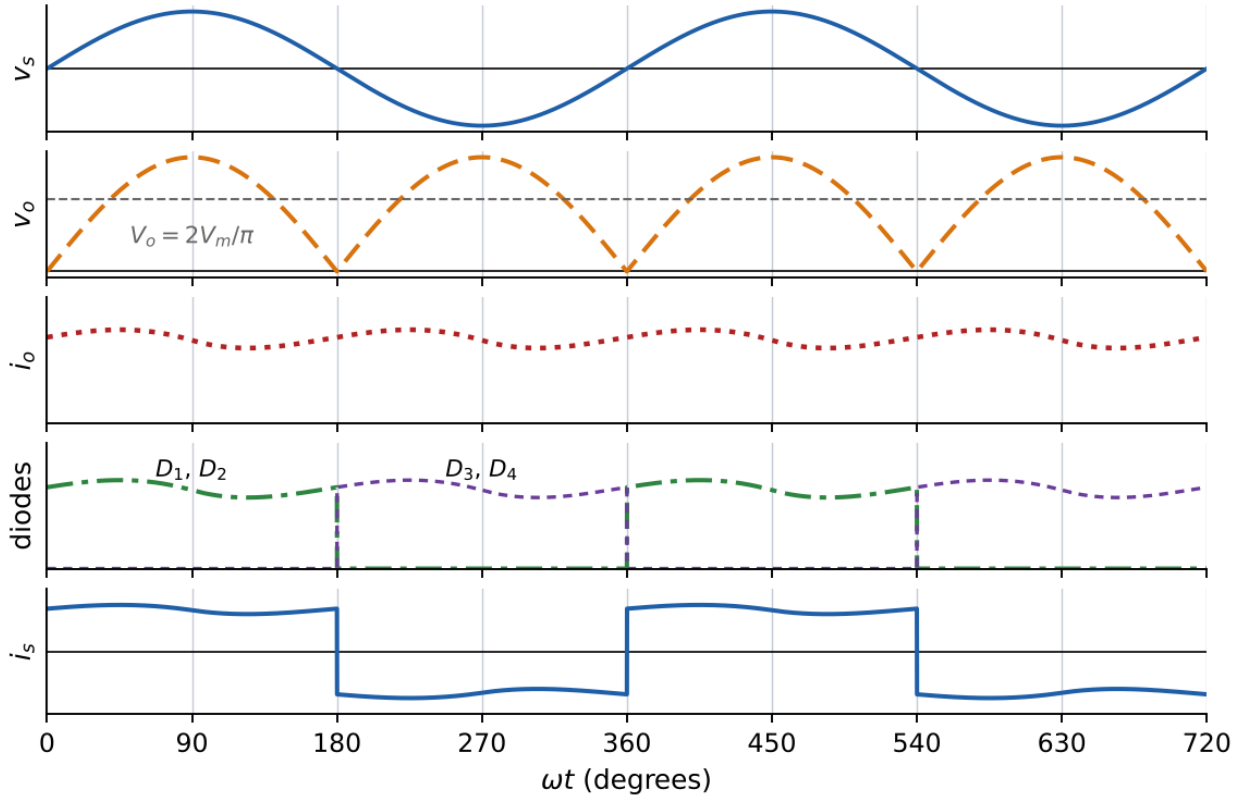


Figure 50: Aligned waveforms for a bridge with an RL load (sketch with ωL/R = 3)

Reading the figure.

- $v_o = |v_s|$, a full-wave rectified sine with average $2V_m/\pi$.
- i_o is a steady current with a small ripple at twice the source frequency.
- Each diode pair carries i_o for one half cycle: D_1, D_2 when $v_s > 0$ (dash-dot green) and D_3, D_4 when $v_s < 0$ (dotted purple).
- The source current is $+i_o$ and then $-i_o$, a nearly square wave.

7.(d) A sinusoidal voltage source of $v(t) = 240\sqrt{2}\sin(2\pi 60t)$ V is applied to a nonlinear load, resulting in a nonsinusoidal current which is expressed in Fourier series form as $i(t) = 8\sin(2\pi 60t) + 4\sin(4\pi 60t)$ A. Determine (a) the power absorbed by the load, (b) the power factor of the load, (c) the distortion factor of the load current, (d) the total harmonic distortion of the load current, (e) crest factor of the load current. Marks: 6

1. Theory

1. Which currents carry power? (Core Idea)

A **nonlinear load** draws a current that is not a sine, even from a sine voltage. Such a current is a fundamental plus harmonics, as the question's Fourier series shows: an 8 A peak term at 60 Hz and a 4 A peak term at 120 Hz.

The rms value of a sine is its peak over $\sqrt{2}$, and the rms values of different harmonics add as squares:

$$I_{rms}^2 = I_1^2 + I_2^2$$

Four numbers describe such a wave. **Power factor** is real power over apparent power, P/S. **Distortion factor** is I_1/I_{rms}. **THD** is the harmonics' rms value over I_1. **Crest factor** is the peak over the rms value.

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Average power comes only from a voltage and a current **of the same frequency**. A 60 Hz voltage times a 120 Hz current averages to zero. So here only the fundamental current carries power (see also 1(a)). Its phase angle with the voltage is zero here.

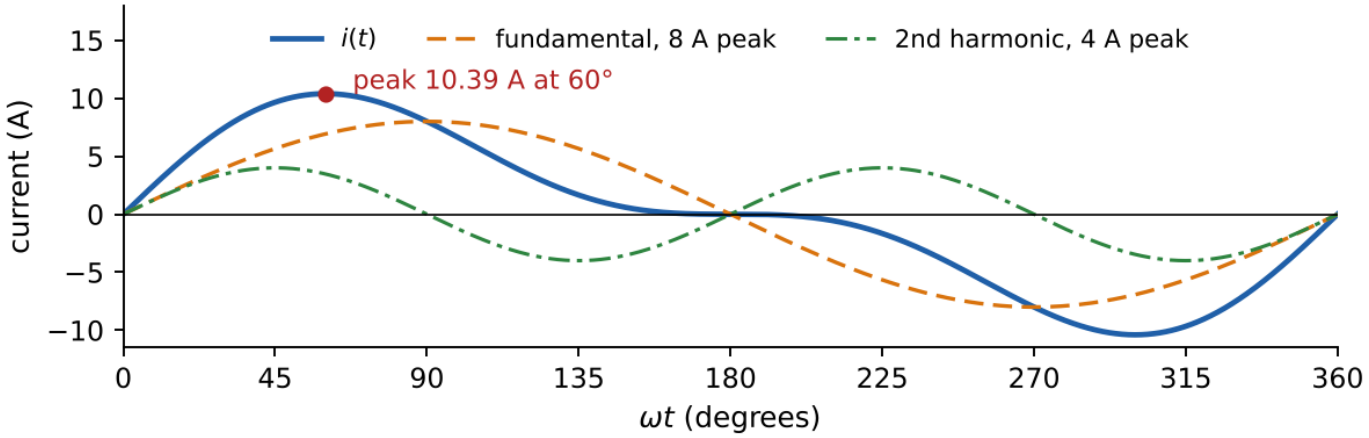


Figure 51: The current (solid blue) with its fundamental (dashed orange) and second harmonic (dash-dot green)

Reading the figure. The two parts add to a tall peak in the first quarter, near 60°. The peak of i is found by setting $di/d\omega t = 0$: $8\cos x + 8\cos 2x = 0$, which gives $x = 60°$.

2. Answer

Given: $V = 240$ V rms; $I_1 = 8/\sqrt{2} = 5.66$ A rms, $I_2 = 4/\sqrt{2} = 2.83$ A rms, in phase with the voltage's zero crossing.

$$I_{rms} = \sqrt{5.66^2 + 2.83^2} = 6.32 \text{ A}$$

(a) Power: only the fundamental counts, with $\theta_1 = 0$:

$$P = VI_1 \cos\theta_1 = 240(5.66) = 1358 \text{ W}$$

$$\boxed{P = 1358 \text{ W}}$$

(b) Power factor:

$$S = VI_{rms} = 240(6.32) = 1518 \text{ VA}, \qquad pf = \frac{1358}{1518} = 0.894$$

$$\boxed{pf = 0.894}$$

(c) Distortion factor:

$$DF = \frac{I_1}{I_{rms}} = \frac{5.66}{6.32} = 0.894$$

$$\boxed{DF = 0.894}$$

(d) THD:

$$THD = \frac{I_2}{I_1} = \frac{2.83}{5.66} = 0.50$$

$$\boxed{THD = 50\ \%}$$

(e) Crest factor: the peak is at $\omega t = 60°$:

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$$I_{peak} = 8\sin 60° + 4\sin 120° = 10.39 \text{ A}, \qquad CF = \frac{I_{peak}}{I_{rms}} = \frac{10.39}{6.32} = 1.64$$

$$\boxed{CF = 1.64}$$

3. Final Concise Answer

(a) $t_{rr} = t_a + t_b$: the time for the reverse current to die after the zero crossing.
(b) I_L: the minimum current to stay on just after firing. I_H: the minimum current to remain on. $I_L > I_H$.
(c) $v_o = |v_s|$; i_o is nearly level; each diode pair conducts for one half cycle; i_s is nearly square.
(d) $P = 1358$ W, $pf = 0.894$, $DF = 0.894$, $THD = 50$ %, $CF = 1.64$.

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8. (a) Draw the circuit diagram, equivalent circuit for switch on and off condition and the waveforms of the inductor current, diode current and load voltage for a typical boost converter. Marks: 5

(b) A buck converter has parameters, Vs=50 V, D=0.4, R=20 Ω, L=400 μH and C=100 μF. The switching frequency is 20 kHz. a) Determine the output voltage, b) Determine the average, minimum and maximum inductor current c) The output ripple voltage. Marks: 7.5

(c) Draw the circuit diagram to control the speed of a separately excited dc motor by chopping field current supply. Explain the operation using required equations. Marks: 2.5

(d) A three phase rectifier is supplied by 480 V rms line to line 60 Hz source. The RL load is 100 Ω resistor in series with 15 mH inductor. Determine the rms load current and rms diode currents. Marks: 5

8.(a) Draw the circuit diagram, equivalent circuit for switch on and off condition and the waveforms of the inductor current, diode current and load voltage for a typical boost converter. Marks: 5

1. Theory

1. What is a boost converter? (Core Idea)

A boost converter takes a dc voltage and gives a **higher** dc voltage. It uses four parts: an inductor L, a switch S that opens and closes very fast, a diode D and a capacitor C. The fraction of each period T for which the switch is closed is the duty ratio D.

2. What does the inductor do?

An inductor stores energy in its magnetic field, and its current cannot jump. Its voltage sets how fast its current changes:

$$v_L = L\frac{di_L}{dt}$$

A positive voltage makes the current rise in a straight line, and a negative voltage makes it fall.

3. What happens in each interval?

Switch closed. The inductor is connected straight across the source, so $v_L = V_s$ and its current rises. It is storing energy. The diode is reverse biased, because the output voltage is higher than the switch's zero volts. The capacitor alone feeds the load, so the output falls a little.

Switch open. The inductor's current cannot stop, so it pushes on through the diode into the capacitor and the load. The inductor's voltage now adds to the source, so the output can be above V_s. Here $v_L = V_s - V_o$, which is negative, so the current falls.

4. Why is the output higher?

In steady state the current ends each period where it began. So the average inductor voltage is zero:

$$V_s DT + (V_s - V_o)(1 - D)T = 0 \quad \Rightarrow \quad V_o = \frac{V_s}{1 - D}$$

D is below 1, so V_o is above V_s. 4(c) designs a boost converter with numbers.

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2. Answer

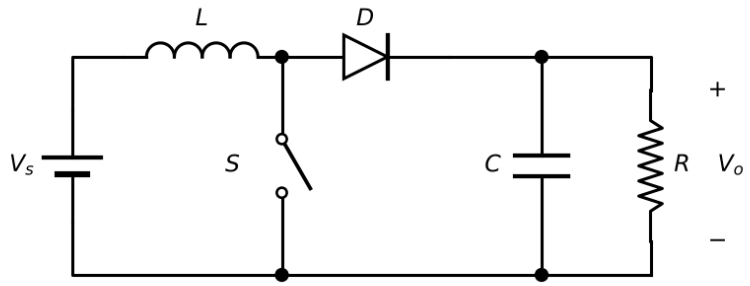


Figure 52: Boost converter circuit

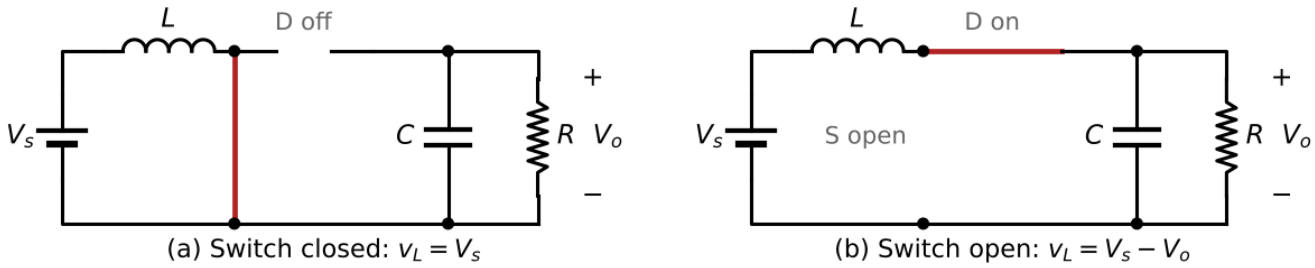


Figure 53: Equivalent circuits of the boost converter: (a) switch closed, (b) switch open. The thick line is the conducting path

Reading the figure. With the switch closed, the diode is reverse biased and the inductor stores energy from the source, while C alone feeds the load. With the switch open, the inductor and source together feed the output through the diode.

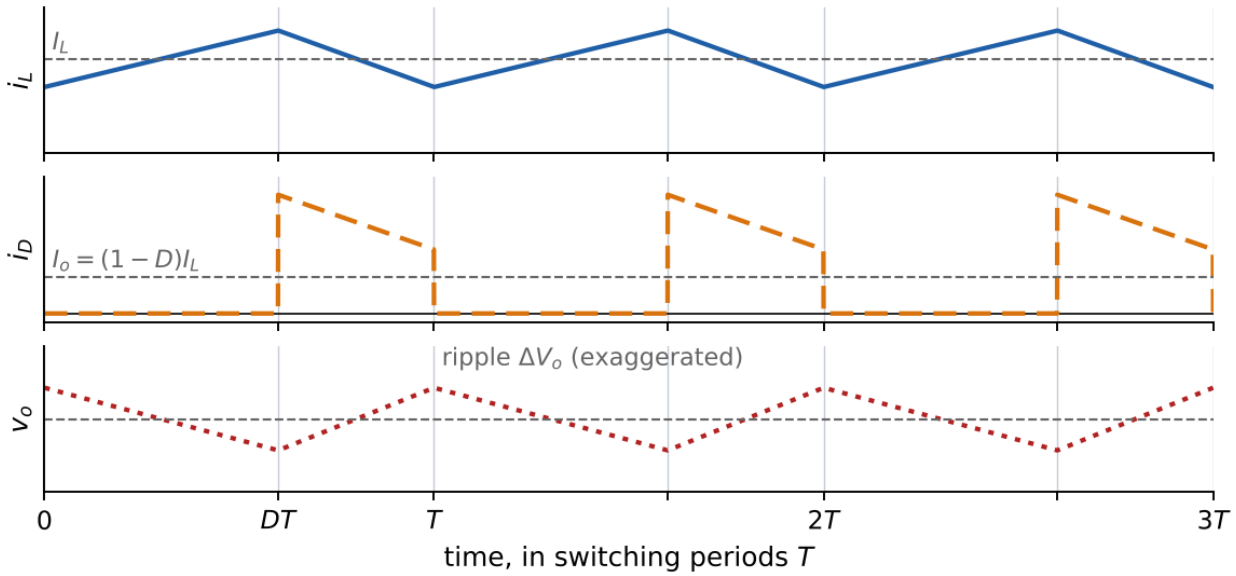


Figure 54: Boost converter waveforms: inductor current (solid blue), diode current (dashed orange) and load voltage (dotted red, ripple exaggerated)

Reading the figure. i_L rises during DT with slope V_s/L and falls during $(1-D)T$ with slope $(V_s - V_o)/L$. The diode carries i_L only in the off time, so its average is the output current, $(1-D)I_L$. The output falls a little when C alone feeds the load, and rises when the diode conducts.

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$$V_o = \frac{V_s}{1-D}$$

8.(b) A buck converter has parameters, Vs=50 V, D=0.4, R=20 Ω, L=400 μH and C=100 μF. The switching frequency is 20 kHz. a) Determine the output voltage, b) Determine the average, minimum and maximum inductor current c) The output ripple voltage. Marks: 7.5

1. Theory

1. What is a buck converter? (Core Idea)

A buck converter gives a dc voltage **lower** than its input. A switch chops the input, and an inductor and capacitor smooth the pulses. If the switch is closed for a fraction D of each period, the average output is

$$V_o = DV_s$$

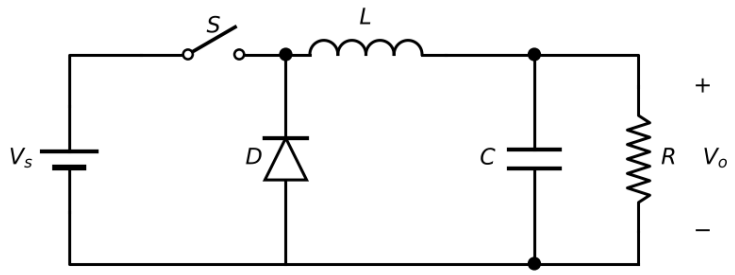


Figure 55: Buck converter

Reading the figure. With the switch closed, the source drives L, and its current rises. With the switch open, the diode carries the inductor's current, and it falls.

2. What sets the inductor current?

The capacitor has no average current in steady state, so the inductor's average current is the load current, $I_L = V_o/R$. With the switch open, the inductor has $-V_o$ across it for a time $(1-D)T$, so its current falls by

$$\Delta i_L = \frac{V_o(1-D)T}{L} = \frac{V_o(1-D)}{Lf}$$

The current swings equally above and below I_L, so $I_{max,min} = I_L \pm \Delta i_L/2$. The derivation is in more detail in 4(b).

3. Where does the output ripple come from?

The output ripple comes from the capacitor. The ripple part of i_L is a triangle, and it flows through C. During half a period it is positive, a triangle of height $\Delta i_L/2$ and base $T/2$. Its charge is $\Delta Q = \frac{1}{2}\cdot\frac{T}{2}\cdot\frac{\Delta i_L}{2} = T\Delta i_L/8$. Since $\Delta V_o = \Delta Q/C$,

$$\Delta V_o = \frac{\Delta i_L}{8Cf}, \qquad \frac{\Delta V_o}{V_o} = \frac{1-D}{8LCf^2}$$

2. Answer

Given: $V_s = 50$ V, $D = 0.4$, $R = 20\ \Omega$, $L = 400\ \mu$H, $C = 100\ \mu$F, $f = 20$ kHz.

a) Output voltage

$$V_o = DV_s = 0.4(50) = 20 \text{ V}$$

$$\boxed{V_o = 20 \text{ V}}$$

b) Inductor current

$$I_L = \frac{V_o}{R} = \frac{20}{20} = 1 \text{ A}$$

$$\Delta i_L = \frac{V_o(1-D)}{Lf} = \frac{20(0.6)}{(400\times10^{-6})(20\times10^{3})} = 1.5 \text{ A}$$

$$I_{max} = 1 + 0.75 = 1.75 \text{ A}, \qquad I_{min} = 1 - 0.75 = 0.25 \text{ A}$$

$$\boxed{I_L = 1 \text{ A}, \quad I_{max} = 1.75 \text{ A}, \quad I_{min} = 0.25 \text{ A}}$$

$I_{min} > 0$, so the current is continuous, as the formulas assume.

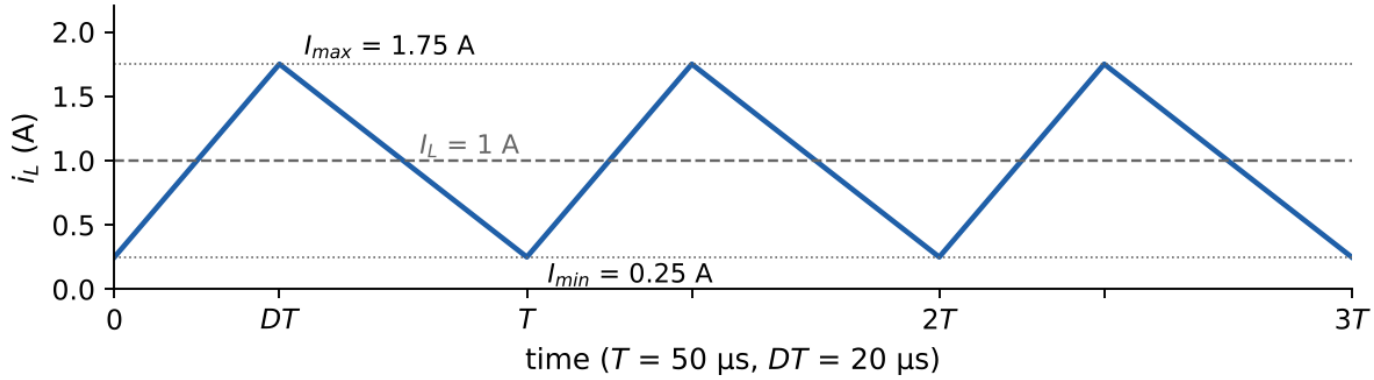


Figure 56: Inductor current of this converter

c) Output ripple voltage

$$\frac{\Delta V_o}{V_o} = \frac{1-D}{8LCf^2} = \frac{0.6}{8(400\times10^{-6})(100\times10^{-6})(20\times10^{3})^2} = 0.00469$$

$$\Delta V_o = 0.00469(20) = 0.094 \text{ V}$$

$$\boxed{\Delta V_o = 0.094 \text{ V} = 0.47\ \%}$$

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8.(c) Draw the circuit diagram to control the speed of a separately excited dc motor by chopping field current supply. Explain the operation using required equations. Marks: 2.5

1. Theory

1. What sets the speed of a dc motor? (Core Idea)

A dc motor turns electrical power into rotation. A separately excited motor has two circuits: the armature, which carries the main current, and the field, which makes the magnetic flux. When the armature turns in the flux, it generates a back voltage E_b proportional to speed and flux. In steady state, this nearly balances the supply. So for a fixed armature voltage, **less flux means more speed**, since the motor must spin faster to generate the same E_b.

2. What does a chopper do?

A chopper is a fast electronic switch on a dc supply. If it is on for a fraction δ of each period, the average voltage it passes is δV_s. The field winding is an inductor, so its current stays steady at the average. This changes the flux with almost no power loss, unlike a series resistor.

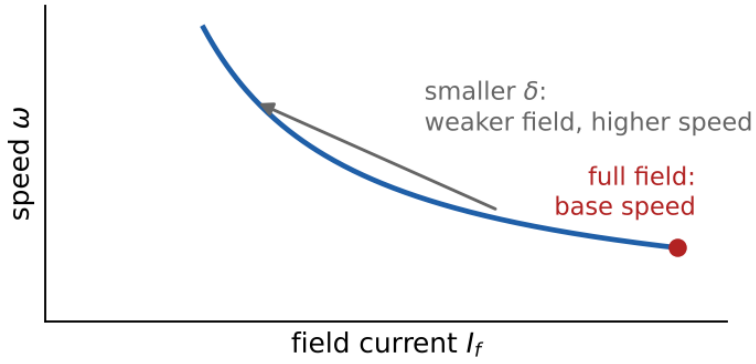


Figure 57: Speed against field current at fixed armature voltage

Reading the figure. The speed rises as the field current falls. The red dot is full field, and moving to the left is field weakening.

2. Answer

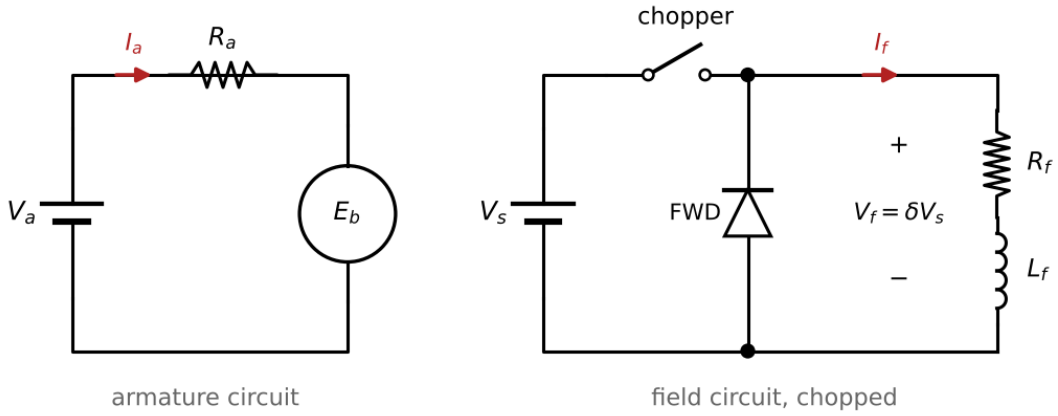


Figure 58: Field chopper control of a separately excited dc motor

Operation: the chopper (switch) connects the field winding to V_s for a fraction δ of each period. The diode carries the field current while the chopper is off. So the average field voltage and current are

$$V_f = \delta V_s, \qquad I_f = \frac{\delta V_s}{R_f}$$

The armature equations are

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$$V_a = E_b + I_a R_a, \qquad E_b = K_v I_f \omega$$

$$\boxed{\omega = \frac{V_a - I_a R_a}{K_v I_f} = \frac{(V_a - I_a R_a) R_f}{K_v \, \delta \, V_s}}$$

So a smaller δ lowers I_f and raises the speed. This gives control above base speed, and the torque $T = K_v I_f I_a$ falls as the field weakens (constant-power region).

8.(d) A three phase rectifier is supplied by 480 V rms line to line 60 Hz source. The RL load is 100 Ω resistor in series with 15 mH inductor. Determine the rms load current and rms diode currents. Marks: 5

1. Theory

1. What is the plan? (Core Idea)

A three-phase rectifier has six diodes. At each moment, the upper diode with the highest voltage and the lower diode with the lowest voltage conduct, so the load sees the largest line-to-line voltage. The pair changes every 60°.

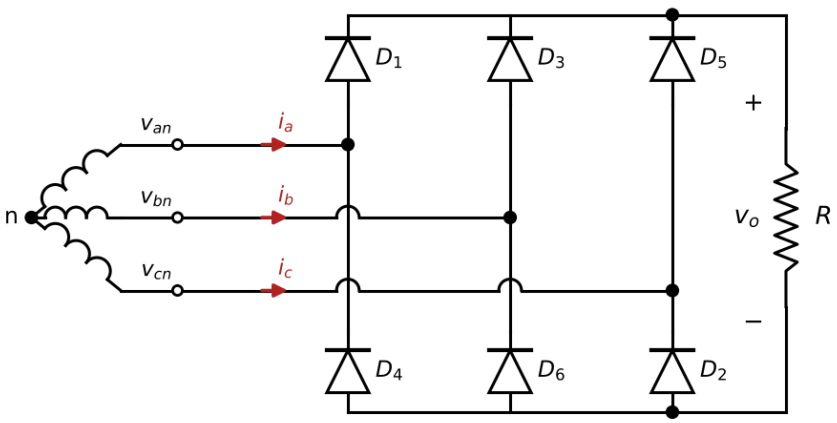


Figure 59: Three-phase diode bridge

Reading the figure. Each line joins the middle of one leg. Here the load is an R and L in series in place of the R shown.

The output repeats six times per period, so it is a dc voltage plus harmonics at $n = 6, 12, 18, \ldots$ With V_m as the peak line-to-line voltage:

$$V_o = \frac{3V_m}{\pi}, \qquad V_n = \frac{2V_o}{n^2 - 1}$$

(This is the formula of 3(d) at $\alpha = 0$.)

The inductor passes dc freely but opposes the harmonics, with reactance $n\omega L$. So find the current at each frequency, then add their squares: rms values of different frequencies add this way.

2. How does each diode share the load current?

Each diode carries the load current for 120° of 360°, that is, one third of the time. The rms value is the root of the mean of the square, so

$$I_{D,rms} = \sqrt{\frac{1}{3}}\, I_{rms}$$

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2. Answer

Given: $V_{L\text{-}L} = 480$ V rms, $V_m = 678.8$ V, $f = 60$ Hz, $R = 100\ \Omega$, $L = 15$ mH.

Step 1: DC part

$$V_o = \frac{3(678.8)}{\pi} = 648.2 \text{ V}, \qquad I_o = \frac{648.2}{100} = 6.48 \text{ A}$$

Step 2: Harmonics

n	V_n (V)	Z_n (Ω)	I_n peak (A)
6	37.0	105.6	0.351
12	9.07	120.9	0.075

Table 7: Output harmonics of the rectifier

The 18th harmonic is 0.028 A, which is negligible.

Step 3: RMS load current

$$I_{rms} = \sqrt{I_o^2 + \left(\frac{I_6}{\sqrt{2}}\right)^2 + \left(\frac{I_{12}}{\sqrt{2}}\right)^2} = \sqrt{6.48^2 + 0.248^2 + 0.053^2} = 6.49 \text{ A}$$

$$\boxed{I_{rms} = 6.49 \text{ A}}$$

Step 4: RMS diode current

$$I_{D,rms} = \frac{I_{rms}}{\sqrt{3}} = \frac{6.49}{\sqrt{3}} = 3.75 \text{ A}$$

$$\boxed{I_{D,rms} = 3.75 \text{ A}}$$

Final Answer: rms load current 6.49 A, rms diode current 3.75 A.

3. Final Concise Answer

(a) See the circuit, the two equivalent circuits and the waveforms above; $V_o = V_s/(1-D)$.
(b) $V_o = 20$ V; $I_L = 1$ A, $I_{max} = 1.75$ A, $I_{min} = 0.25$ A; $\Delta V_o = 0.094$ V.
(c) A chopper sets $I_f = \delta V_s/R_f$, and $\omega = (V_a - I_a R_a)/(K_v I_f)$, so a lower δ gives a higher speed.
(d) $I_{rms} = 6.49$ A, $I_{D,rms} = 3.75$ A.

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FORMULA SHEET

EEE339

Power Electronics

Formula Sheet

Session 2020–21 paper

Every formula behind the Final Examination 2023 and Term Tests 1–2

TOPICS

1 Distortion and power quality — 8 formulas

2 Power devices — 6 formulas

3 Single-phase rectifiers — 10 formulas

4 Three-phase rectifiers — 8 formulas

5 Dc-dc converters — 14 formulas

6 Inverters — 10 formulas

7 Dc motor by field chopper — 3 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

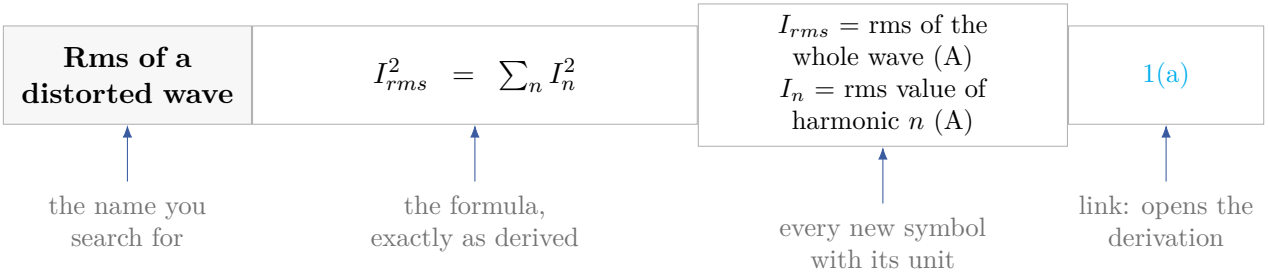
Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Distortion and power quality 4

2 Power devices 5

3 Single-phase rectifiers 5

4 Three-phase rectifiers 6

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7 Dc motor by field chopper 9

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Conventions (Hart; Rashid for devices): V_m is the peak source voltage, and for three-phase circuits the peak line-to-line voltage; V_n and I_n are peak values unless marked rms, except in Section 1, where they are rms; D is the duty ratio and α the delay angle; ideal parts and continuous inductor current.

Columns: Title · Equation · Notation · Ref. (reference).

1. Distortion and power quality:

Title	Equation	Notation	Ref.
Rms of a distorted wave	$I_{rms}^2 = \sum_n I_n^2$	I_{rms} = rms of the whole wave (A) I_n = rms value of harmonic n (A)	1(a)
★ **Distortion factor**	$DF = \dfrac{I_1}{I_{rms}}$	DF = distortion factor I_1 = rms fundamental (A)	1(a)
★ **Total harmonic distortion**	$THD = \dfrac{\sqrt{I_{rms}^2 - I_1^2}}{I_1}$	THD = total harmonic distortion harmonics over fundamental	1(a)
★ **DF and THD**	$DF = \sqrt{\dfrac{1}{1+THD^2}}$		1(a)
Power factor	$pf = \dfrac{P}{S} = DF\cos\theta_1$	pf = power factor P = average power (W) S = apparent power, $V_{rms}I_{rms}$ (VA) θ_1 = angle of the fundamental current	1(a)
One harmonic only	$I_n = THD\, I_1$		1(d)
Crest factor	$CF = \dfrac{I_{peak}}{I_{rms}}$	CF = crest factor I_{peak} = peak of the wave (A)	7(d)
Power with a sine source	$P = V_1 I_1 \cos\theta_1$	V_1 = rms source voltage (V)	7(d)

Table 1: Distortion and power quality

2. Power devices:

Title	Equation	Notation	Ref.
Recovery time	$t_{rr} = t_a + t_b$	t_{rr} = reverse recovery time (s) t_a = time to the reverse peak (s) t_b = fall time after the peak (s)	1(b)
Peak reverse current	$I_{RR} = t_a \dfrac{di}{dt}$	I_{RR} = peak reverse current (A) di/dt = rate of fall of current (A/s)	1(b)
★ **Storage charge**	$Q_{RR} \approx \dfrac{1}{2}\dfrac{di}{dt} t_{rr}^2$	Q_{RR} = stored charge removed during recovery (C) t_b small, $t_{rr} \approx t_a$	1(c)
★ **Peak from the charge**	$I_{RR} = \sqrt{2Q_{RR}\dfrac{di}{dt}}$		1(c)
★ **Two-transistor model**	$I_A = \dfrac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$	I_A = anode current (A) α_1, α_2 = common-base gains of the two transistors I_G = gate current (A) I_{CBO1}, I_{CBO2} = leakage currents of the two transistors (A)	2(a)
Turn-on condition	$\alpha_1 + \alpha_2 \approx 1$	regeneration starts	2(a)

Table 2: Power devices

3. Single-phase rectifiers:

Title	Equation	Notation	Ref.
Impedance and angle	$Z = \sqrt{R^2 + (\omega L)^2},\ \theta = \tan^{-1}\dfrac{\omega L}{R}$	Z = load impedance (Ω) R = load resistance (Ω) L = load inductance (H) ω = angular frequency, $2\pi f$ (rad/s) θ = load angle (rad)	2(b)
★ **RL load current**	$i = \dfrac{V_m}{Z}\sin(\omega t - \theta) + Ae^{-\omega t/\omega\tau}$	i = load current (A) V_m = peak source voltage (V) t = time (s) A = constant set by $i(\alpha) = 0$ (A) $\tau = L/R$ (s) valid for $\alpha \le \omega t \le \beta$	2(b)
Constant	$A = -\dfrac{V_m}{Z}\sin(\alpha - \theta)e^{\alpha/\omega\tau}$	α = delay angle (rad)	2(b)
Average, half-wave	$I_o = \dfrac{1}{2\pi}\displaystyle\int_\alpha^\beta i\, d(\omega t)$	I_o = average load current (A) β = extinction angle, $i(\beta) = 0$	2(b)
Rms, half-wave	$I_{rms}^2 = \dfrac{1}{2\pi}\displaystyle\int_\alpha^\beta i^2\, d(\omega t)$	I_{rms} = rms load current (A)	2(b)
Load power	$P = I_{rms}^2 R$	P = load power (W)	2(b)
Half-wave output	$V_o = \dfrac{V_m}{2\pi}(\cos\alpha - \cos\beta)$	V_o = average output voltage (V)	2(b)

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Title	Equation	Notation	Ref.
★ **Controlled bridge, R load**	$V_o = \frac{V_m}{\pi}(1+\cos\alpha)$		2(c)
Rms, controlled bridge	$V_{rms} = \frac{V_m}{\sqrt{2}}\sqrt{1-\frac{\alpha}{\pi}+\frac{\sin 2\alpha}{2\pi}}$	V_{rms} = rms output voltage (V)	2(c)
Diode bridge	$V_o = \frac{2V_m}{\pi}$	RL load keeps the same V_o	7(c)

Table 3: Single-phase rectifiers

4. Three-phase rectifiers:

Title	Equation	Notation	Ref.
★ **Diode bridge**	$V_o = \frac{3V_m}{\pi} = 0.955V_m$	V_o = average output voltage (V) V_m = peak line-to-line voltage (V)	3(a)
★ **Controlled bridge**	$V_o = \frac{3V_m}{\pi}\cos\alpha$	α = delay angle (rad)	3(d)
Harmonics (controlled)	$V_n = \frac{3V_m}{\pi}\frac{2}{n^2-1}\sqrt{\cos^2\alpha + n^2\sin^2\alpha}$	V_n = peak voltage of harmonic n (V) n = harmonic order (6, 12, 18, ...)	3(d)
Harmonics (diode)	$V_n = \frac{2V_o}{n^2-1}$	$\alpha = 0$	8(d)
Harmonic current	$I_n = \frac{V_n}{\sqrt{R^2+(n\omega L)^2}}$	I_n = peak current of harmonic n (A) R = load resistance (Ω) ω = angular frequency of the source (rad/s) L = load inductance (H)	3(d)
Rms load current	$I_{rms}^2 = I_o^2 + \sum_n \left(\frac{I_n}{\sqrt{2}}\right)^2$	I_{rms} = rms load current (A) I_o = average load current, V_o/R (A)	8(d)
Rms diode current	$I_{D,rms} = \frac{I_{rms}}{\sqrt{3}}$	$I_{D,rms}$ = rms diode current (A) each diode on for 120°	8(d)
Line current	$i_a = i_{D1} - i_{D4}$	i_a = line current of phase a (A) i_{D1}, i_{D4} = currents of the two diodes on phase a (A)	3(a)

Table 4: Three-phase rectifiers

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5. Dc-dc converters:

Title	Equation	Notation	Ref.
Linear regulator	$V_o = V_{ref}\left(1 + \frac{R_1}{R_2}\right)$	V_o = output voltage (V) V_{ref} = reference voltage (V) R_1, R_2 = divider resistors (Ω)	4(a)
Regulator loss	$P_{loss} = (V_s - V_o)I_L,\ \eta \approx \frac{V_o}{V_s}$	P_{loss} = power lost in the regulator (W) V_s = input voltage (V) I_L = load current (A) η = efficiency	4(a)
★ **Buck output**	$V_o = DV_s$	D = duty ratio	4(b)
Buck ripple current	$\Delta i_L = \frac{V_o(1-D)}{Lf}$	Δi_L = peak-to-peak inductor ripple current (A) L = inductance (H) f = switching frequency (Hz)	4(b)
★ **Diode current**	$I_D = (1-D)I_L$	I_D = average diode current (A) I_L = average inductor current (A)	4(b)
Switch current	$I_S = DI_L$	I_S = average switch current (A)	4(b)
Inductor extremes	$I_{max,min} = I_L \pm \frac{\Delta i_L}{2}$	$I_{max,min}$ = largest and smallest inductor current (A)	8(b)
★ **Buck ripple voltage**	$\frac{\Delta V_o}{V_o} = \frac{1-D}{8LCf^2}$	ΔV_o = peak-to-peak output ripple (V) C = output capacitance (F)	8(b)
★ **Boost output**	$V_o = \frac{V_s}{1-D}$		4(c)
Boost inductor current	$I_L = \frac{V_o^2}{V_s R}$	R = load resistance (Ω)	4(c)
Boost ripple current	$\Delta i_L = \frac{V_s D}{Lf}$		4(c)
★ **Boost ripple voltage**	$\frac{\Delta V_o}{V_o} = \frac{D}{RCf}$		4(c)
SEPIC	$V_o = V_s\frac{D}{1-D}$	$V_{C1} = V_s$	4(d)
Cuk	$V_o = -V_s\frac{D}{1-D}$	$V_{C1} = V_s + \lvert V_o\rvert$	4(d)

Table 5: Dc-dc converters

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6. Inverters:

Title	Equation	Notation	Ref.
Square-wave harmonics	$V_n = \frac{4V_{dc}}{n\pi}$	V_n = peak voltage of harmonic n (V) V_{dc} = dc source (V) n = harmonic order, odd	5(d)
★ **Square-wave THD**	$THD = 48.3\ \%$	THD = total harmonic distortion $V_{1,rms} = 0.900V_{dc}$	5(d)
Cascade output	$v_o = v_1 + v_2 + v_3$	v_o = output voltage (V) v_1, v_2, v_3 = outputs of the three cells (V) k = number of cells; levels $= 2k + 1$	5(b)
Modulation ratios	$m_a = \frac{V_{sine}}{V_{tri}},\ m_f = \frac{f_{carrier}}{f}$	m_a = amplitude ratio V_{sine} = peak of the reference sine (V) V_{tri} = peak of the carrier triangle (V) m_f = frequency ratio $f_{carrier}$ = carrier frequency (Hz) f = output (fundamental) frequency (Hz)	5(c)
★ **PWM fundamental**	$V_1 = m_a V_{dc}$	V_1 = peak fundamental output voltage (V) $m_a \leq 1$	5(c)
Harmonic current at the carrier	$I_n = \frac{V_n}{n\omega L}$	I_n = peak current of harmonic n (A) ω = fundamental angular frequency (rad/s) L = load inductance (H) inductive at high n	5(c)
Resonance	$\omega_0 = \frac{1}{\sqrt{LC}}$	ω_0 = tuned frequency (rad/s) C = filter capacitance (F)	6(a)
★ **Series filter reactance**	$X_n = \omega_0 L\left(n - \frac{1}{n}\right)$	X_n = filter reactance at harmonic n (Ω)	6(a)
Load harmonic	$\frac{V_{o,n}}{V_1} = \frac{1}{n}\frac{R}{\sqrt{R^2 + X_n^2}}$	$V_{o,n}$ = load voltage at harmonic n (V) R = load resistance (Ω) square-wave bridge	6(a)
★ **Capacitor peak**	$V_{C,max} \approx \frac{I_1}{\omega_0 C}$	$V_{C,max}$ = peak capacitor voltage (V) I_1 = peak fundamental load current (A)	6(a)

Table 6: Inverters

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7. Dc motor by field chopper:

Title	Equation	Notation	Ref.
Field current	$I_f = \frac{\delta V_s}{R_f}$	I_f = field current (A) δ = chopper duty ratio V_s = supply voltage (V) R_f = field resistance (Ω)	8(c)
Armature equation	$V_a = E_b + I_a R_a$	V_a = armature voltage (V) I_a = armature current (A) R_a = armature resistance (Ω) E_b = back voltage, $K_v I_f \omega$ (V)	8(c)
★ **Speed**	$\omega = \frac{V_a - I_a R_a}{K_v I_f}$	ω = speed (rad/s) K_v = motor constant	8(c)

Table 7: Dc motor by field chopper

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8. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Distortion factor	$DF = \frac{I_1}{I_{rms}}$
Total harmonic distortion	$THD = \frac{\sqrt{I_{rms}^2 - I_1^2}}{I_1}$
DF and THD	$DF = \sqrt{\frac{1}{1+THD^2}}$
Storage charge	$Q_{RR} \approx \frac{1}{2}\frac{di}{dt}t_{rr}^2$
Peak from the charge	$I_{RR} = \sqrt{2Q_{RR}\frac{di}{dt}}$
Two-transistor model	$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1-(\alpha_1+\alpha_2)}$
RL load current	$i = \frac{V_m}{Z}\sin(\omega t - \theta) + Ae^{-\omega t/\omega\tau}$
Controlled bridge, R load	$V_o = \frac{V_m}{\pi}(1+\cos\alpha)$
Diode bridge	$V_o = \frac{3V_m}{\pi} = 0.955V_m$
Controlled bridge	$V_o = \frac{3V_m}{\pi}\cos\alpha$
Buck output	$V_o = DV_s$
Diode current	$I_D = (1-D)I_L$
Buck ripple voltage	$\frac{\Delta V_o}{V_o} = \frac{1-D}{8LCf^2}$
Boost output	$V_o = \frac{V_s}{1-D}$
Boost ripple voltage	$\frac{\Delta V_o}{V_o} = \frac{D}{RCf}$
Square-wave THD	$THD = 48.3\ \%$
PWM fundamental	$V_1 = m_a V_{dc}$
Series filter reactance	$X_n = \omega_0 L\left(n - \frac{1}{n}\right)$
Capacitor peak	$V_{C,max} \approx \frac{I_1}{\omega_0 C}$
Speed	$\omega = \frac{V_a - I_a R_a}{K_v I_f}$

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9. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
I_{rms}	rms of the whole wave (A)	Rms of a distorted wave
I_n	rms value of harmonic n (A)	Rms of a distorted wave
DF	distortion factor	Distortion factor
I_1	rms fundamental (A)	Distortion factor
THD	total harmonic distortion	Total harmonic distortion
pf	power factor	Power factor
P	average power (W)	Power factor
S	apparent power, $V_{rms}I_{rms}$ (VA)	Power factor
θ_1	angle of the fundamental current	Power factor
CF	crest factor	Crest factor
I_{peak}	peak of the wave (A)	Crest factor
V_1	rms source voltage (V)	Power with a sine source
t_{rr}	reverse recovery time (s)	Recovery time
t_a	time to the reverse peak (s)	Recovery time
t_b	fall time after the peak (s)	Recovery time
I_{RR}	peak reverse current (A)	Peak reverse current
di/dt	rate of fall of current (A/s)	Peak reverse current
Q_{RR}	stored charge removed during recovery (C)	Storage charge
I_A	anode current (A)	Two-transistor model
α_1, α_2	common-base gains of the two transistors	Two-transistor model
I_G	gate current (A)	Two-transistor model
I_{CBO1}, I_{CBO2}	leakage currents of the two transistors (A)	Two-transistor model
Z	load impedance (Ω)	Impedance and angle
R	load resistance (Ω)	Impedance and angle
L	load inductance (H)	Impedance and angle
ω	angular frequency, $2\pi f$ (rad/s)	Impedance and angle
θ	load angle (rad)	Impedance and angle
i	load current (A)	RL load current
V_m	peak source voltage (V)	RL load current
t	time (s)	RL load current
A	constant set by $i(\alpha) = 0$ (A)	RL load current
τ	L/R (s)	RL load current
α	delay angle (rad)	Constant
I_o	average load current (A)	Average, half-wave
β	extinction angle, $i(\beta) = 0$	Average, half-wave
I_{rms}	rms load current (A)	Rms, half-wave

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Symbol	Meaning	First used in
P	load power (W)	Load power
V_o	average output voltage (V)	Half-wave output
V_{rms}	rms output voltage (V)	Rms, controlled bridge
V_m	peak line-to-line voltage (V)	Diode bridge
V_n	peak voltage of harmonic n (V)	Harmonics (controlled)
n	harmonic order (6, 12, 18, …)	Harmonics (controlled)
I_n	peak current of harmonic n (A)	Harmonic current
$I_{D,rms}$	rms diode current (A)	Rms diode current
i_a	line current of phase a (A)	Line current
i_{D1}, i_{D4}	currents of the two diodes on phase a (A)	Line current
V_{ref}	reference voltage (V)	Linear regulator
R_1, R_2	divider resistors (Ω)	Linear regulator
P_{loss}	power lost in the regulator (W)	Regulator loss
V_s	input voltage (V)	Regulator loss
I_L	load current (A)	Regulator loss
η	efficiency	Regulator loss
D	duty ratio	Buck output
Δi_L	peak-to-peak inductor ripple current (A)	Buck ripple current
f	switching frequency (Hz)	Buck ripple current
I_D	average diode current (A)	Diode current
I_L	average inductor current (A)	Diode current
I_S	average switch current (A)	Switch current
$I_{max,min}$	largest and smallest inductor current (A)	Inductor extremes
ΔV_o	peak-to-peak output ripple (V)	Buck ripple voltage
C	output capacitance (F)	Buck ripple voltage
V_{dc}	dc source (V)	Square-wave harmonics
v_o	output voltage (V)	Cascade output
v_1, v_2, v_3	outputs of the three cells (V)	Cascade output
k	number of cells	Cascade output
m_a	amplitude ratio	Modulation ratios
V_{sine}	peak of the reference sine (V)	Modulation ratios
V_{tri}	peak of the carrier triangle (V)	Modulation ratios
m_f	frequency ratio	Modulation ratios
$f_{carrier}$	carrier frequency (Hz)	Modulation ratios
f	output (fundamental) frequency (Hz)	Modulation ratios
V_1	peak fundamental output voltage (V)	PWM fundamental
ω_0	tuned frequency (rad/s)	Resonance
C	filter capacitance (F)	Resonance
X_n	filter reactance at harmonic n (Ω)	Series filter reactance

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Symbol	Meaning	First used in
$V_{o,n}$	load voltage at harmonic n (V)	Load harmonic
$V_{C,max}$	peak capacitor voltage (V)	Capacitor peak
I_1	peak fundamental load current (A)	Capacitor peak
I_f	field current (A)	Field current
δ	chopper duty ratio	Field current
V_s	supply voltage (V)	Field current
R_f	field resistance (Ω)	Field current
V_a	armature voltage (V)	Armature equation
I_a	armature current (A)	Armature equation
R_a	armature resistance (Ω)	Armature equation
E_b	back voltage, $K_v I_f \omega$ (V)	Armature equation
ω	speed (rad/s)	Speed
K_v	motor constant	Speed

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